

Competing for Loan Informal Seniority: Theory and Evidence*

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Abstract

Using administrative data covering the universe of retail loans in Brazil, we document two facts that challenge standard models of non-exclusive lending. First, universal default is rare: 84% of borrowers who default on one credit card continue to repay other cards. Second, borrowers systematically prioritize repayment, favoring cards with higher credit limits, lower interest rates, or bundled with other products. Motivated by these facts, we develop a model of non-exclusive lending in which lenders compete for repayment priority through contract terms and cross-selling. We show that there can be over- or underprovision of relationship benefits and that equal-recovery bankruptcy policies may reduce welfare. Our back-of-the-envelope calculation implies that the credit limit one year after origination are lower by 7% relative to the first best. Imposing equal recovery would lower credit limits by 2.5%.

Keywords: debt dilution, non-exclusive contracts, selective default, bankruptcy law

JEL Classifications: G21, G51, D62, K35

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1 Introduction

Individuals frequently hold multiple credit cards, student loans, mortgages, or auto loans, often financed by different lenders. On the one hand, such borrowing patterns may reflect financial flexibility: borrowers are not locked into past choices and can shop across institutions for more attractive products. On the other hand, non-exclusive lending can give rise to debt dilution: by originating new loans, lenders may impose a default externality on their competitors, as additional borrowing can affect a borrower's ability and incentives to repay existing obligations held by other institutions (e.g., [Bizer and DeMarzo, 1992](#); [Green and Liu, 2021](#)).

Although the importance of debt dilution is well established in corporate finance theory, little is known about its empirical relevance or how it operates in the retail market. Unlike corporate credit, which typically includes cross-default triggers, households can be delinquent on a specific debt without repercussions from their other lenders.¹ Retail borrowers can thus choose which creditors to repay. A key obstacle to studying non-exclusive lending in this context is data availability: the vast majority of credit registers for retail loans—such as, for instance, the Federal Reserve Board's FR Y-14M dataset—use a different borrower identifier for different loan types issued by different lenders. As a result, it is often impossible to use credit registers to reconstruct a customer's loan portfolio.

To the best of our knowledge, this paper is the first to analyse, both empirically and theoretically, how non-exclusive lending affects default decisions, contract terms, and welfare in the retail loan market. Our analysis leverages administrative data from the Central Bank of Brazil. Unlike the vast majority of credit registries, we observe all loans held by a given borrower across all financial institutions, including mortgages, car loans, consumer loans, and credit cards.

In the first part of the paper, we show that selective default is quantitatively relevant and analyse its determinants. To do so, we reconstruct each borrower's complete loan portfolio and document new facts about repayment behaviour in the presence of multiple lending relationships. We first show that, contrary to what has been assumed in theoretical models developed for the corporate loans market, borrowers in financial distress do not default on all their debt. For instance, in 76% of cases, borrowers with two active credit cards default only on one of them. Second, we show that borrowers prioritize the repayment of more favorable credit contracts: conditional on selective default, they are more likely to continue servicing cards with higher available limits, lower interest rates, those issued by digital banks, and those provided by lenders that

¹See, for instance, the Federal Reserve of the United States post: <https://libertystreeteconomics.newyorkfed.org/2025/03/when-the-household-pie-shrinks-who-gets-their-slice/>. The 1999 Visa Bankruptcy Survey mentions that 66.7% of credit card loans were charged off for reasons other than bankruptcy.

also supply other financial products, such as auto loans. We refer to the loan being prioritized relative to others as having informal seniority.

In the second part of the paper, we study the implications of competition for informal seniority. To do so, we relax the simultaneous default and equal-recovery assumption made in models of non-exclusive lending (e.g., [Bizer and DeMarzo, 1992](#); [Green and Liu, 2021](#)) by allowing borrowers to prioritize the repayment of contracts with more favorable terms. In the model, lenders impose default externalities on their competitors through two distinct channels. The first is the standard debt-dilution channel: originating a new loan increases overall indebtedness and raises default probabilities across all outstanding loans. The second operates through repayment prioritisation: issuing a new loan affects which obligations the borrower chooses to repay. We show that distinguishing between these channels is important, as they generate different positive and normative implications.

The key insight of our model is as follows. We study an environment in which lenders can pay a cost to increase customers' continuation value from repaying their debt. In the baseline model, we interpret these relationship benefits as improving the quality of the bank mobile application or the credit card reward system; but our results also extend to settings in which the investment takes the form of cross-selling or offering more favourable terms (credit limits and interest rates) in the future. Following the industrial organisation literature, we refer to such investments as investments in credit card quality or investment in relationship benefits.²

We show that there can be over- or underprovision of relationship benefits relative to the first-best due to two opposing forces. First, the standard debt dilution channel implies underprovision. Due to the default externality, lenders expect borrowers to leverage too much relative to the first-best, and thus to default too frequently.³ Holding default probabilities fixed, this reduces the expected duration of the lending relationship and thus the marginal benefit of relationship benefits for the borrower, thereby weakening incentives to invest in card quality. Second, competition for repayment priority leads to overprovision. When borrowers prioritize repaying the loan with the highest continuation value, each lender has incentives to invest more than their competitors. This competition can lead to an arms race, resulting in overprovision of relationship benefits.

In that setup, we show that standard bankruptcy regulation features, such as cross-default provisions and equal recovery for loans of the same formal seniority, can reduce welfare. By imposing the same recovery rate on non-performing loans, the regulator cancels the overprovision channel. This regulation can reduce welfare when the over-

²See, for instance, [Berry and Waldfogel \(2010\)](#).

³As in [Bizer and DeMarzo \(1992\)](#), a lender internalizes that originating a new loan increases the borrower's default probability on its own loan, but does not internalize the impact on the default probabilities of loans held by other lenders.

provision channel is mild relative to the underprovision one. In practice, the strength of the overprovision channel depends on the default sensitivities to relationship benefits and the lenders' profits conditional on survival, while the underprovision channel depends on the equilibrium average default probability in the data relative to the first-best. Using our model together with our empirical estimates, a sufficient-statistic approach implies that the relationship benefits (in our quantification, the credit limit one year after origination) are lower by 7% relative to the first-best. The underprovision channel dominates and lower credit limits by 9.5%, the overprovision partially mitigates it by increasing it by 2.5%. Imposing equal recovery would eliminate the overprovision channel and thus lower credit limits by 2.5%.

We begin our empirical analysis by documenting three facts suggesting that debt dilution is a first-order force in the credit card market. First, multi-lender borrowing is pervasive: among 85 million credit card users in June 2022, 39.4 million (46.5%) maintained active cards issued by two or more institutions, collectively accounting for over 70% of the total outstanding credit card balance. Second, borrowing through credit cards is common. Approximately 50% of borrowers revolve their balance at the end of the month.⁴ Finally, default rates are high: during the sample period, 9% of credit card borrowers default on at least one card, defined as failing to make the minimum payment for 90 days or more.

We then derive new facts about repayment behavior in the credit card market. First, we highlight that borrowers with multiple active cards rarely default on all their active cards. Indeed, among those who default on at least one card, only 16% default on all of their active cards, meaning that 84% continue to make payments on at least one card. This pattern is not driven by lenders preemptively cancelling credit cards. Defaulting on one card has little effect on the credit limits or interest rates of other cards, whereas limits and monthly usage drop to zero on cards that borrowers do not repay.

Second, we show that the likelihood of defaulting on at least one card increases with the number of cards held. For example, in a given month, 7% of individuals holding a single credit card are in default (more than 90 days past due), compared with 17% of individuals holding four or more cards. This positive correlation may reflect adverse selection, whereby *ex ante* riskier borrowers are more likely to hold multiple cards, or moral hazard, whereby holding additional cards causally increases default risk. Understanding which channel explains the correlation is central, as when recovery rates are close to zero, moral hazard is the key driver of the debt dilution problem.⁵

To estimate the causal effect of obtaining an additional card on default and to measure the extent of debt dilution, we exploit a novel, plausibly exogenous variation in the

⁴About 20% of monthly credit card purchases are not repaid at the end of the month due to revolving balances and card-linked buy-now-pay-later arrangements.

⁵When recovery rates are not equal to zero, debt dilution can happen by lowering the recovery rate, even if issuing a new loan does not affect default.

number of cards owned within a difference-in-differences framework. The variation arises when banks form partnerships with a firm. The arrangement requires employees to receive their wages through the partner bank, thereby increasing the likelihood that they obtain a card from that bank. We compare two groups of borrowers employed at the same firm around the time the partner bank changed: those who already had a relationship with the partner bank before the contract (control) and those who did not (treatment). The identification assumption is that, absent the partner bank's change, the two groups of workers would have continued to follow parallel trends. Thus, the observed relative change in default probability between them can be attributed to the increased likelihood of the treatment group obtaining a new card. We find that the relative default probability of the treated borrowers one year after the treatment increases by 1.1 percentage points. Given that treated borrowers acquire, on average, 0.11 new card lenders, obtaining a new card lender increases default rates by 10 percentage points.⁶

Finally, provide suggestive evidence that lenders can compete to gain loan seniority (i.e., be the bank the borrower prioritizes repaying) using contract terms and cross-selling other financial products. We document this pattern by looking for each individual who is in financial trouble which card they choose to repay. Estimating regressions of default choices among selective defaulters holding multiple credit cards, we find that, on average, these borrowers default on 45% of their cards and systematically prioritize repayment of cards with higher credit limits (by 4 percentage points), those originated by digital banks (by 14 percentage points), and those issued by lenders that also provide other financial products (e.g., a 14-percentage-point higher repayment probability when the lender also holds the borrower's mortgage and a 5-percentage-point higher probability when the lender holds the borrower's auto loan).

These estimates should be interpreted with caution, as they are not recovered using quasi-experimental variation. However, the richness of our data allows us to account for a wide range of confounding factors by exploiting variation arising from selective defaulters. The coefficient on, for instance, credit limit is recovered by examining the default decision of the same individual holding multiple cards with different credit limits. This feature allows us to have borrower-time and, exploiting multiple observations, bank-time fixed effects, allowing us to capture differences in debt collection technology. Any remaining bias in the estimated coefficients of contract terms would have to arise from unobserved product-level characteristics correlated with both repayment priority and contract design (e.g., rewards programs associated with premium cards, which also tend to have higher credit limits).

Even if our regression coefficients were to be biased, these findings still indicate that

⁶The impact on default probability is of a similar magnitude to the one found in [De Giorgi et al. \(2023\)](#) for retail borrowers in Mexico.

default choices across lenders are not random but instead correlate with lender characteristics. Lenders can plausibly shape those characteristics, consistent with competition for repayment priority. Our empirical results also point to an additional rationale for banks to cross-sell financial products: cross-selling effectively confers implicit seniority in repayment. This mechanism differs from existing explanations for cross-selling, which emphasise account stickiness or the value of lending relationships for future business.⁷

Guided by those key features of the credit card market, we develop a model to analyse the implications of competition for loan seniority. The baseline version of the model features two decision periods. In the first period, lenders simultaneously offer loan contracts, and borrowers may accept any subset of these offers. The key friction is that they cannot commit to exclusive contracts. In the second period, borrowers choose which loan to repay, subject to an income constraint.

For exposition purposes, we refer to the contract as a credit card contract. A credit card contract typically includes a credit limit for the initial period, an interest rate, and other card features that benefit the borrower in the second period. We refer to these features collectively as card quality or relationship benefits. In practice, the quality of a card can refer to its rewards system, the growth of its credit limit, or to other features associated with the card, such as the bank's mobile application or the promise of obtaining another financial product (e.g., a mortgage). We explore each of these interpretations by extending our baseline model to multiple periods. We show that the key implications in these extensions can be derived from the baseline two-period model by examining a one-dimensional card quality variable whose value depends on the borrower's repayment behavior in the second period.

In our model, borrowers' card choice is driven by the following trade-off. By holding multiple cards, borrowers can increase leverage and consume more in the first period, at the cost of potentially being able to repay only a subset of their obligations. Lenders optimally cancel the credit card if the borrower does not repay their balance, and borrowers derive some utility from the relationship benefits in the second period if they repay the credit card.

Lenders can choose the loan interest rate and the credit limit, and can pay a cost to increase the borrower's perceived value of repaying the card (i.e., the card quality). When issuing a loan, lenders recognise that the borrower may also obtain credit from competing lenders and, therefore, may be able to repay only a subset of their outstanding obligations.

Due to a default externality, lenders have incentives to originate too much, as they do not internalise the cost they impose on their competitors. As a result, borrowers' default probabilities are inefficiently high, and contract terms deviate from their first-

⁷See, for instance, [Drucker and Puri \(2005\)](#), [Puri and Rocholl \(2008\)](#), and [Basten and Juelsrud \(2023\)](#).

best level. The direction of this distortion is ambiguous: lenders either over- or underprovide relationship benefits. Incentives to overprovide arise because of an arms race: offering a higher quality relative to other banks increases the probability that the borrowers prioritize repaying this card. Incentives to underprovide arise from lenders expecting borrowers to default more frequently due to the default externality, thereby benefiting less often from the credit card's quality.

Due to these competing channels, standard personal bankruptcy procedures, which treat loans of the same contractual seniority equally (i.e., making them share the same recovery rate), can have unintended consequences. Indeed, by setting the same recovery rate for each loan, those regulations shut down the competition for the loan seniority channel. However, this overinvestment channel can compensate for the distortions created by the underprovision channel. As a result, in situations where the overprovision channel is not too strong, such regulation can be welfare-decreasing.

Finally, we combine our estimates with a generalised version of the model to quantify the distortions in credit limits arising from non-exclusivity, using a sufficient-statistic-inspired approach. We show that, within a broad class of models, the deviation of card quality from the first best can be expressed as a weighted sum of two components: the ratio of the sensitivity of survival to contract terms, which captures the overprovision channel, and the survival probability observed in the data, which captures the underprovision channel. This sum is normalized by the survival probability in the first best. To operationalize this decomposition, we use our estimates of the sensitivity of default to credit limits together with average default rates to quantify the first two components. For the survival probability in the first best, we use our difference-in-differences estimates to construct a back-of-the-envelope measure of the average default rate among borrowers who hold two credit cards in the data but would optimally hold only one in the first best.

The rest of the paper is organized as follows. [Section 2](#) provides a literature review. [Section 3](#) describes the institutional setting and the data. [Section 4](#) presents the descriptive statistics and stylized facts. [Section 5](#) reports the difference-in-differences results. [Section 6](#) introduces the model as well as the equilibrium and welfare analysis. Finally, [Section 7](#) concludes.

2 Related Literature

We connect with several strands of the literature. First, we build on papers that study non-exclusive lending. The contributions in this field are mostly theoretical (e.g., [Bizer and DeMarzo, 1992](#); [Parlour and Rajan, 2001](#); [DeMarzo and He, 2021](#); [Green and Liu,](#)

2021)).⁸ A common modelling assumption in those papers is that default and recovery rates are the same for all loans with the same formal seniority. We contribute to the literature by looking at a situation in which borrowers do not universally default, and by allowing lenders to compete for repayment (i.e., loan informal seniority) through contract terms.⁹ The closest theoretical paper is [Donaldson et al. \(2025\)](#), which studies the borrower’s choice between debt with and without anti-dilution covenants. The key trade-off is as follows. Choosing an anti-dilution covenant prevents borrowers from overinvesting. However, this commitment comes at the cost of being unable to borrow when they have a profitable investment opportunity later on. In our paper, we focus on a different trade-off (i.e., how lenders design contracts so that the borrower prioritises repaying them), and its impact on bankruptcy regulation.¹⁰ Compared to [Donaldson et al. \(2025\)](#), none of the channels we discuss (the over- and under-provision channels) is inherently bad or good; they both distort contract terms relative to the first best, but having both is useful, as one can compensate for the distortion generated by the other.

Our paper also relates to the theoretical literature on bankruptcy regulation and the choice of default versus bankruptcy. A standard rationale for bankruptcy procedures is to mitigate coordination failures among creditors and to prevent inefficient collection efforts and premature liquidation of assets (e.g., [Aghion et al., 1992](#); [White, 2007](#)). In particular, the theoretical literature justifies cross-default provision as a way to eliminate priority races across lenders (see, for instance, [Ayotte et al., 2025](#)). We show that this priority race can increase welfare by leading to the provision of higher-quality banking services.

Our paper contributes to the empirical strands of the literature studying credit card borrowing, bankruptcy, and cross-selling. Recent papers on credit cards have focused on the pricing of credit cards—see, for instance, [Ru and Schoar \(2016\)](#), [Stango and Zinman \(2016\)](#), and [Drechsler et al. \(2025\)](#) for reduced-form analysis, or [Galenianos and Gavazza \(2022\)](#) and [Nelson \(2025\)](#) for structural models—through a competition, behavioral, or asymmetric information angle. Our paper is closer to [Ponce et al. \(2017\)](#) or [Gathergood et al. \(2019\)](#), which shows that borrowers use the rule of thumb when repaying their debt. Our results focus on delinquent loans and suggest that when the costs are higher (i.e., losing the credit cards), the borrower behavior becomes more strategic.

⁸We focus on the debt dilution aspect of non-exclusivity. Other studies have highlighted benefits of non-exclusivity, such as the reduction of hold-up problems (e.g., [Ioannidou and Ongena, 2010](#)) and the provision of insurance against idiosyncratic shocks to lenders’ liquidity (e.g., [Detragiache et al., 2000](#)).

⁹Previous models in the literature are thus better suited to the commercial loan market, where defaulting on one loan typically triggers a bankruptcy procedure across all other loans, whereas our model is better suited to the retail market, where such bankruptcies are often too costly for lenders to use widely.

¹⁰We show that this can lead to under- or over-provision of relationship benefits (credit limit, interest rates, probability of selling a mortgage). Because of debt dilution, borrowers value less those relationship benefits (underprovision), but increasing relationship benefits lead the borrower to prioritise repaying them (overprovision).

The literature on delinquent loans in household finance largely focuses on personal bankruptcy (see, e.g., [Antill and Hunter, 2023](#); [Indarte and Kanz, 2024](#); [Argyle et al., 2025](#); [Ben-Ishai et al., 2026](#)), which typically features cross-default provisions. We contribute to the literature by examining defaults outside personal bankruptcy, which constitute the vast majority of credit card defaults and are characterised by the presence of selective default.¹¹ While selective default is acknowledged by the regulation¹² it is relatively understudied. The empirical literature on selective default include [Andersson et al. \(2013\)](#), [Degryse et al. \(2016\)](#), [Schlegl et al. \(2019\)](#) and [De Giorgi et al. \(2023\)](#). [Andersson et al. \(2013\)](#) use OCC data to document a pecking order in the retail loan market: borrowers tend to prioritise repaying their mortgage over their credit card debt. We complement this paper by looking at repayment priority within loans of the same formal seniority (i.e., credit cards), studying how selective defaults relate to contract terms broadly defined, and analysing informal seniority implications for optimal bankruptcy regulations. [Degryse et al. \(2016\)](#) focuses on corporate loans and uses a difference-in-difference approach to show that lenders' willingness to lend decreases when a firm obtains loans from other banks. [Schlegl et al. \(2019\)](#) studies the pecking order for debt issued by the government: the government prioritise repaying the IMF, then banks, then other governments. [De Giorgi et al. \(2023\)](#) is closer to our paper and focuses on retail loans. They use data from a private bank and discontinuities in the bank's approval rule to show that issuing a new credit card increases the default probability on existing loans. Their estimates are of similar magnitude to those from our difference-in-differences design around changes in firm-bank relationships. We add to these empirical papers by documenting and analysing the implications of a new channel through which lenders impose a default externality on others: the competition for loan seniority channel.

By showing that other loan products affect credit card repayment decisions, we also contribute to the banking literature on cross-selling (e.g., [Robles-Garcia et al., 2022](#); [Basten and Juelsrud, 2023](#)). The existing literature has focused on account stickiness ([Puri and Rocholl, 2008](#)) or strengthening of the relationship with borrowers ([Drucker and Puri, 2005](#)). In our setting, we highlight a source of complementarity across products that arises from cross-selling, leading to repayment prioritisation. To the best of our knowledge, this source of economies of scope has not been analysed empirically and theoretically.

¹¹The 1999 Visa Bankruptcy Survey mentions that 66.7% of credit card loans were charged off for reasons other than bankruptcy. [Dawsey and Ausubel \(2002\)](#) shows that 50.7% of all written-off accounts (and 45.8% of credit losses in dollars) were lost without the debtor filing for bankruptcy.

¹²For instance, the CARD Act (2009) restricts one credit card lender from repricing a consumer who defaults on another lender's card

3 Institutional Setting and Data

3.1 Data

We assemble data from different sources to construct a monthly panel data set of retail loans for all borrowers in Brazil. We focus our main analysis on 2019-2024 as we have better information on credit card operations during that period.

We obtain monthly loan-level data from the credit registry of the Central Bank of Brazil (*Sistema de Informações de Créditos*, SCR). The dataset contains the universe of bank loans—mortgages, personal credit, car loans, credit card loans, payroll-deductible loans, and others—above 200 BRL (around 35 USD). For each term loan, we observe outstanding loan amounts, interest rates, maturity, defaults, anonymized bank and borrower identifiers consistent across products, and the date of origination. For credit cards, we observe monthly credit card usage, revolving balances, amounts due each month, and credit limits.¹³ If a borrower holds multiple cards with the same bank in a given month (e.g., two cards from different card networks), we observe aggregate amounts across all of their cards. We also observe whether the borrower does not make the minimum payment due on their card at the end of the month.

To get information on the borrower characteristics, we supplement credit information with matched employer-employee data from the Ministry of Labor’s *Relação Anual de Informações Sociais* (RAIS). These data provide information on the wage, employer, and selected sociodemographic characteristics, including age, sex, and occupation type. For informal workers, we use the lender’s assessment of their income. After 2023, we also have data on the borrower’s credit score.

To obtain a tractable sample size, we draw a representative sample of the borrower population using a stratified sampling strategy to ensure sufficient coverage of borrowers experiencing financial distress. Within each cohort, defined at the calendar month level, we partition the population of borrowers into two groups and draw random samples. The first group comprises individuals who were current on all cards until failing to make a payment on at least one card in the cohort month. We designate these individuals, who may be transitioning into default (more than 90 days past due), as *borrowers nearing distress*. The second group comprises all remaining borrowers. Our focus on credit cards in the debt dilution analysis, and hence our sample construction, is motivated by the descriptive statistics reported in the next section (Section 3.2). We provide further details on the sampling procedure and sample characteristics in Appendix A1.

¹³Credit card interest rate information is available only when the borrower maintains a revolving balance. The accuracy of credit card limit data improved sharply after May 2021. In certain specifications, we will limit the data to the period following this month.

3.2 The Credit Card Industry in Brazil

In this subsection, we use aggregate, publicly available data to present descriptive statistics on the Brazilian credit card market and, where possible, compare them with those of the United States. These statistics suggest that debt dilution is a first-order force in the credit card market: individuals hold cards issued by multiple institutions and frequently borrow and default.

Prevalence of multi-lender borrowing. The credit card industry has expanded rapidly in Brazil. After a stagnant period, the number of active credit cards doubled and the total amount of credit card loans grew by 73% between 2018 and 2023 (Appendix Figure A1). The growth in the number of cards was driven by two factors: the addition of new users and the acquisition of cards from different banks by existing users. According to Central Bank of Brazil (2022), the number of credit card users increased by 31% between June 2019 and June 2022, from 65 million to 85 million (52% of the population aged 15 years or older).¹⁴ Over the same period, Panel A of Appendix Figure A2 shows that the share of users with active credit cards from at least two distinct financial institutions increased from 38% to 47%. Panel B shows that the share of outstanding credit card balances held by these users rose from 58% to 70%, consistent with higher average credit card usage among multi-lender users relative to single-lender users. Panel C shows that a substantial portion of the growth in credit card usage is associated with digital banks, which reduced the costs of applying for new cards.¹⁵ In contrast, in the United States, 82% of consumers used a credit card in 2023, and 76.5% of consumers held two or more cards.¹⁶

Credit card borrowing. Borrowing with a credit card is common. In Panel A of Appendix Figure A3, we decompose the total credit card balance into three parts: non-interest-bearing purchases, revolving debt, and buy now, pay later. Non-interest-bearing purchases refer to transactions made during the billing cycle that are fully due on the next due date and incur no interest charges if paid by that date; revolving debt refers to the accumulation of debt resulting from not paying the full balance of the monthly credit card bill; and buy now, pay later debt refers to card transactions where payment is split into installments, with or without interest charges. The composition of total credit card debt has remained stable, with non-interest-bearing purchases accounting for 80% of total debt, revolving debt for 10%, and buy now, pay later for 10%. Using our micro-level data, we show that 50% of borrowers revolve at the end of the month. This number is similar to the one in the United States (see, for instance, Lee

¹⁴We use population figures from the 2022 census.

¹⁵Buchak et al. (2018) and Fuster et al. (2019) document faster application processes among fintech lenders in the mortgage market.

¹⁶Data are drawn from the 2023 *Survey and Diary of Consumer Payment Choice* conducted by the Federal Reserve Bank of Atlanta. The information on the number of cards is not fully informative about multi-lender borrowing, since a consumer may hold multiple cards issued by the same financial institution.

and Maxted, 2025).

Credit card loans are a substantial part of households' balance sheets. In Panel B of Appendix Figure A3, we show that the participation of credit card debt in total household debt increased from 12.7% to 15% between 2018 and 2023.¹⁷ In the United States, credit cards accounted for 6% of total household debt in 2023.¹⁸

Default. Using our micro-level dataset (see Appendix Table A2), we show that 7% of credit cards (approximately 11% of credit card users) are in default in a given month, where default is defined as failing to make the minimum payment for 90 days. Conditional on borrowing (revolving and BNPL), the number goes to 15%. In the United States, default—defined as a credit card loan being charged off in a given year—occurs at a rate of around 5% (Drechsler et al., 2025) and rises to as high as 10% among borrowers with low FICO scores.

Appendix Figure A4 shows that the share of total balances in default for revolving credit card debt exceeded 50% in 2023, while the corresponding share for *buy now, pay later* balances reached 10%. The figure also shows that, although credit cards account for about 15% of total credit, they represent more than 50% of household debt overdue by more than 15 days, a share that has been increasing since 2018.

Household debt collection. The default procedure in Brazil is broadly similar to that in the United States.¹⁹ Following a missed credit card payment, lenders typically send internal reminders via email or phone. During this stage, the Consumer Protection Code restricts collection practices, prohibiting harassment and abusive exposure. Creditors generally wait at least 30 days before reporting the delinquency to credit bureaus. Subsequently, lenders may outsource or sell delinquent receivables to collection agencies or initiate judicial collection. In practice, few loans are recovered, as legal proceedings are costly and recovery rates are low.

4 Summary Statistics and Stylized Facts

In this section, we provide descriptive statistics and stylized facts obtained from the sample described in Section 3.1, covering the period from June 2019 to June 2022.

¹⁷Credit cards are the most common loan contract, accounting for more than 50% of total household loan contracts.

¹⁸New York Fed Consumer Credit Panel.

¹⁹While Brazil, like many emerging economies, does not have a personal bankruptcy law (Indarte and Kanz, 2024), this institutional difference is less consequential in practice, as the vast majority of household debt resolutions occur out of court through collections or private renegotiation rather than formal bankruptcy proceedings (Dawsey et al., 2013). Moreover, even in settings with a formal bankruptcy procedure, competition for seniority may occur prior to filing.

4.1 Summary Statistics

In [Table 1](#), we report summary statistics for groups of individuals owning different numbers of active cards issued by different institutions. We report these statistics separately for *borrowers nearing distress* (defined in [Section 3.1](#)) and for a random sample of the overall borrower population.

While 40% of the overall population holds credit cards from more than one financial institution, the incidence is significantly higher among *borrowers nearing distress*, at 53%. Within this group, 27% hold cards from two institutions, 14% from three, 6% from four, 3% from five, and another 3% from six or more institutions. [Figure 3](#) confirms the trend documented in [Section 3.2](#), namely that the number of active card lenders per borrower has been increasing since 2019.

Individuals with a larger number of credit card lenders are more likely to hold formal employment, have a higher monthly income, and are more financially integrated. For example, 36% of those with a single card lender are formally employed, compared to 46% among those with three card lenders. Wages exhibit an approximately inverted U-shaped relationship with the number of card lenders. Average monthly earnings rise from 3,757 BRL among individuals with a single card to 4,171 BRL among those with three cards, and subsequently decline to 3,812 BRL for individuals with four cards. Individuals with a greater number of card lenders exhibit a higher debt-service-to-income ratio (18%, 38%, and 59% for those with one, three, and five cards, respectively), as well as higher average limit utilization across their cards (54%, 58%, and 64%). They also rely more heavily on other forms of credit. For example, 7%, 16%, and 24% of individuals with one, three, or five cards, respectively, hold an auto loan, while the corresponding figures for consumer loans are 14%, 27%, and 42%.

Individuals in the *borrowers nearing distress* group tend to have lower incomes (earning roughly 1,000 BRL less per month) and are more likely to work informally (by about 5–10 percentage points). They are also more leveraged, exhibiting higher credit card limit utilization and a greater propensity to hold other types of loans.

In [Table 2](#), we report default rates in the month before the cohort by category of number of card lenders. We define default as not making the minimum payment on the card at the end of the month for more than 90 days. By construction, the share of borrowers with at least one card in default in the *borrower nearing distress* subsample is zero across all categories. In the general population, this share increases monotonically with the number of card lenders (from 7% among people owning one card to 15% among people owning four). However, there is no correlation between the number of credit cards and default on other loans. We interpret this as suggestive evidence that debt dilution is more substantial in the credit card market. Yet, in the *borrowers nearing distress* group, borrowers with one card lender have a higher probability of defaulting

on other loans compared to borrowers with more than one card lender. The latter is consistent with the fact that this population subsample uses their credit cards to repay other loans.

4.2 Four Facts About Default in the Credit Card Market

In this section, we provide four new stylized facts about default behavior in the credit card market. First, we show that many borrowers default on only a subset of their cards. Second, we show that the default probability of *borrowers nearing distress* six months after missing a payment increases with the number of cards, even after controlling for observable determinants of default. Third, we show that lenders do not cancel the credit limits of borrowers who defaulted on a competitor's card. Finally, we show that borrowers prioritize repayment of cards with more favorable terms and of cards issued by banks with which they have other credit relationships.

Fact 1: Many borrowers default on only a subset of their cards.

To derive the first fact, we focus on the *borrowers nearing distress* group. These individuals were in good standing with all their cards in month $t - 1$ and then miss a payment (overdue for more than 15 days) in time t . We investigate whether these individuals default on all or a subset of their loans six months later, defining a loan as in default if it is more than 90 days overdue. We use this measure of default to ensure that our results are not driven by borrowers who forget to repay on time. [Figure A5](#) shows the default incidence for alternative horizons. The results of the paper are not driven by the choice to measure default six months after the borrower first misses a payment, as the group of borrowers is relatively stable after 30 days.

In [Table 3](#), we show that 46% of *borrowers nearing distress* become more than 90 days overdue on at least one card, and that this share increases monotonically with the number of card lenders. This group of individuals can be decomposed into two subgroups: *universal defaulters*, who default on all cards (for instance, $11/45 = 24\%$ of the defaulters with two card lenders and $5/47 = 11\%$ among those with three card lenders), and *selective defaulters*, who default on some cards while continuing to repay others (76% and 89%, respectively).

While the term selective default could apply to individuals who repay a subset of their credit card loans, they may default on the card for which they lack sufficient resources to repay, suggesting that the default decision is driven solely by liquidity constraints rather than a deliberate choice. In this case, selective defaulters would default on the cards that constitute the majority of their debt, implying that the distribution of the share of the total balance in default would be concentrated above the 50% threshold. However, as shown in Appendix [Figure A8](#), the distribution does not follow this

pattern. Specifically, there is a large group of selective defaulters who default on cards accounting for less than 50% of their balance, suggesting that they had sufficient resources to pay these cards but chose not to.

Motivated by this fact, we create a more conservative measure of selective default. We identify borrowers who default on the cards that account for less than 50% of their total credit card debt. In [Table 3](#), we show that selective defaulters (21%) are comprised of two groups: those that default on less than 50% of their credit card balance (9% of borrowers), and those that default on more than 50% of their credit card balance (12% of borrowers). That is, *at least* $43\% = 9/21$ of selective defaulters made a deliberate repayment choice. The proportion of these borrowers also increases with the number of cards.

In [Table 4](#), we show that selective defaulters typically hold more cards, have a greater number of bank relationships, and a higher debt-service-to-income ratio than universal defaulters and no-defaulters. While selective defaulters have higher incomes and credit card balances compared to universal defaulters, their income and balances are lower than those of non-defaulters. Regarding other loan types, selective defaulters tend to have more outstanding loans than universal defaulters.

In [Figure A6](#), we show that all default measures are increasing over time, potentially due to the expansion in the number of credit cards. In [Appendix Figure A7](#), we present the same figures for individuals with at least two card lenders. In this subsample, the incidence of selective default is significantly higher, as we exclude individuals with only one card, who can only default universally. The figures confirm that the incidence of selective default is increasing over time.

Fact 2: The share of borrowers who default on at least one card increases with the number of card lenders, even after controlling for observable characteristics.

In [Table 2](#) and [Table 3](#), we show that default rates (at the borrower level) increase monotonically with the number of card lenders. For instance, six months after missing a payment, the probability that *borrowers nearing distress* default on at least one card is 43% for borrowers with a single card lender, 45% for those with two card lenders, and 47% for those with three card lenders. This relationship may be driven by factors such as income, liquidity, and leverage. In [Table 5](#), we try to control for these variables and additional covariates in the subsample of formal workers, for whom salary information is available. The positive association between borrower-level defaults and the number of card lenders persists even after controlling for wage, the debt-service-to-income ratio (a proxy for liquidity), the total-debt-to-wage ratio (a proxy for leverage), and other characteristics, including age, gender, and tenure at the last job.

Fact 3: Lenders do not cancel the credit limit of borrowers who defaulted on a competitor's card.

In [Figure 4](#), we show that defaults entail the loss of the defaulted card, as the limit available and monthly card usage drop to zero. However, select defaulters keep the limit and usage amounts of the cards they repay. In other words, as long as banks remain the prioritized lender, they continue to supply credit to selective defaulters.

In [Appendix Figure A9](#), we also show that defaulting on a credit card decreases the borrower's credit scores significantly. In an event study, we show that the probability of obtaining a new loan (including revolving card loans) also drops significantly after a default.

In [Section 6](#), we rationalize lenders' behavior in our model as follows. From a lender's perspective, observing a borrower default on another card while continuing to be repaid is not necessarily bad news. Such behavior reveals informal seniority relative to competing lenders. In addition, the loss of access to the defaulted card and the deterioration in the borrower's ability to obtain new credit increase the cost of default. Taken together, these mechanisms may reduce the likelihood of default on the repaid card.²⁰ This view is also expressed by market participants.²¹

Fact 4: Borrowers prioritize repaying cards with better terms.

To understand the characteristics of the bank-borrower relationship that predict default (or repayment), we use the sample of selective defaulters that repay a subset of their loans and data at the borrower-cohort-bank level. We estimate the following regression:

$$y_{icb} = \beta X_{icb} + \alpha_{ic} + \epsilon_{icb} \quad (1)$$

where y_{icb} is a binary variable that takes the value one if, six months after becoming overdue for more than 15 days on any card, individual i of cohort c defaults (i.e., is more than 90 days overdue) on the credit card issued by bank b . The vector X_{icb} contains borrower-bank characteristics measured in the month when the borrower becomes overdue for more than 15 days: limit available, limit utilization, global limit, loan balances, dummies for other active loan types, dummies for non-performance of other loan types, and dummies for bank types (digital or state-owned). We include

²⁰The lender that continues to be repaid may also gain market power as the borrower's number of banking relationships declines.

²¹The former CFO of Itaú, the largest private bank in Brazil in terms of total assets, mentions in an [interview](#) with Euromoney that: "Everyone in Brazil has four credit cards, and if they don't have money to buy groceries, they use all these cards.[...] But they pay the one from the bank they have the relationship with. Banks don't care if you have a negative score if you keep paying them. So, being the main bank determines the cost of credit. People underestimate the huge relationship between being the main bank and default risk. If you are the principal relationship bank, they might default on the other banks but you will be the last one."

borrower-cohort fixed effects (α_{ic}).

We report descriptive statistics of the regression variables in [Table 6](#). Selective defaulters default on 45% of their banks on average. In 69% of card relationships, borrowers used up all their credit card limits (100% of the limit); in 11%, more than 90% and less than 100%; in only 20% of card relationships, borrowers used less than 80% of the limit. 23% of relationships are with digital banks, 11% with state-owned banks, and the remainder (66%) with other banks. 6% of relationships contain an active payroll loan, 1% an active mortgage, 2% an auto loan, 12% a personal (consumer) loan, and 16% other loan types.

We report regression results in [Table 7](#). Borrowers tend to default more on cards for which there is less limit available. In particular, default probabilities increase sharply (by 40 percentage points) on the card for which there is no limit available (100% utilization). Recalling that the mean of the dependent variable is 45%, the limit available seems to be an important driver of repayment choice.

Borrowers tend to default less on banks where they have a positive global limit, which includes limits for other loan types, such as overdraft. In the specification using data from after May 2021, when information on limits is more reliable, the results show that borrowers are 14 percentage points less likely to default on digital banks and 13 percentage points more likely to default on government-owned banks. This pattern suggests that borrowers may value their relationships with digital banks more than with other types of banks, or that digital banks have more effective debt collection infrastructure.

In general, borrowers tend to default less on banks that offer other loan products: payroll (-7 percentage points), mortgage (-14 percentage points), auto loan (-5 percentage points), and other loans (-7 percentage points). However, default is more likely if the borrower has personal loans with the bank (3 percentage points). These results suggest that offering one loan type may increase the profitability of another, as it strengthens the bank's relationship with the borrower and improves its position in the repayment hierarchy. If a borrower has already defaulted on other products with a given bank, the borrower is substantially more likely to default on that bank's credit card than on a credit card issued by another bank.

In [Appendix Table A5](#), we also examine whether interest rates are a driver of default decisions. Interest rate information is available only for a subset of contracts, which reduces the sample size. We find that borrowers are 25 percentage points more likely to default on the card with the highest interest rate.

Omitted variable bias: The above results enlighten a correlation between relative contract terms and default probability. One should be cautious when interpreting the coefficients due to omitted variables. As we look at the decision of which loan to default on at the individual level, the key concern is unobserved product characteristics

that varies across credit cards sold by the same bank. For instance, the correlations between default probabilities and contract terms may be due to an omitted variable that varies at the product-bank level. Heterogeneity in debt collection procedure concerns is thus partially mitigated by the use of lender time fixed effects, as these features are largely fixed at the bank level. Yet it is possible that a bank offers, for instance, a higher credit limit on credit cards with a high-rewards program. This would create a positive correlation between credit limits and rewards systems. In that case, although the coefficients would not reflect the credit limit's causal impact, they still indicate that the lender can leverage certain contract characteristics to gain informal seniority. Similarly, the correlation between having other products and default could be due to better borrowers obtaining a card with a higher reward and a mortgage offer, creating a correlation between unobserved credit card characteristics and getting a mortgage. It is possible that the digital bank fixed effect comes from debt collection; we account for that possibility in the theoretical part of the paper.

Overall, while one should be cautious about interpreting the results as the causal impact of each characteristic, the results provide suggestive evidence that lenders can use some characteristics to gain informal seniority. The theoretical part of the paper analyses its implications and whether the type of characteristics used has an impact on welfare and the way one should regulate this market.

Additional facts about the credit card limits. We also report additional facts about credit limits. The aggregate credit limit, summing all credit cards, is around 3 times the borrower's income. The average credit limit per card is 1.5 times the borrower's income, regardless of the number of cards the borrower owns. In addition, lenders do not reduce the credit limit on existing cards after the borrower gets a new card and increase the credit limit over time, provided the borrower does not default. This suggests that lenders consider the possibility of the borrower getting additional cards when setting their initial limit.

5 Difference-in-Differences

In [Table 5](#), we show a positive relationship between the number of card lenders and default probabilities at the borrower level, even after controlling for observable default determinants. However, this relationship may arise from both unobserved borrower heterogeneity—for example, higher-risk borrowers may be more likely to obtain additional cards (adverse selection)—and the causal effect of forming new card relationships (moral hazard or burden of payment). Identifying the relative contribution of

these channels is crucial, as the latter mechanism has a direct effect on debt dilution.²²

Using quasi-experimental variation, we show that the debt dilution channel is quantitatively important: getting a new card increases the default probability on any card one year after origination by 10 percentage points.

To determine the causal impact of obtaining a new card, we employ a difference-in-differences approach. We achieve this by leveraging certain institutional features of the Brazilian market. In particular, banks commonly form partnerships with firms to obtain the rights to process employee salary payments. The arrangement effectively mandates that employees receive their wages through the partner bank, increasing the likelihood that employees will obtain a card from that bank. We thus compare two groups of borrowers who worked at the same firm around the time the partner bank changed. The first group already had a relationship with the partner bank before it won (control), while the second did not (treatment). The identifying assumption is that the relative change in default probability between the two groups is due to the impact of the treatment group getting a new card. Appendix [Figure A10](#) summarizes the research design.

Using the credit registry of the Central Bank of Brazil, we construct a worker-level panel around the month in which a new bank is assigned to manage a firm's payroll.²³ We retain workers who remain employed at the same firm throughout the entire window, are active credit card users, and rely exclusively on this type of loan.²⁴ We examine events (i.e., changes in banks managing payroll payments) occurring within the twelve months between May 2017 and April 2018.²⁵ Appendix [Table A6](#) reports the characteristics of treated and control individuals in the month prior to the new bank assuming management of payroll. Control and treated individuals are similar in terms of the number of card lenders and debt-to-wage ratio: on average, control individuals have 1.5 card lenders and a debt-to-wage ratio of 1.5, compared to 1.2 card lenders

²²When the recovery rate conditional on default is zero, debt dilution operates via the impact on loan origination on default probabilities. When the loan recovery rate is different from zero conditional on default, debt dilution can also occur even without an impact on default rates.

²³The Central Bank uses an algorithm to identify changes in the bank managing a firm's payroll by detecting structural breaks in the volume of wage-backed payroll loans originated by each bank to the firm's employees.

²⁴Restricting the sample to borrowers who rely exclusively on credit cards is not ideal, as it conditions on post-treatment borrowing choices. Nonetheless, we make this restriction to reduce concerns that the treatment alters the demand for other loan types, particularly wage-backed payroll loans. Restricting the sample to workers who remain at the same firm throughout the observation window allows us to indirectly control for changes in wages (for instance, when examining the debt-to-wage ratio) and to account for firm-by-time fixed effects.

²⁵We base our sample selection on two main considerations. First, in May 2019, a change occurred in the reporting standards for credit card loans, making April 2018 the last month for which we can observe a long enough post-event window (12 months) under a consistent reporting regime. Second, we focus on a period prior to the rise of digital banks as significant market players. Following their entry, the cost of obtaining a new credit card declined substantially, reducing the likelihood that our treatment would remain effective.

and a debt-to-wage ratio of 1.3 among treated individuals. However, the groups differ slightly in baseline default rates: 5% of control individuals are delinquent on some card and 4% are in default on at least one card, compared to 2% and 2%, respectively, among treated individuals. Given that differences in baseline default rates may influence future borrowing trajectories, our regression specification allows for heterogeneous trends by baseline default status. The majority of individuals in our sample (80%) belong to the treatment group; only 20% already held a card from the bank that subsequently assumed management of the firm's payroll.

We estimate the following event-study regression to capture the dynamics of the effects:

$$y_{it} = \sum_{\tau=1}^{\tau=6} \beta_{-\tau} \times Treat_{i,t-\tau} + \sum_{\tau=0}^{\tau=15} \beta_{\tau} \times Treat_{i,t+\tau} + \alpha_i + \alpha_{f(i)t} + \alpha_{d(i)t} + \epsilon_{it} \quad (2)$$

where i indexes the individual, t the calendar month, and $f(i)$ the firm where individual i works. The variable $Treat_{i,t-\tau}$ is a dummy variable that takes the value one if individual i belongs to the treatment group and the new bank starts managing firm $f(i)$'s payroll τ months after t . The variable $Treat_{i,t+\tau}$ is a dummy variable that takes the value one if individual i belongs to the treatment group and the new bank starts managing firm $f(i)$'s payroll τ months before t . We control for time-invariant worker heterogeneity by including individual fixed effects, α_i , and we control for time-varying shocks to all workers of firm $f(i)$ by including firm-by-time fixed effects, $\alpha_{f(i)t}$. Because baseline differences in default status may shape subsequent borrowing trajectories, we interact time dummies with an indicator for whether borrower i had a default on any card in the month prior to the new bank assuming management of firm $f(i)$'s payroll, $\alpha_{d(i)t}$. We cluster standard errors at the firm level.

We also estimate the following specification at the individual-month level:

$$y_{it} = \beta_1 \times Post_Treat_{it}^{[0-6]} + \beta_2 Post_Treat_{it}^{[7-15]} + \alpha_i + \alpha_{f(i)t} + \alpha_{d(i)t} + \epsilon_{it} \quad (3)$$

where the variable $Post_Treat_{it}^{[0-6]}$ is a dummy equal to one if individual i belongs to the treatment group and period t falls within zero to six months after the date on which a new bank assumes management of firm $f(i)$'s payroll. The variable $Post_Treat_{it}^{[7-15]}$ is a dummy equal to one if individual i belongs to the treatment group and period t falls within seven to fifteen months after the date on which a new bank assumes management of firm $f(i)$'s payroll.

Figure 5 plots the estimated coefficients from Equation 2. On average, individuals in the treated group obtain 0.11 more card lenders than those in the control group. The take-up is gradual and peaks about 10 months after a new bank begins managing salaries. The debt-to-wage ratio also increases gradually. Interestingly, the debt-

to-wage ratio begins to rise before the number of cards increases, suggesting that the prospect of obtaining a new card influences the utilization of existing cards. We also observe that the delinquency (more than 15 days overdue) and default (more than 90 days overdue) rates increase. These effects start about 12 months after a new bank begins managing salaries. Across all outcomes, treated and control borrowers exhibit parallel trends in the pre-treatment period.

Table 8 reports the estimated coefficients from Equation 3 and shows similar patterns. We find that the treatment leads to an increase in the number of cards held, the total debt-to-wage ratio, and the rates of delinquency and default. On average, treated individuals increase the number of card lenders by 0.11, and their probability of default rises by 1.1 percentage points. Consequently, obtaining a new card lender raises default rates by 10 percentage points. Under the standard assumption that the distributions of unobserved borrower characteristics are similar in the treated and control groups, the treatment effect can be interpreted as the causal effect of getting a credit card.

Appendix Table A6 shows that only a small fraction of treated borrowers use a card from the new bank at some point of the post-treatment period (163 out of 2,864, or 5.7%). We refer to these individuals as compliers. Given the small number of compliers, the overall treated-versus-control comparison is largely driven by treated borrowers who do not take up a new card (non-compliers). Therefore, Appendix Figure A11 reports results separately for compliers and non-compliers. While all three groups—compliers, non-compliers, and control individuals—exhibit parallel pre-treatment trajectories, treatment effects are substantially larger for compliers. In particular, compliers experience earlier and sharper increases in delinquency and default rates.

Validity of the research design: The identification assumption is that the relative difference in default rates between the two groups of borrowers working in the same firm is due to the causal impact of getting a new card rather than any other shock that affects the two groups differently. One potential concern is that the two groups of borrowers differ in unobservable characteristics (e.g. savings, creditworthiness). This can explain why they have different numbers of credit cards, which translates into them having different sensitivities to income shocks. Those concerns would materialise into a different parallel trend. Figures A11 show that the groups have parallel trends, making the above concerns less likely to hold in practice. Another concern is that the borrowers might be on a different borrowing trajectory, meaning that the borrowers treated would have defaulted later anyway (6 months after treatment), which drives the results. Selection into treatment: not a problem as long as the treated and control group follows parallel trend. In addition, the fact that our results align with those of comfort us in thinking that this is not the case. Finally, our results recover the local

average treatment effect. It is possible that those coefficients are an upper bound on the causal impact of getting a credit card. Banks that are more likely to benefit from doing the payroll are more likely to be picked, given that they are, on average, worse at screening or monitoring employees, which gives us a lower bound. Similarly, people select into treatment (compliers); this is not an issue under the standard assumption that the unobserved characteristics are the same in the treated (compliers and non-compliers) and non-treated groups.

6 Model

Section 4.2 provides suggestive evidence that lenders can use credit card terms (e.g., credit limit) and cross-sell other products to gain informal seniority. To study the impact of competing for informal seniority on contract terms and welfare, we develop a model with non-exclusive loan contracts. We first develop a parsimonious two-period model to clarify the underlying economic mechanisms, and then generalise it in our quantification exercise.

The model features two periods with two decision dates. On the initial date, lenders simultaneously offer credit card contracts, and borrowers choose the set they accept. The key friction is that borrowers cannot commit to exclusive contracts. At the second date, borrowers decide which credit cards to repay, subject to a resource constraint.

Let $\mathcal{C}$ denote the set of credit cards held by a borrower, with individual cards indexed by $c \in \mathcal{C}$. Each contract specifies a loan amount L_c , which is used for consumption during the first period, and a face value R_c due at the beginning of the second period. Conditional on repaying the debt, the borrower obtains the continuation value $V(q_c)$, where q_c denotes the vector of contract characteristics affecting future utility. We refer to q_c as relationship benefits. Those benefits can be interpreted as credit card rewards, financial tracking provided via a mobile app, or any other loan terms offered in the second period.

All contract terms (L_c, R_c, q_c) are set by lenders at the initial date. By improving contract terms, such as lowering interest rates or increasing relationship benefits q_c , a lender can change their position in the borrower's repayment hierarchy. Lenders can thus impact the likelihood that their loan is repaid in states where the borrower does not fully honour all obligations.

6.1 Borrower

This section describes borrowers' choice of credit cards and default. We proceed in two steps. We first characterize the borrower's repayment decision, taking the set of contracts $\mathcal{C}$ as given. We then analyze the borrower's contract choice $\mathcal{C}$ at the initial

date.

6.1.1 Default

At the second date, a borrower defaults on the credit card c whenever the repayment r_c falls short of the face value R_c . The borrower chooses repayments r_c subject to a resource constraint $\sum_{c \in \mathcal{C}} r_c \leq W$. Conditional on repayment, the borrower obtains utility $V(q_c) + \epsilon_c - R_c$, where ϵ_c is an idiosyncratic, card-specific shock to continuation value realized at the second date. Conditional on not repaying the card the borrowers loses utility $V(q_c) + \epsilon_c - R_c$ and gets v^d instead.

Formally, the borrower chooses r_c to maximise its utility conditional on the set of credit cards owned $\mathcal{C}$:

$$V(\mathcal{C}, \epsilon) := \max_{r_c} \sum_{c \in \mathcal{C}} \{ \mathbf{1}_{\{r_c \geq R_c\}} [V(q_c) + \epsilon_c - R_c] + \mathbf{1}_{\{r_c < R_c\}} v^d \} \quad (4)$$

$$s.t. \sum_{c \in \mathcal{C}} r_c \leq W \quad (5)$$

The key trade-off faced by the borrower is as follows. By repaying the credit card c , the borrower benefits from its net continuation value $V(q_c) - R_c + \epsilon_c - v^d$. However, repaying a card reduces the resources the borrower can allocate to other loans.

Introducing the idiosyncratic shock ϵ_c allows us to make the default function continuous in contract terms, and to flexibly parametrise default sensitivities. To illustrate our point, let us consider the case in which the borrower can repay only one card (i.e., $R_c = W, \forall c$). Let θ_c denote the survival probability, defined as $\theta_c := E_\epsilon[\mathbf{1}_{\{r_c^*(\epsilon) \geq R_c\}}]$; where $r_c^*(\epsilon)$ is the optimal repayment strategy function.

In that case, the borrower's problem 4 implies that $r_c = R_c$ for the loan that has the maximum $V_c - R_c - v^d + \epsilon_c$. When ϵ_c are independent and uniformly distributed, the expected repayment probability on card c is thus $\theta_c = \Pi_{j \in \mathcal{C}} Pr(V_c - R_c - v^d + \epsilon_c > V_j - R_j - v^d + \epsilon_j)$. It is equal to one over the number of credit cards ($\frac{1}{|\mathcal{C}|}$) when all contracts are the same and incentives to repay are high enough (i.e., $V_c - R_c + \epsilon_c > v^d, \forall \epsilon_c$).

In our example, the variance of the ϵ parameterises the default sensitivity. For instance, assuming ϵ_c are uniformly distributed on $(0, \sigma)$, the default sensitivity to contract terms q are $\partial_q \theta = (\frac{V'}{\sigma})^{|\mathcal{C}|}$ for a lender that takes competitors' contract as given. We discuss the implications of introducing utility shocks ϵ_c for the demand of credit cards in the next section.

6.1.2 Number of credit cards

At the first date, the borrower chooses the set of cards $\mathcal{C} \in \mathcal{F}$ to maximize their utility $U(\mathcal{C})$ net of any costs associated with getting credit cards $C(|\mathcal{C}|)$. $\mathcal{F}$ is the power set of

all credit cards available in the market. Formally, we have:

$$\max_{\mathcal{C} \in \mathcal{F}} U(\mathcal{C}) - C(|\mathcal{C}|) \quad (6)$$

$U(\mathcal{C}) := u(L) + \beta E_\epsilon[V(\mathcal{C}, \epsilon)]$ is the borrower expected utility at date one. $u(L)$ is the utility coming from credit limits, where $L := (L_c)_{c \in \mathcal{C}}$ are the vector of credit limits the borrower can use in the first period. $\beta E_\epsilon[V(\mathcal{C}, \epsilon)]$ is the expected utility in period two, defined in equation 4, discounted by β .

$C(|\mathcal{C}|)$ is the cost of getting $|\mathcal{C}|$ credit cards. It captures, for instance, non-monetary costs such as the mental effort of managing multiple cards or the search costs. We interpret $C(|\mathcal{C}|)$ as a cost, but it can also be interpreted as additional benefits from obtaining multiple cards that $U(\mathcal{C})$ does not capture.

To illustrate the different economic channels driving the demand for multiple credit cards, let us consider that the borrower gets offered two similar contracts ($L_1 = L_2$, $q_1 = q_2$). Each of them maxes out their available income (i.e., $R_c = W$). By accepting two cards instead of one, the borrower receives a higher credit limit in the first period but must default on at least one of them in the second period. Formally, the borrower chooses two cards instead of one when:

$$U(\{1, 2\}) - C(2) \geq U(\{1\}) - C(1) \quad (7)$$

$$\iff \underbrace{\Delta u(L)}_{\text{additional loan size}} + \underbrace{\delta(2)}_{\text{love for variety}} \geq \underbrace{\Delta C(2)}_{\text{additional non-monetary cost}} + \underbrace{2[1 - \theta]v^d}_{\text{higher default probability}} \quad (8)$$

Equation 8 illustrates the different costs and benefits of getting an additional card. $\Delta u(L)$ is the utility increase from higher credit limits ($\Delta u(L) = u(L_1, L_2) - u(L_1)$). The term $\delta(2) = E[\sum_{c \in \{1, 2\}} E[\mathbf{1}_{r_c^*(\epsilon) > R_c} \epsilon_c] - E[\mathbf{1}_{r_1^*(\epsilon_1) > R_1} \epsilon_1]]$ captures other expected benefits of getting a new card. We refer to it as the love for variety. This channel comes from the ϵ terms in the second period: A higher number of cards increases the probability of getting a higher continuation value due to getting a higher ϵ draw.²⁶ $\Delta C(2) := C(2) - C(1)$ is the cost of getting an additional card. Finally, $2[1 - \theta]v^d$ is the loss in utility coming from having to default on one of the credit cards.

One central assumption is that borrowers take the credit card offers as given and cannot commit to exclusive contracts. Commitment is relevant in our model because borrowers have incentives to accept too many cards. For instance, even when the first-best contract is the exclusive one, it may not be sustainable as an equilibrium. The reason is that if each lender offers contracts with those terms (L^E, R^E, q^E) but without exclusivity, borrowers would have an incentive to accept more than one when default

²⁶Our modeling of default choices with ϵ introduces a love for variety. One can use the costs function C to cancel those effects so that ϵ just affects the default sensitivities without affecting the demand for cards.

costs are low and the borrowers value getting multiple cards (see equation 8).²⁷ By getting too many cards, the borrower will default too often, making the contract terms (L^E, R^E, q^E) unprofitable for the lender.

6.2 Lenders

This section describes how lenders set their contract terms. There is an arbitrary number of lenders. Each lender b chooses the contract terms (L, R, q) that maximise their expected profits. The profit is the probability that the borrower chooses a credit card from their bank times the expected profit on that card. We denote $D(U(\mathcal{C}_b); \mathcal{C}_{-b})$ the probability that the borrower chooses card b , taking as given the set of cards offered by competitors $(\mathcal{C}_{-b})$. Conditional on being chosen, profits per card are $\theta(q, R, \mathcal{C}_b)R - mc(L, q)$. The term $\theta(q, R, \mathcal{C})R$ is the expected amount the lender receives in the second period, while $mc(L, q)$ is the cost of issuing a credit card. $D(U(\mathcal{C}_b); \mathcal{C}_{-b})$ and $\theta(q, R, \mathcal{C}_b)$ are derived from the borrower's maximization problem. We often use the notation θ_b instead of $\theta(q, R, \mathcal{C})$ for notational convenience.

Formally, lender b chose contract terms (R, L, q) , taking as given the set of cards $\mathcal{C}_{-b}$. We have:

$$\max_{R, L, q} D(U(\mathcal{C}); \mathcal{C}_{-b}) [\theta(q, R, \mathcal{C}_b)R - mc(L, q)] \quad (9)$$

The first order conditions of the lender problem can be rewritten the following way:²⁸

$$L : u'(\mathbf{L}) = \frac{mc_L}{1 + \frac{\partial_R \theta_b}{\theta_b}} \quad (10)$$

$$q : \frac{mc_q - \partial_q \theta R}{\beta \theta_b V'(q)} = \frac{mc_L}{u'(\mathbf{L})} \iff \beta V'(q) \{ \partial_V \theta_b R + \frac{mc_L}{u'(\mathbf{L})} \theta_b \} = \partial_q mc(q) \quad (11)$$

$$R : \theta R = mc(L, q) + [1 + \frac{\partial_R \theta_b}{\theta_b}] \frac{D(U(\mathcal{C}); \mathcal{C}_{-b})}{\partial_U D(U(\mathcal{C}); \mathcal{C}_{-b})} \quad (12)$$

Proof: See Appendix B. The default sensitivity-to-interest-rate function $(\partial_R \theta_b)$ can be discontinuous; we discuss this in detail in Appendix B.

Equation 10 captures the cost and benefits of increasing the credit limit in the first period. $u'(\mathbf{L})$ is the marginal benefit for the borrower. mc_L is the marginal cost for the lender. $1 + \frac{\partial_R \theta}{\theta}$ captures that an increase in loan size is compensated by an increase in

²⁷In practice, credit card contracts are non-exclusive. The reason may be litigation costs, as courts would likely view it as restricting consumer freedom, or borrowers not linking exclusivity so that it is not offered in a competitive market.

²⁸There can be jumps in the sensitivity to interest rates at point where increasing the interest rate change the set of credit cards one can repay, we report here the derivatives outside of those points.

the debt face value, which in turn elevates default probabilities and thereby increases the cost of issuing credit.

Equation 11 states that the marginal rate of transformation between loan size and relationship benefits should be the same. To get intuition, let us consider that $\partial_R \theta$ is equal to zero so that $\frac{mc_L}{u'(\mathbf{L})}$ is equal to one. In that case, equation 11 states that the marginal benefits of increasing the relationship benefits for the borrower $\beta \theta V'(q)$ should be equal to the marginal cost for the lender (i.e., mc_q net of $\partial_q \theta R$). $\partial_q \theta R$ is the benefits coming from selective default: taking the competitors behaviour as given, increasing q stir customers into repaying bank b .

Equation 12 states that the bank sets the face value R as a markup-up $([1 + \frac{\partial_R \theta}{\theta}] \frac{D(u(\mathcal{C}); \mathcal{C}_{-b})}{\partial_U D(u(\mathcal{C}); \mathcal{C}_{-b})})$ over the cost of banking ($mc(L, q)$). The markup is the inverse demand semi-elasticity $\frac{D(u(\mathcal{C}); \mathcal{C}_{-b})}{D_u(u(\mathcal{C}); \mathcal{C}_{-b})}$ scaled down by $\frac{\theta_R}{\theta}$. $\frac{\theta_R}{\theta}$ is due to the fact that lenders internalise that higher interest rates increase the borrower's default probability.

6.3 Equilibrium Contracts and Welfare Analysis

The key friction in our model stems from the fact that borrowers cannot commit to borrowing exclusively from a single lender. In this section, we highlight how the above-mentioned friction impacts contract terms and welfare in our model and what type of policy intervention can restore the first-best. To gain intuition, we focus on a particular parametrisation of the model that yields closed-form equilibrium contracts. We also provide a numerical analysis of alternative functional form specifications in Appendix ??.

6.3.1 Equilibrium Contracts

We assume that borrowers' utility and marginal costs are linear in loan size, $u(\mathbf{L}) = \alpha \sum_c L_c$ and $mc(L, q) = mc_L \cdot L + mc(q)$. For the credit market to exist, we consider that borrowers' valuation of loan size is sufficiently high that lending generates a positive surplus (i.e., $\alpha > mc_L$), and that the cost of default is sufficiently high so that borrowers want to repays at least one card (i.e., $V_c - R_c + \epsilon_c > v^d, \forall \epsilon_c$).

Finally, to analyse non-exclusivity, we assume that the net benefits of holding credit cards are sufficiently large for the borrower to optimally hold multiple cards (i.e., $|\mathcal{C}^*| > 1$). The optimal number of cards $|\mathcal{C}^*|$ is determined endogenously and satisfies $\alpha L \leq F(|\mathcal{C}^*| + 1)$, where $F(|\mathcal{C}^*| + 1) := \Delta C(|\mathcal{C}^*| + 1) - \delta(|\mathcal{C}^*| + 1)$ is the net cost of getting a card (defined in equation 8).

Under these assumptions, one can show that it is optimal (see Appendix C) for each lender to maximise the borrower's debt capacity, which we denote by $R = W$. A borrower that gets multiple credit cards thus repays only one (i.e., $r_c^*(\epsilon) = R_c$ for the card with the highest ϵ realisation, and zero for all others). The symmetric equilibrium

is characterised by the following contracts and default probabilities:

$$\theta_c^* = \frac{1}{|\mathcal{C}^*|} \quad (13)$$

$$R_c^* = W \quad (14)$$

$$L_c^*(q_c^*) = \overbrace{\theta_c^* \frac{R_c - c(q_c^*)}{mc_L}}^{\text{break-even condition}} - \overbrace{\varepsilon^{-1}}^{\text{mark-down}} \quad (15)$$

$$q_c^* := \{q : \beta V'(q^*)[\partial_q \theta_c^* W + \theta_c^* \frac{mc_L}{\alpha}] = c'(q^*)\} \quad (16)$$

Proof: See Appendix C. Because the borrower's utility is discontinuous in R , there may be multiple equilibria. We discuss this in detail in Appendix C. We show that the equilibrium analysed can be made unique when the cost of defaulting, v^d , is decreasing in the fraction of the credit card repaid.

Equations 15 and 16 coincide with those derived in the lender problem section (Section 6.2). The key difference is that, with a binding interest rate, the credit limit serves as the instrument through which surplus is allocated between borrowers and lenders. The markdown ε^{-1} is the inverse demand semi-elasticity $\frac{D(U(\mathcal{C}); \mathcal{C}_{-b})}{\partial_U D(U(\mathcal{C}); \mathcal{C}_{-b})}$. For instance, when demand has a logit form,²⁹ the equilibrium semi-elasticity is $\frac{1}{\sigma(1 - \frac{1}{|\mathcal{C}^*|})}$ with $\sigma \in \mathbf{R}^+$ driving the demand-semi elasticity. When σ tends to infinity, each lender is a perfect substitute for the others, and the markup tends to zero.

Our specification with linear utilities allows us to fix the marginal rate of transformation of loan ($\frac{mc_L}{\alpha}$) in equation 16. This parametrisation isolates the role of informal seniority in shaping relationship benefits q , which is the central object of the analysis. It also delivers tractability and facilitates a transparent comparison between the social planner's allocation and the decentralized equilibrium. Yet, the parametrisation abstracts from the role of the default elasticity with respect to the interest rate, $\frac{\theta_R}{\theta}$, which affects markups and loan-size distortions (see equations 10 and 12). We incorporate these additional channels in Appendix F, where we present a numerical analysis with nonlinear preferences and show that our results hold.

6.3.2 Welfare

We now examine how competition for informal loan seniority affects welfare. In what follows, we characterise the constrained efficient allocation and compare it to the decentralised equilibrium.

²⁹For instance, when lenders expect the borrower i gets $|\mathcal{C}^*|$ cards, they compete to be one of the card, and each card-lender is associated with an additional borrower utility shock $\sigma^{-1}\varepsilon_{ic}$, we get a semi-elasticity of $\frac{1}{\sigma(1 - \frac{1}{|\mathcal{C}^*|})}$.

Let us start by constructing the utilitarian welfare function, defined as the sum of the borrowers' and lenders' utilities. Denoting $\beta V'(q) = \gamma$, the borrower utility in the symmetric equilibrium is:

$$u^*(|\mathcal{C}^*|) := \alpha L^*(q^*)|\mathcal{C}^*| + \gamma q^* - F(|\mathcal{C}^*|) \quad (17)$$

αL is the utility coming from each loan. It is multiplied by $|\mathcal{C}^*|$ as borrowers have multiple loans. γq is the utility coming from relationship benefits. They expect to repay only one loan on average, so they get the benefit from $|\mathcal{C}^*|$ loans with probability $\frac{1}{|\mathcal{C}^*|}$.

Assuming the costs of lending are quadratic $c(q) = \frac{c}{2}q^2$, the utilitarian welfare function is thus:

$$W^*(q^*) := (\alpha - mc_L)L(q^*)|\mathcal{C}^*| + [\gamma - \frac{c}{2}q^*]q^* + \delta(|\mathcal{C}^*|) \quad (18)$$

The loan size $L(q^*) := \theta[\frac{W-c(q^*)}{mc_L}]$ is given by the lender break even condition. $(\alpha - mc_L)L(q^*)|\mathcal{C}^*|$ is the social surplus generated by lending in the first period, and $[\gamma - \frac{c}{2}q^*]q^*$ is the one generated by relationship benefits.

Now, let us use the welfare function to characterise the optimal level of relationship benefits (q^{SB}) by taking the first order condition of the welfare function $W^*(q^*)$. The key trade-off is the following. Providing more relationship benefits comes at the cost of reducing the credit limit so that the lender's break-even condition is satisfied (equation 18). Neglecting the effect on the number of credit cards $|\mathcal{C}^*|$ for expositional clarity, the optimal level of benefits q^{SB} is:

$$q^{SB} = \frac{\gamma mc_L}{\alpha c} \quad (19)$$

Equation 19 illustrate that the social planner chooses to provide more relationship benefits when the borrower values it relatively more than loans (i.e., $\frac{\gamma}{\alpha}$ high), and when it is relatively cheaper to offer (i.e., $\frac{mc_L}{c}$ high).

We can compare the social planner allocation q^{SB} to the market allocation. Using equation 16, and denoting the default sensitivity by $d := \partial_q \theta_c^*$, the decentralized equilibrium level of relationship benefits q^* is:

$$q^* = \frac{\gamma}{c} \left[\overbrace{dW}^{\text{overprovision}} + \overbrace{\frac{mc_L}{\alpha} \frac{1}{|\mathcal{C}^*|}}^{\text{underprovision}} \right] \quad (20)$$

Contrasting the social planner allocation (19) to the market outcome (20) highlights two distinct channels through which non-exclusivity distorts relationship benefits. These channels are summarized by the terms $\frac{mc_L}{\alpha} \frac{1}{|\mathcal{C}^*|}$ and dW .

To isolate the first channel, set the default sensitivity to zero so that only $\frac{mc_L}{\alpha} \frac{1}{|\mathcal{C}^*|}$

remains. In this case, relationship benefits are underprovided ($q^* < q^{SB}$), since the survival probabilities are lower than one $\theta = \frac{1}{|C^*|} < 1$. The intuition is that lenders anticipate that borrowers hold too many credit cards in a non-exclusive environment. This raises default probabilities ($\theta = \frac{1}{|C^*|}$), which reduces the marginal value of relationship benefits from the borrower's perspective relative to the exclusive benchmark (θV). As a result, lenders have weaker incentives to provide such benefits.

We now turn to the second channel, dW . When this term is large, relationship benefits are overprovided ($q^* > q^{SB}$). The reason is that each lender has an incentive to increase q in order to raise the likelihood of being repaid in states of partial default. Because all lenders engage in this behavior, competition for repayment priority generates an arms race in q , leading to overprovision. This channel is stronger when default sensitivities (d) are high or when profits conditional on repayment (W) are large. This channel is absent from standard models of debt dilution, which impose equal recovery and cross-default so that d is zero.

6.3.3 Bankruptcy Regulation: Equal Recovery Policies

The previous section illustrated that there could be over- or underprovision of relationship benefits due to two forces and that default sensitivities d can affect their relative strength. In this section, we analyse whether personal bankruptcy regimes that impose equal recovery rates and cross-default provisions can mitigate those distortions and thus increase welfare. We focus on such policies for two reasons. First, they eliminate the competition for informal seniority (i.e., the overprovision channel), allowing us to understand its impact. Second, those policies are often used in bankruptcy regulations, so it is natural to assess their unintended effects once we allow for informal seniority.

First, let us start by noting that a social planner who could set the default sensitivity so that the under- and over-provision channels exactly cancel each other could replicate the efficient allocation described in the previous section:

$$d^{SB} : \frac{\gamma}{c} [dW + \frac{mc_L}{c} \frac{1}{|C^*|}] = \frac{\gamma mc_L}{c\alpha} \quad (21)$$

In practice, the social planner might not be able to set default sensitivities to a particular number so that it satisfies equation 21. Instead, they can set it to zero by imposing an equal recovery rate and cross-default. Using equation 18 and comparing the welfare for various levels of default sensitivities, we show that such a policy would decrease welfare when the overprovision channel is mild relative to the underprovision channel. This is more likely to be the case when the default sensitivity d is low in the centralised equilibrium, when banks' profits conditional on survival are high W when the average default is high (i.e., $\frac{1}{|C^*|}$ low). The following statement and Figures 1

and 2 summarise when such a policy is likely to be desirable.

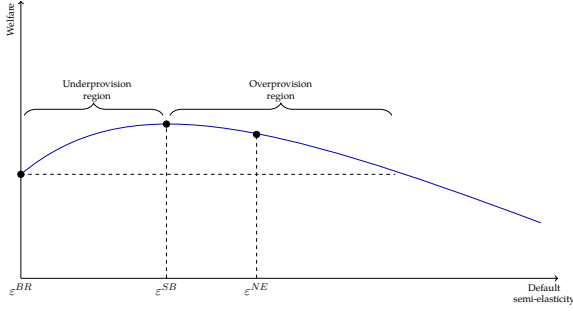


Figure 1: $(\varepsilon^{SB}, \varepsilon^{NE}, \varepsilon^{BR})$ are the optimal, current, and under-regulation semi-elasticities.

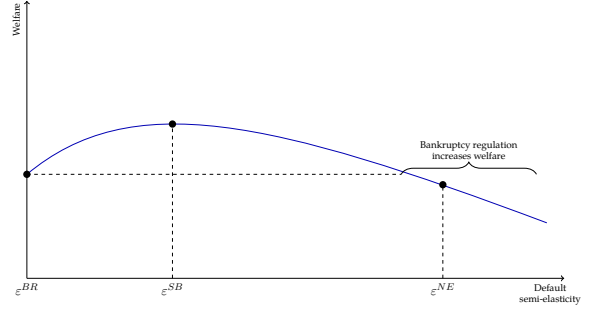


Figure 2: Example where the bankruptcy regulation is welfare increasing.

Bankruptcy Regulation: Exclusive contract. For the interested reader, we analyse in the Appendix M when the exclusive contract equilibrium is more desirable than the non-exclusive one. The key trade-off is the following. Non-exclusive contracts offer borrowers a higher product variety and the possibility to mix and match products (i.e., higher F), however, it comes at the cost of a distortion in contract terms (over- or underinvestment). There is an additional benefit to offering non-exclusive contracts when gaining seniority involves offering another loan: competition for loan seniority provides incentives to lower the markup on the product.

6.4 Discussion about the Modelling Assumptions

In this section, we discuss our key modeling assumption and how they impact our results.

6.4.1 Contract terms

Who set the contract terms: We assume that lenders choose contract terms (L, R, q) . In practice, lenders set the credit limit and borrowers choose how much to borrow. Under perfect information, the equilibrium allocation in our model is the same as that in which lenders can use nonlinear pricing and let the borrower choose how much to borrow. The same allocation can also be achieved with linear pricing with a credit limit at L_c .

What are the relationship benefits: The relationship benefits can have multiple interpretations. Appendix I discusses the micro-foundations for $V_c(q)$ across the three channels that allow lenders to gain seniority: by offering rewards, modifying the card terms conditional on repayments, or cross-selling products. We show that the three channels are equivalent, as they yield a V_c function that is increasing in $q_c \in \mathbb{R}$.

One central result of our paper is the tension between the under- and over-provision channels. For those two channels to move in opposite directions, we assumed that a higher relationship benefits increase both the borrower's utility ($\partial_q V$) and the repayment probability ($\partial_q \theta$).

One can argue that those two channels go in the same direction when relationship benefits are interpreted as the lender's debt collection process. Debt collection may not directly affect borrowers' utility (it may be a shrouded attribute or merely a nudge to repay a given bank), affect it positively (when being nice to the borrower is the way to get repaid), or negatively (when the bank harasses the borrower). When debt collection negatively affects borrowers' utility, both the arms race and the underprovision channel move in the same direction (as $\partial_q V < 0$ and $\partial_q \theta > 0$). This distinction matters for welfare and the impact of imposing the same recovery rate for all loans. Thus, assessing empirically how sensitivity to debt collection affects demand and default is important. In interviews with market participants, they pointed out that due to consumer protection, recovery is mainly a matter of being nice to the borrower and reaching out to them once they miss a payment. However, a formal study of debt collection is outside the scope of our paper.

Cancelling relationship benefits upon default: We assume that the borrowers do not get access to relationship benefits conditional on default, so that borrowers choose to fully repay only one credit card. This behaviour is consistent with the data. We show in Appendix G that this behaviour arises in a model in which lenders can set q for the default state as well. The intuition is that lenders optimally set relationship benefits to a minimum in default states to prompt customers to repay them in full.

6.4.2 Demand

One central element for the underinvestment channel is that demand sensitivities to relationship benefits depend on default probabilities. In our model, the marginal sensitivity to relationship benefits is equal to $\theta_c V'(q)$. $\theta_c V'(q)$ captures the fact that when borrowers are more likely to default, they value a marginal increase in quality q less as they will benefit from it less often. The same formula would arise in any model where the survival probability is derived from a maximisation problem, so the envelope theorem applies.

6.4.3 Supply

Demand function from lenders' perspective: We assume that each lender can observe the set of contracts offered and predict the set the borrower will accept. This assumption is made for tractability and is consistent with the fact that, in practice,

lenders do not update their contract terms after when the borrowers gets a new credit card.³⁰

Dynamics: We abstract from the dynamics aspect of lending in our model by having the lender simultaneously offer loan contracts in the first period. This is justified by the fact that selective defaults do not appear to be driven by relationship length in the data. The equilibrium in our model is the same as the one in which lenders sequentially lend and have rational expectations about the other lenders' behaviour. In Appendix XX, we allow for lenders to offer a new loan contract after the borrowers default in the first period. Once default happens, the surviving lender knows he has informal seniority and can extract most of the surplus from lending. The under- and over-provisioning channel is robust to this extension.

6.4.4 Equilibrium and Welfare

Effect on the number of credit cards : For expositional clarity, we hold the number of credit cards constant in our previous welfare analysis. However, when the borrower's utility increases due to the policy, they may respond by requesting more cards. As a result, the average default rate increases, strengthening the underprovision channel. This strengthens the effect of the policy when we are in an overprovision region. However, this runs counter to the policy's intended effect when we are in the underprovision region. When the sensitivity of credit card demand is high, the policy can decrease welfare.

Priority rules: [Moore and Hart \(1995\)](#) shows that a priority rule can mitigate the debt dilution problem. However, as noted in [Bizer and DeMarzo \(1992\)](#), priority will have no effect if junior issuances increase the likelihood of default on senior claims due to, for instance, moral hazard.

In our model, we do not model moral hazard. Imposing a priority rule such that the "first lender to issue" should be repaid first would thus restore efficiency. While we do not explicitly model who is the first lender, we can accommodate our model as follows to include this priority rule. After the borrower accepts the credit cards, lenders learn whether they are the first cardholder (i.e., the one with priority).³¹ If they have priority, lenders must commit to the contract they offered. If they do not have seniority, lenders do not have to commit. In that case, it is optimal for them to cancel the credit limit, as they would make no profit. As a result, the exclusive equilibrium contract can be

³⁰To microfound the fact that the demand is continuous, one can assume that each card $c \in \mathcal{C}$ represents a card type, that borrowers will get only one card of each type and that a group of lenders compete on each card type.

³¹In practice, this can reflect the fact that many borrowers make simultaneous credit card applications.

sustained.

Implementing such a regulation in practice might be challenging, as formal bankruptcy procedures that would guarantee priority are rarely used in the credit card context due to the significant fixed costs of litigation and the low recovery rate. An alternative would thus be to solve the problem outside of bankruptcy. For instance, by imposing a cap on the aggregate contract characteristics, such as the aggregate debt-to-income ratio (similarly to a covenant banks set for corporate loans). This regulation is costly to implement, as it may impose unnecessary restrictions if not fine-tuned to cater to borrowers' and lenders' heterogeneous needs.

Given those limitations, we limited ourselves to analysing the current regulation, under which all credit cards should have the same recovery rate in bankruptcy. While this regulation is not used in practice as most bankruptcies in the retail market are done out of court, our analysis is relevant as policymakers are trying to lower the cost of going to court so that more formal bankruptcy regulations are implemented.

6.5 Quantification

In this section, we use our model together with our empirical estimates to quantify the distortions to credit limits after origination (i.e., q) due to non-exclusivity. We employ a parsimonious approach: we use a general version of our model to derive the ratio of the loan size in the data to that in the first best. We write this ratio as a function of observables over the first-best survival probabilities. Instead of simulating the first-best survival probabilities using a structural model, which would be untractable, we use our difference-in-differences estimates to proxy for them.

We first extend our model to feature multiple periods and derive the lender's first-order condition for the relationship benefits for lenders who revolve their credit card expenses. The key difference with the two-period model is that the debt face value R is replaced by the lender's continuation value of lending, denoted π . We thus have:

$$q : \beta V'(q) \{ \partial_{\beta V(q)} \theta_b(q, R, \mathcal{C}) \pi + mc_L \theta_b \} = \partial_q mc(q) \quad (22)$$

We now assume log utilities for $V(q) = \alpha \log(\sum_c q_c)$ and linear costs, and write the ratio of equation 22 for the aggregate credit limit (q^{Data}) relative to the first best aggregate credit limit (q^{FB}). Under the first best, we consider that $\partial_{\beta V(q)} \theta_b(q, R, \mathcal{C})$ is equal to zero, as, conditional on the debt face value, credit limit arguably only affects repayment through selective default, we thus have:

$$\frac{q^{FB}}{q^{Data}} = \frac{\partial_q \theta_b \pi + mc_L \theta_b^{Data}}{mc_L \theta_b^{FB}} \quad (23)$$

To quantify $\frac{q^{FB}}{q^{Data}}$, we thus need: the survival probabilities per period in the first-best (θ_b^{FB}) and in the data (θ_b^{Data}), the profits per period π conditional on survival, the survival sensitivities to contract terms $\partial_q \theta_b$ and the marginal cost of lending mc_L .

We estimated that $\partial_q \theta_b \approx 1.5 \times 10^{-5}$. With log utilities, this implies a coefficient of $\partial_{V(q)} \theta_b 1.5 \times 10^{-5} * \frac{q^{Data}}{\beta \alpha} = \frac{1.5}{1.13} \times 10^{-5}$. $\frac{q^{Data}}{\beta \alpha}$ is the borrower's willingness to pay for an additional real, which we assume is around 1.13 (given the interest rate in Brazil).

We have $\pi = r \cdot L$ with the net profitability conditional on survival per year being 17% as in Drechsler et al. (2025)³² and the average borrowing amount is 10000 reais per card as in our data. We get that $\partial_q \theta_b(q, R, C)\pi$ is around $0.17 * 10000 * 1.5 \cdot 10^{-5} = 2.5\%$ in the data.

θ_b is the survival rate, which is around 10% for borrowers with two cards and around 1% for the same borrowers with one card only, according to our difference-in-differences estimates. Considering that the marginal cost of lending is about the return on deposits (i.e., 1+13%), we get that $mc_L \theta_b$ is 0.96 in the data and 1.06 in the first-best.

The relative credit credit limit is $\frac{L^{data}}{L^{FB}} = \frac{0.96+0.025}{1.06} \approx 0.93$ so the aggregate credit limit are 7% lower than in the first best. Imposing a regulation requiring equal recovery would further reduce the credit limit by 2.5%.

This exercise allows one to get a sense of credit-limit distortions. However, they offer only a partial view of the distortions, as we do not analyse the effect on other dimensions of the relationship benefits, such as cross-selling or debt collection processes. Doing this would require estimates of the causal impact of cross-selling on default sensitivities and average default probabilities, which is outside the scope of our paper.

7 Conclusion

In this paper, we use a rich administrative dataset of household credit in Brazil to document new facts about the credit card market. We show that it is common for borrowers to hold multiple active cards issued by different lenders. We find that default rates rise with the number of lenders, and that, when facing financial distress, borrowers tend to prioritize repaying certain banks over others. Furthermore, we show that prioritized banks keep supplying credit to individuals even after they default on other banks.

While borrowing from multiple lenders may reflect increased competition and greater access to differentiated loan products and services, it also presents challenges, as externalities between lenders can lead to inefficiencies. Banks may respond to it by competing for repayment priority. We study these market inefficiencies through the lens of a model featuring non-exclusivity and repayment decisions.

³²The net returns are 6.7% to which we add the net credit loss rate of 10%

Although our focus is on the credit card market, the idea of lenders endogenously competing for seniority applies more broadly to contexts in which claimants compete for scarce resources that are not allocated according to rules set by regulation (e.g., bankruptcy law) or contractual agreements. In many countries, the lack of formal household debt resolution mechanisms means that providers of unsecured loan products, beyond just credit cards, also compete for repayment priority. Similarly, before filing for bankruptcy, both firms and individuals may strategically select which creditors to prioritize. Consequently, our framework provides valuable insights into the implications of non-exclusivity in a variety of other contexts.

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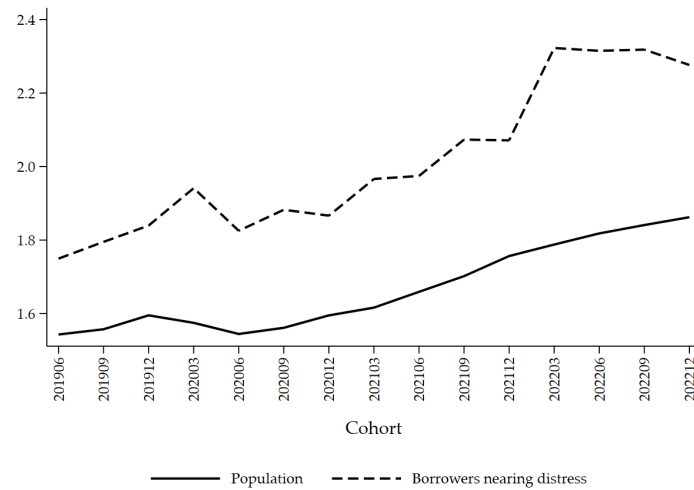
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Figures

Figure 3: Evolution of borrowing from multiple card lenders



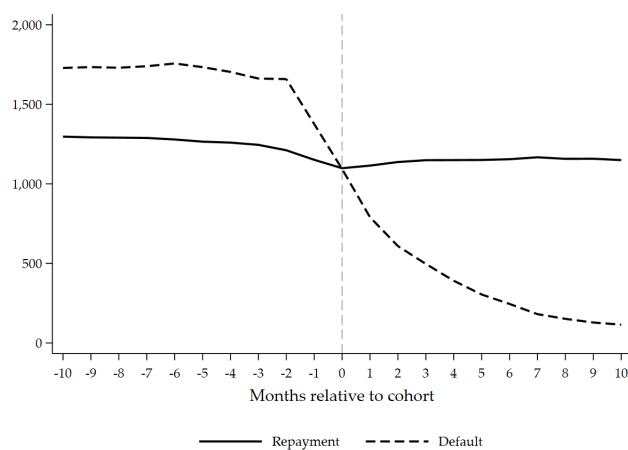
(a) Average number of credit card lenders



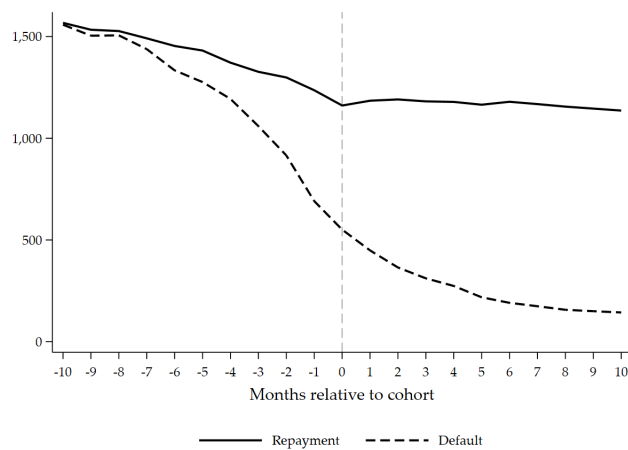
(b) Share of card borrowers with two or more credit card lenders

Notes: The unit of observation is the borrower-cohort. Panel (a) shows the evolution of the average number of credit card lenders per borrower. Panel (b) shows the evolution of the share of borrowers with two or more credit card lenders. Population statistics are computed using inverse probability weighting.

Figure 4: Evolution of credit card limits and usage among selective defaulters: paid versus unpaid cards



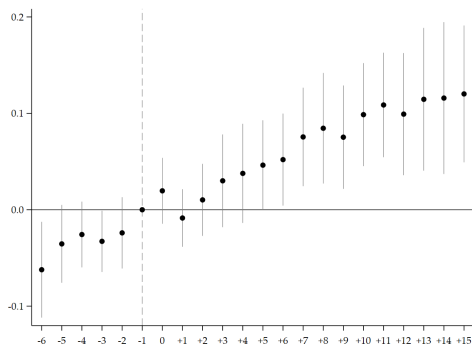
(a) Selective defaulters' monthly card usage: paid versus unpaid cards



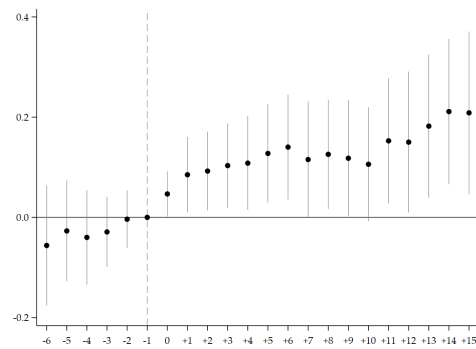
(b) Selective defaulters' limit available: paid versus unpaid cards

Notes: The sample consists of *borrowers nearing distress* who are selective defaulters—individuals repaying at least one card while defaulting on another (payments over 90 days past due). Month 0 is defined as the month in which a borrower becomes more than 15 days overdue on any card.

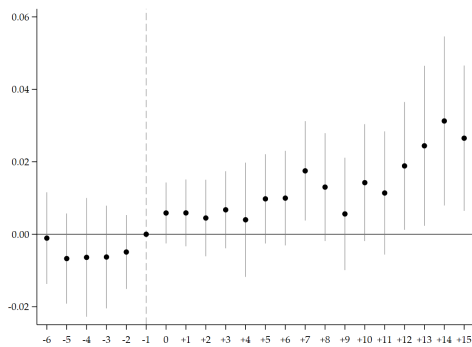
Figure 5: Difference-in-Differences



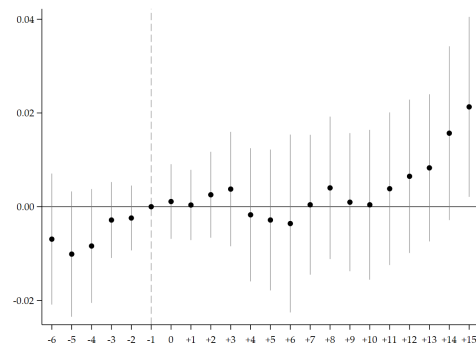
(a) Number of cards



(b) Debt-to-wage ratio



(c) Delinquency rate



(d) Default rate

Notes: The figures present the results from estimating Equation 2. We report 95% confidence intervals calculated using standard errors clustered at the firm level. Delinquency (default) is a dummy variable equal to one if the individual has any debt payment overdue by more than 15 (90) days.

Figure 6: $(\varepsilon^{SB}, \varepsilon^{NE}, \varepsilon^{BR})$ are the optimal, current and under regulation semi-elasticities

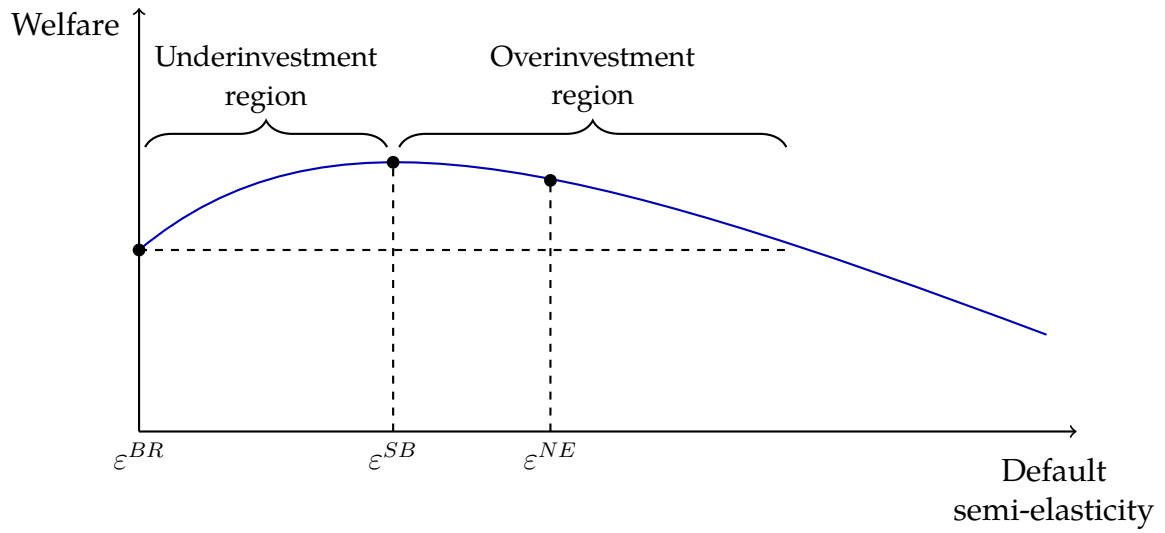
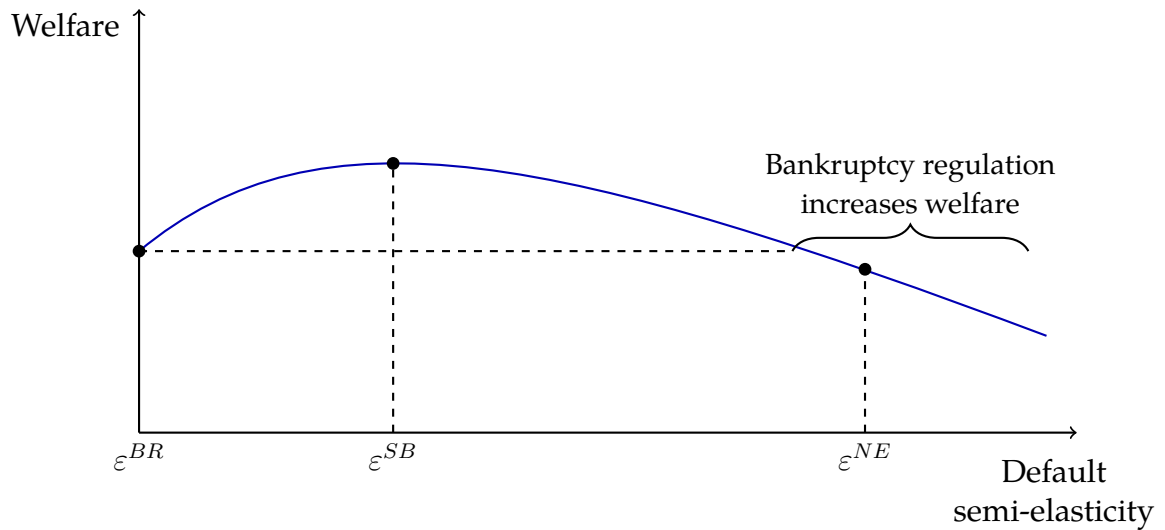


Figure 7: Example where the bankruptcy regulation is welfare increasing



Tables

Table 1: Mean of selected variables by number of credit card lenders

	Number of card lenders					
	1	2	3	4	5	6+
<i>Panel A: Random sample (population)</i>						
Wage (formal)	3,757	4,124	4,171	3,812	3,841	3,920
Wage (reported)	3,214	3,704	3,766	3,607	3,737	3,663
Debt service to income	0.18	0.27	0.38	0.48	0.59	0.85
Formal sector	0.36	0.43	0.46	0.46	0.47	0.47
Age	45	44	44	44	43	41
Total balance card	2,631	5,386	7,985	10,300	13,335	20,069
Average limit utilization (%)	54	55	58	61	64	69
Share cards 100% utilization	0.22	0.22	0.24	0.26	0.29	0.34
Total card limit available	5,129	9,247	11,206	11,793	12,688	13,086
Dummy auto loan	0.07	0.12	0.16	0.20	0.24	0.26
Dummy mortgage	0.06	0.09	0.10	0.11	0.11	0.11
Dummy other loans	0.27	0.35	0.43	0.48	0.53	0.60
Dummy payroll loan	0.17	0.21	0.23	0.26	0.26	0.25
Dummy personal loan	0.14	0.20	0.27	0.34	0.42	0.53
Fraction of borrowers	60%	24%	9%	4%	2%	1%
<i>Panel B: Borrowers nearing distress</i>						
Wage (formal)	2,151	2,943	3,320	3,399	3,548	3,396
Wage (reported)	2,121	2,778	3,054	3,159	3,272	3,374
Debt service ratio	0.26	0.41	0.54	0.65	0.77	1.00
Formal sector	0.29	0.38	0.42	0.44	0.44	0.45
Age	39	42	42	42	42	41
Total balance card	1,558	4,234	6,950	9,702	12,621	19,724
Average limit utilization (%)	83	77	75	76	76	78
Share cards 100% utilization	0.74	0.53	0.47	0.45	0.44	0.45
Total card limit available	548	2,431	3,901	4,639	5,233	6,271
Dummy auto loan	0.06	0.11	0.16	0.20	0.22	0.24
Dummy mortgage	0.03	0.06	0.08	0.09	0.10	0.11
Dummy other loans	0.35	0.45	0.52	0.57	0.63	0.69
Dummy payroll loan	0.12	0.22	0.25	0.26	0.28	0.26
Dummy personal loan	0.21	0.30	0.36	0.43	0.48	0.59
Fraction of borrowers	47%	27%	14%	6%	3%	3%

Notes: The unit of observation is the borrower-cohort. In Panel A, we compute population averages using inverse probability weighting. In Panel B, we compute sample means for the group of *borrowers nearing distress*. *Wage (formal)* refers to wages recorded in administrative data (RAIS) and are only available for workers employed in formal jobs. *Wage (reported)* refers to wages reported by banks. *Formal sector* is a dummy variable that takes the value one if the individual has a formal job. *Debt service to income* is the ratio of debt payments to income. All monetary values are in Brazilian reais (BRL).

Table 2: Mean of default variables by number of credit card lenders

	Number of card lenders					
	1	2	3	4	5	6+
<i>Panel A: Random sample</i>						
Default payroll loan	0.04	0.04	0.03	0.03	0.03	0.02
Default mortagage	0.01	0.01	0.01	0.01	0.01	0.02
Default auto loan	0.04	0.03	0.03	0.02	0.03	0.04
Default personal loan	0.11	0.10	0.10	0.10	0.11	0.13
Default other loans	0.10	0.08	0.09	0.09	0.09	0.11
Default credit card	0.07	0.10	0.13	0.15	0.17	0.21
<i>Panel B: Borrowers nearing distress</i>						
Default payroll loan	0.07	0.04	0.03	0.03	0.02	0.03
Default mortagage	0.05	0.02	0.01	0.01	0.00	0.01
Default auto loan	0.09	0.05	0.03	0.03	0.02	0.02
Default personal loan	0.09	0.08	0.06	0.05	0.05	0.05
Default other loans	0.11	0.07	0.05	0.04	0.04	0.07
Default credit card	0.00	0.00	0.00	0.00	0.00	0.00

Notes: The unit of observation is the borrower-cohort in the month before the cohort. In Panel A, we compute population averages using inverse probability weighting. In Panel B, we compute sample means for the group of *borrowers nearing distress*.

Table 3: Default rates by number of cards of *borrowers nearing distress*

	(1)	(2)	(3)	(4)	(5)	(6)
Number of	Repay	Default on at	Default on		Selective	
card lenders	all cards	least one card	all cards	Selective	Largest	Smallest
1	0.57	0.43	0.43	-	-	-
2	0.55	0.45	0.11	0.34	0.20	0.13
3	0.53	0.47	0.05	0.41	0.22	0.19
4	0.51	0.49	0.04	0.45	0.23	0.22
5	0.46	0.54	0.03	0.51	0.25	0.26
6+	0.36	0.64	0.02	0.62	0.35	0.27
Total	0.54	0.46	0.24	0.21	0.12	0.09

Notes: The unit of observation is the borrower-cohort for the sample of individuals classified as *borrowers nearing distress*. Columns (1) and (2) partition this group into borrowers who repay all credit cards six months after the cohort month and those who default on at least one card six months after the cohort month, respectively. Columns (3) and (4) further classify borrowers who default on at least one card into those who default on all cards (*universal defaulters*) and those who default on at least one card while continuing to repay at least one other card (*selective defaulters*), respectively. Columns (5) and (6) further split the latter group—*selective defaulters*—into borrowers who default on cards accounting for more than 50 percent and less than 50 percent of their total credit card balance, respectively, measured in the month of the missed payment.

Table 4: Mean of selected variables by default category of borrowers nearing distress

	No	Selective		Universal
	default	Largest balance	Smallest balance	default
Number of banks	4.1	4.6	4.8	3.6
Number of card lenders	2.9	3.2	3.3	2.4
Number of digital banks	0.6	1.0	0.9	0.7
Share card balance digital banks	0.15	0.15	0.17	0.16
Number of state-owned banks	0.6	0.7	0.7	0.6
Wage (formal)	3,559	2,735	3,083	2,245
Wage (reported)	3,203	2,719	2,855	2,365
Debt service to income	0.46	0.68	0.56	0.48
Formal job	0.42	0.40	0.41	0.29
Age	43	39	42	36
Dummy payroll loan	0.25	0.20	0.29	0.12
Dummy mortgage	0.08	0.07	0.08	0.05
Dummy auto loan	0.15	0.15	0.14	0.10
Dummy personal loan	0.31	0.42	0.41	0.36
Dummy other loan types	0.50	0.55	0.53	0.46
Dummy default payroll loan	0.01	0.01	0.01	0.01
Dummy default mortgage	0.00	0.00	0.00	0.00
Dummy default auto loan	0.00	0.01	0.01	0.01
Dummy personal loan default	0.02	0.03	0.04	0.03
Dummy other loan types	0.03	0.05	0.04	0.04
Total card balance	6,432	9,166	7,279	5,295
Total credit balance	38,455	35,735	37,930	20,201
Age bank relationship (years)	8.1	6.2	6.8	4.8

Notes: The unit of observation is the borrower-cohort for the sample of individuals in the *borrowers nearing distress* group with two or more card lenders. *Selective* means that the borrower defaults on at least one card while repaying at least one other card. *Selective largest balance* means that borrowers defaulted on the cards that accounted for more than 50% of their total card balance measured in the month when they missed a payment. *Selective smallest balance* means that borrowers defaulted on the cards that accounted for less than 50% of their total card balance measured in the month when they missed a payment. *Wage (formal)* refers to wages recorded in administrative data (RAIS) and only available for workers employed in formal jobs. *Wage (reported)* refers to wages reported by banks. Formal sector is a dummy that takes the value 1 if the individual has a formal job. *Debt service to income* is the ratio of debt payments to income. All monetary values are in Brazilian reais (BRL).

**Table 5: Default rates by number of cards (for borrower nearing distress):
controlling for observables**

	(1)	(2)	(3)	(4)	(5)	(6)
2 card lenders	0.03*** (0.006)	0.04*** (0.006)	0.04*** (0.006)	0.03*** (0.006)	0.04*** (0.006)	0.03*** (0.006)
3 card lenders	0.06*** (0.011)	0.07*** (0.011)	0.08*** (0.011)	0.06*** (0.009)	0.07*** (0.011)	0.05*** (0.009)
4 card lenders	0.09*** (0.011)	0.10*** (0.010)	0.12*** (0.010)	0.09*** (0.011)	0.10*** (0.011)	0.08*** (0.012)
5 card lenders	0.12*** (0.017)	0.14*** (0.015)	0.15*** (0.015)	0.11*** (0.014)	0.13*** (0.016)	0.10*** (0.014)
6+ card lenders	0.22*** (0.017)	0.23*** (0.017)	0.24*** (0.016)	0.19*** (0.016)	0.21*** (0.018)	0.17*** (0.016)
Age		-0.004*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
Female		-0.038*** (0.003)	-0.042*** (0.004)	-0.044*** (0.004)	-0.042*** (0.003)	-0.043*** (0.004)
Wage			-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)
Tenure			-0.000** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
Debt service to income				0.083*** (0.012)		0.076*** (0.012)
Total debt to income					0.002*** (0.000)	0.001*** (0.000)
Constant	0.460*** -0.005	0.609*** (0.006)	0.604*** (0.006)	0.594*** (0.007)	0.603*** (0.006)	0.595*** (0.007)
Observations	74,468	74,448	74,448	73,792	74,448	73,792
R-squared	0.153	0.161	0.164	0.174	0.167	0.175

Notes: The sample consists of *borrowers nearing distress* with a formal job. The dependent variable is an indicator equal to one if the borrower defaults on any card six months after the cohort month. All independent variables are measured at the cohort month. *Age* is measured in years, and *female* is a dummy variable equal to one if the worker is female. We only include borrowers with a formal job in the cohort month. *Tenure* refers to the length of time (in months) the worker has been in the job, and *wage* is the monthly salary (in thousands of BRL). *Debt service to income* is defined as total monthly debt payments divided by wage, while *total debt to income* refers to total outstanding debt divided by wage. The data are at the borrower-cohort level. All regressions include cohort and county fixed effects. Standard errors, reported in parentheses, are clustered at the cohort level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6: Descriptive statistics of regression variables

	(1)	(2)	(3)	(4)	(5)	(6)
	Mean	SD	p25	p50	p75	N
Default	0.45	0.5	0	0	1	174,893
Positive global limit available	0.16	0.37	0	0	0	174,893
Limit utilization 50% - 60%	0.02	0.15	0	0	0	174,893
Limit utilization 60% - 70%	0.02	0.15	0	0	0	174,893
Limit utilization 70% - 80%	0.03	0.17	0	0	0	174,893
Limit utilization 80% - 90%	0.03	0.18	0	0	0	174,893
Limit utilization 90% - 100%	0.11	0.31	0	0	0	174,893
Limit utilization 100%	0.69	0.46	0	1	1	174,893
Digital bank	0.23	0.42	0	0	0	174,893
State-owned bank	0.11	0.31	0	0	0	174,893
Payroll loan	0.06	0.23	0	0	0	174,893
Mortgage	0.01	0.1	0	0	0	174,893
Auto loan	0.02	0.14	0	0	0	174,893
Personal loan	0.12	0.32	0	0	0	174,893
Other loan types	0.16	0.37	0	0	0	174,893
Default payroll loan	0	0.03	0	0	0	174,893
Default mortgage	0	0.01	0	0	0	174,893
Default auto loan	0	0.03	0	0	0	174,893
Default personal loan	0.01	0.07	0	0	0	174,893
Default other loan types	0.01	0.08	0	0	0	174,893
Positive card limit	0.31	0.46	0	0	1	174,893
Largest positive card limit available	0.16	0.36	0	0	0	174,893
Largest card balance	0.31	0.46	0	0	1	174,893
2nd largest positive card limit available	0.08	0.28	0	0	0	174,893
2nd largest card balance	0.31	0.46	0	0	1	174,893

Notes: Data at the borrower-bank-cohort level for the sample of *borrowers nearing distress* with two or more card lenders and that default on a subset of their cards. All variables are binary and measured in the cohort month. We report the mean in column (1), the standard deviation in column (2), the 25th percentile in column (3), the median in column (4), the 75th percentile in column (5), and the number of observations in column (6).

Table 7: Predictors of selective default

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Unrestricted sample				Sample with interest rate information			
	All cohorts		Post May 2021		All cohorts		Post May 2021	
<i>Group 1: Card characteristics</i>								
Card with the largest contractual limit	-0.06*** (0.00)	-0.05*** (0.00)	-0.08*** (0.01)	-0.08*** (0.01)	-0.03*** (0.01)	-0.04*** (0.01)	-0.05*** (0.01)	-0.06*** (0.01)
Card with the largest balance	0.21*** (0.00)	0.20*** (0.00)	0.23*** (0.01)	0.21*** (0.01)	0.21*** (0.01)	0.20*** (0.01)	0.22*** (0.01)	0.20*** (0.01)
Card with the oldest relationship	0.01*** (0.00)	-0.03*** (0.00)	-0.00 (0.00)	-0.03*** (0.00)	0.03*** (0.00)	-0.03*** (0.00)	0.01*** (0.00)	-0.03*** (0.01)
Card with the largest interest rate					0.31*** (0.00)	0.26*** (0.00)	0.32*** (0.00)	0.27*** (0.00)
<i>Group 2: Holdings of other loan types</i>								
Payroll loan	-0.11*** (0.01)	-0.13*** (0.01)	-0.13*** (0.01)	-0.13*** (0.01)	-0.09*** (0.01)	-0.13*** (0.01)	-0.11*** (0.01)	-0.13*** (0.01)
Mortgage	-0.17*** (0.04)	-0.16*** (0.04)	-0.22*** (0.05)	-0.20*** (0.04)	-0.13** (0.05)	-0.14*** (0.05)	-0.18*** (0.06)	-0.18*** (0.05)
Mortgage x state-owned bank	0.29*** (0.04)	0.18*** (0.04)	0.29*** (0.05)	0.19*** (0.05)	0.14** (0.05)	0.13** (0.05)	0.14** (0.06)	0.14** (0.06)
Auto loan	-0.07*** (0.01)	-0.05*** (0.01)	-0.08*** (0.01)	-0.06*** (0.01)	-0.08*** (0.01)	-0.06*** (0.01)	-0.10*** (0.02)	-0.08*** (0.02)
Personal loan	0.03*** (0.01)	0.02*** (0.01)	0.03*** (0.01)	0.03*** (0.01)	0.01* (0.01)	0.02** (0.01)	0.01* (0.01)	0.03*** (0.01)
Other loans	-0.14*** (0.00)	-0.07*** (0.00)	-0.13*** (0.01)	-0.05*** (0.01)	-0.09*** (0.01)	-0.07*** (0.01)	-0.08*** (0.01)	-0.04*** (0.01)
<i>Group 3: Default on other products</i>								
Default payroll	0.18*** (0.05)	0.17*** (0.05)	0.29*** (0.07)	0.27*** (0.07)	0.27*** (0.06)	0.24*** (0.06)	0.31*** (0.08)	0.28*** (0.07)
Default mortgage	-0.04 (0.17)	-0.03 (0.18)	-0.04 (0.24)	-0.00 (0.25)	-0.01 (0.20)	-0.00 (0.20)	0.19 (0.27)	0.20 (0.27)
Default auto loan	0.21*** (0.06)	0.24*** (0.06)	0.23*** (0.08)	0.28*** (0.07)	0.13* (0.08)	0.16** (0.08)	0.20** (0.09)	0.22** (0.09)
Default personal loan	0.20*** (0.02)	0.19*** (0.02)	0.19*** (0.03)	0.19*** (0.03)	0.16*** (0.03)	0.17*** (0.03)	0.16*** (0.04)	0.17*** (0.04)
Default other loans	0.20*** (0.02)	0.17*** (0.02)	0.16*** (0.03)	0.13*** (0.03)	0.25*** (0.03)	0.20*** (0.03)	0.22*** (0.04)	0.18*** (0.04)
<i>Group 4: Bank type</i>								
Digital bank	-0.17*** (0.00)		-0.15*** (0.00)		-0.12*** (0.01)		-0.11*** (0.01)	
State-owned bank	0.18*** (0.01)		0.17*** (0.01)		0.24*** (0.01)		0.25*** (0.01)	
Observations	164,613	164,570	112,634	112,593	101,989	101,950	68,905	68,869
R-squared	0.190	0.272	0.199	0.287	0.276	0.342	0.290	0.358
Borrower FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	Yes	No	Yes	No	Yes	No	Yes

Notes: The table reports coefficient estimates from Equation 1. The sample is comprised of *borrowers nearing distress* who have two or more credit card lenders, default on at least one card, and repay at least one card. The data are at the borrower-cohort-bank level. All variables are binary. All regressions include borrower-cohort fixed effects. Robust standard errors are reported in parentheses. *Selective: less than 50%* means that borrowers defaulted on the cards that accounted for less than 50% of their total card balance measured in the month when they missed a payment. *Selective: more than 50%* means that borrowers defaulted on the cards that accounted for more than 50% of their total card balance measured in the month when they missed a payment. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8: Difference-in-differences approach

	(1)	(2)	(3)	(4)
	Number of cards	Card debt to income	Delinquency	Default
<i>Post_Treat</i> ^[0–6]	0.05*** (0.02)	0.092*** (0.033)	0.009** (0.004)	0.007 (0.005)
<i>Post_Treat</i> ^[7–15]	0.11*** (0.03)	0.137** (0.053)	0.019** (0.008)	0.011* (0.006)
Constant	1.3*** (0.03)	1.3*** (0.05)	0.029*** (0.010)	0.025*** (0.009)
Observations	67,132	67,132	67,132	67,132
R-squared	0.775	0.841	0.648	0.651

Notes: The table reports results from the estimation of Equation 3. Delinquency (default) is a dummy variable equal to one if the individual has any debt payment overdue by more than 15 (90) days. In all regressions, we include individual fixed effects, firm-by-month fixed effects, and baseline default status interacted with time dummies. Standard errors, reported in parentheses, are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Internet Appendix

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A Empirical part

A1 Sampling

For a given month t (cohort), we take a 1% random sample of individuals who were in good standing with all their credit card loans in month $t - 1$ (i.e., no card with payments overdue by more than 15 days) but became more than 15 days overdue on at least one card in month t . We refer to these individuals who are potentially transitioning into default (i.e., more than 90 days overdue) as *borrowers nearing distress*. To reduce the total sample size, we apply this procedure every three months, starting in June 2019 and ending in December 2022—a total of 15 cohorts. For each individual in cohort t , we also collect credit information for the months $m \in \{t-12, t-11, \dots, t-1, t, t+1, \dots, t+11\}$.

For each cohort, we also take a 0.025% random sample of individuals who are not *borrowers nearing distress*. These individuals include those who were not in good standing with all their cards at $t - 1$ and those who were in good standing with all their cards at $t - 1$ and remained in good standing at month t . We refer to these individuals as *all other borrowers*. The sampling procedure is described in Appendix Table A1. We apply inverse probability weighting to estimate statistics for the population of credit card users. The final data is at the cohort-individual-bank-month level.

In Appendix Table A2, we report the default and delinquency rates for both groups in the month preceding the cohort. For the *all other borrowers* group, 7% of individuals have at least one card with payments overdue by more than 90 days (default), and 10.3% of borrowers have at least one card with payments overdue by more than 15 days (delinquent). By design, both the delinquency and default rates for the *borrowers nearing distress* group are zero. In Appendix Table A3, we report a total of 524,533 observations at the cohort-by-individual level, with 252,844 observations belonging to the *borrowers nearing distress* group. Consistent with the growth of the credit card industry, more recent cohorts are larger than the older ones.

Table A1: Sampling procedure for cohort t

	Repayment Standing at t-1	Repayment Standing at t	Share sampled
Borrowers nearing distress	Good	Bad	1%
All other borrowers	Good	Good	
All other borrowers	Bad	Good	0.025%
All other borrowers	Bad	Bas	

Notes: The table describes our sampling procedure. A borrower is classified as being in good repayment standing if all active cards exhibit payment delays of no more than 15 days. Conversely, an individual is classified as being in bad repayment standing if at least one active card has a payment overdue by more than 15 days. The sample is taken at time t at the individual level, conditional on the individual having at least one active card.

Table A2: Default and delinquency incidence at $t - 1$ (month before the cohort)

	Borrowers nearing distress		All other borrowers	
	Card level	Borrower level	Card level	Borrower level
<i>Panel A: All card users</i>				
Delinquency rate	0	0	10.3	13.2
Default rate	0	0	7.0	9.2
<i>Panel B: Card users with 2+ cards</i>				
Delinquency rate	0	0	10.4	17.6
Default rate	0	0	7.0	12.2

Notes: A credit card loan is classified as delinquent if its payments are more than 15 days overdue and as in default if its payments are more than 90 days overdue. A borrower is considered delinquent (in default) if at least one of their credit cards has payments that are more than 15 (90) days overdue.

Table A3: Sample evolution

Cohort	Borrowers nearing distress	All other borrowers	Total
201906	15,186	15,799	30,985
201909	15,435	16,186	31,621
201912	13,348	16,476	29,824
202003	18,293	16,469	34,762
202006	10,128	16,390	26,518
202009	10,323	16,514	26,837
202012	11,281	16,931	28,212
202103	17,696	17,047	34,743
202106	15,804	18,025	33,829
202109	17,780	18,726	36,506
202112	22,750	19,668	42,418
202203	22,817	20,130	42,947
202206	20,407	20,667	41,074
202209	21,499	21,125	42,624
202212	20,097	21,536	41,633
Total	252,844	271,689	524,533

Notes: The table shows the evolution of the number of unique card users per cohort and group classification.

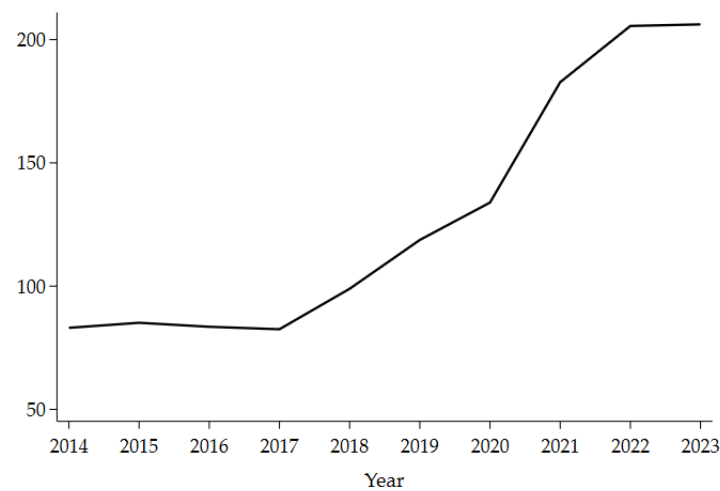
Table A4: Size of groups based on the number of credit card lenders

Number of card lenders	Frequency			Shares	
	Borrowers nearing distress	All other borrowers	Total	Borrowers nearing distress	All other borrowers
1	117,788	163,114	280,902	47%	60%
2	68,462	65,961	134,423	27%	24%
3	34,509	25,363	59,872	14%	9%
4	16,432	9,908	26,340	6%	4%
5	7,793	4,033	11,826	3%	1%
6+	7,860	3,310	11,170	3%	1%
Total	252,844	271,689	524,533	100%	100%

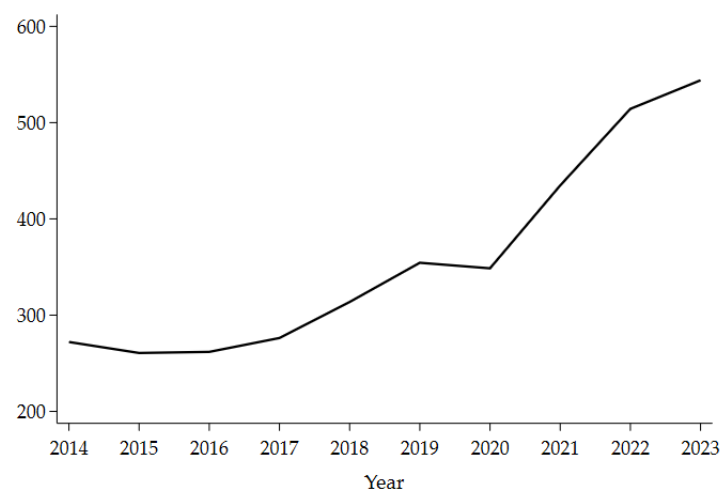
Notes: Borrowers are categorized based on the number of credit card lenders. The data are at the borrower-cohort level.

A2 Institutional setting: additional details from aggregate data

Figure A1: Evolution of the number of credit cards and credit card balance



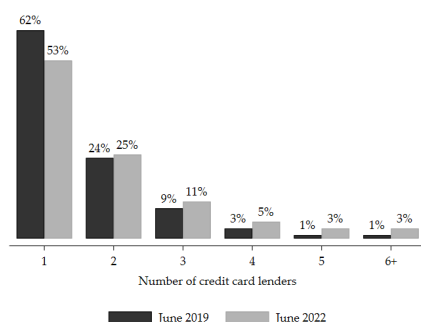
(a) Total number of active credit cards (in millions)



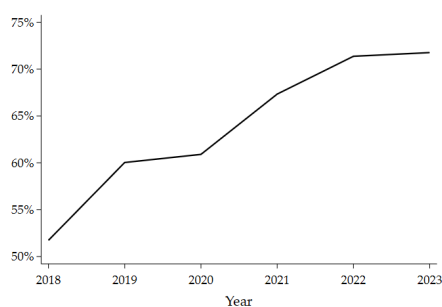
(b) Total credit card loans (inflation-adjusted values in billions BRL)

Notes: The figure reports the total number of active credit cards and credit cards outstanding balance per year. Monetary values are expressed in constant 2023 Brazilian reais and adjusted by the consumer price index (*Índice Nacional de Preços ao Consumidor Amplo*, IPCA). Source: Central Bank of Brazil (*Estatísticas de Meios de Pagamentos* and *IF.data*).

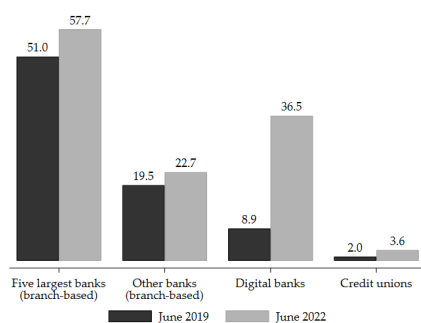
Figure A2: Evolution of borrowers with more than one card lender and growth of digital banks



(a) Share of credit card users by number of credit card lenders, 2019 and 2022



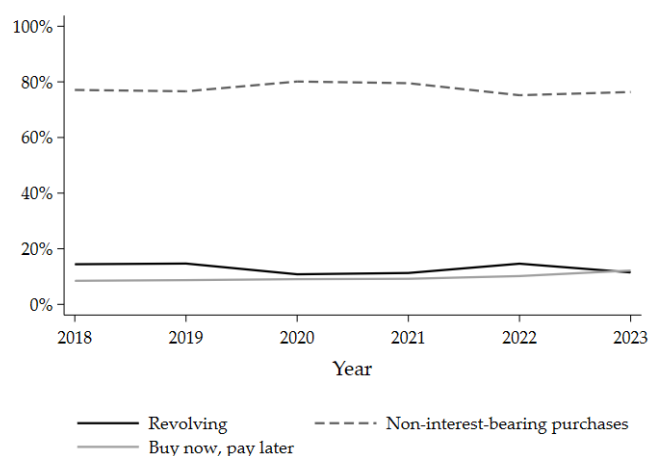
(b) Share of total credit card balance held by borrowers with two or more card lenders



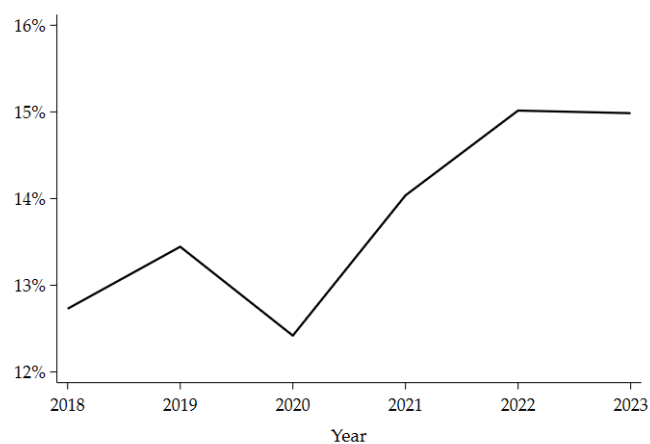
(c) Number of credit card users by bank type category, 2019 and 2022

Notes: Panel (a) shows the distribution of credit card users by the number of card lenders in 2019 and 2022. Panel (b) reports the share of total outstanding credit card balances held by users with two or more card issuers. Panel (c) presents the evolution of the number of credit card users per bank type category. All statistics are computed using data from [Central Bank of Brazil \(2022\)](#).

Figure A3: Composition of credit card balance and participation of credit card debt in total household debt



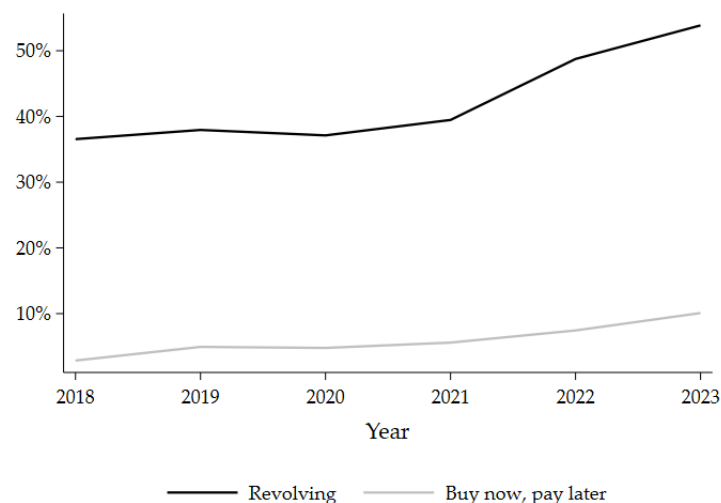
(a) Composition of credit card balance



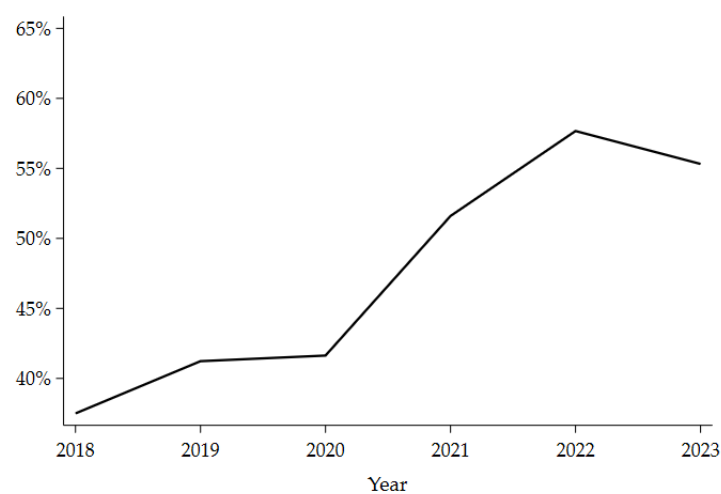
(b) Participation in total household debt

Notes: Panel (a) shows the composition of credit card balances. *Revolving* balances arise when cardholders do not pay the full statement balance by the due date and therefore accumulate interest-bearing debt. *Non-interest-bearing purchases* are transactions made during the billing cycle that are fully due on the next payment date and incur no interest if paid on time. *Buy now, pay later* (BNPL) balances correspond to credit card transactions repaid in installments, with or without interest. Panel (b) shows the participation of credit card debt in total household debt. The data are from the Central Bank of Brazil's [SCR data](#) and [Time Series Management System](#).

Figure A4: Participation of credit card debt in total household debt default



(a) Share of total balance in default

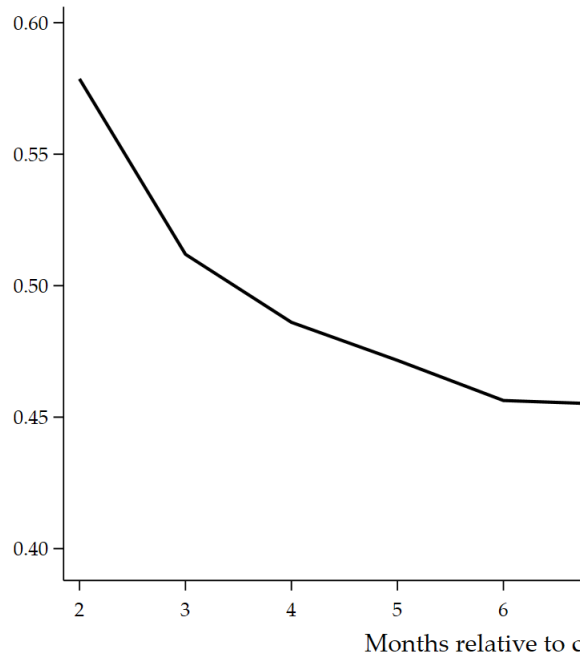


(b) Participation in total amount overdue for more than 15 days

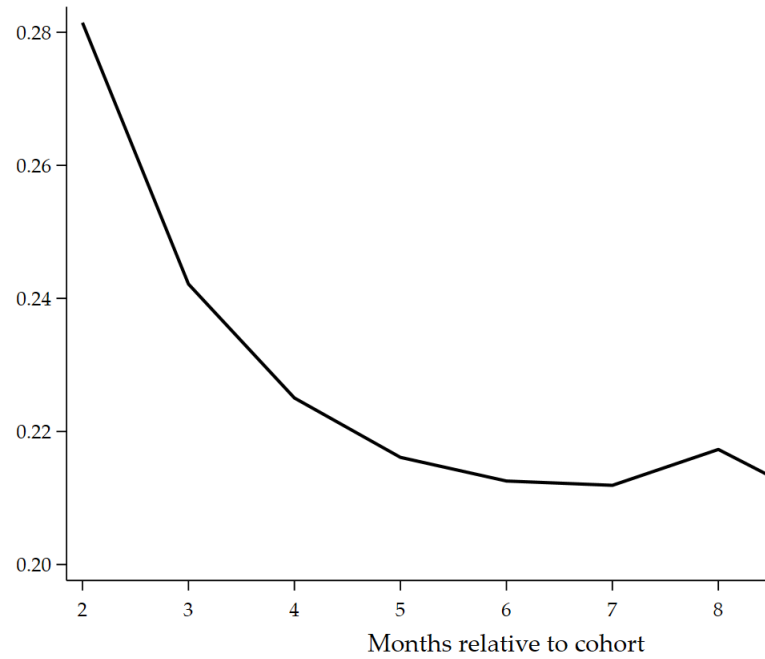
Notes: Panel (a) shows the share of balances in default for the *revolving* and *buy now, pay later* categories. Default is defined as payments more than 90 days past due. Panel (b) shows the share of credit card debt overdue by more than 15 days in total household debt overdue by more than 15 days. We use data from the Central Bank of Brazil's [SCR.data](#) and [Time Series Management System](#).

A3 Default and selective default evolution t months after becoming 15+ days overdue

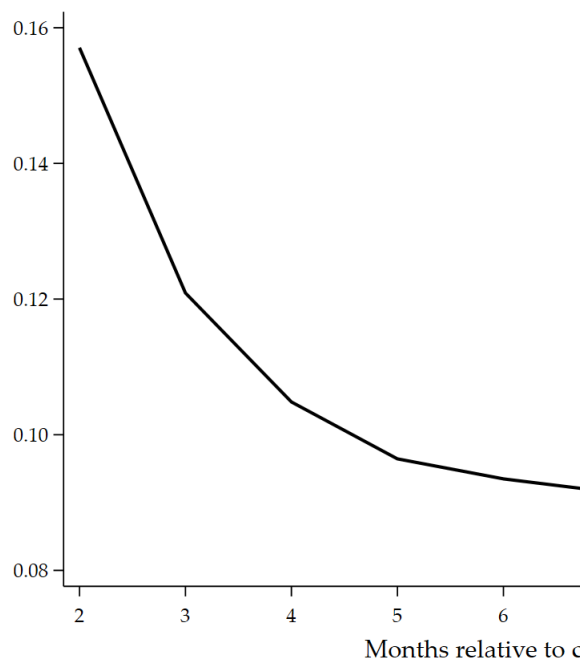
Figure A5: Default and selective default evolution t months after becoming 15+ day overdue



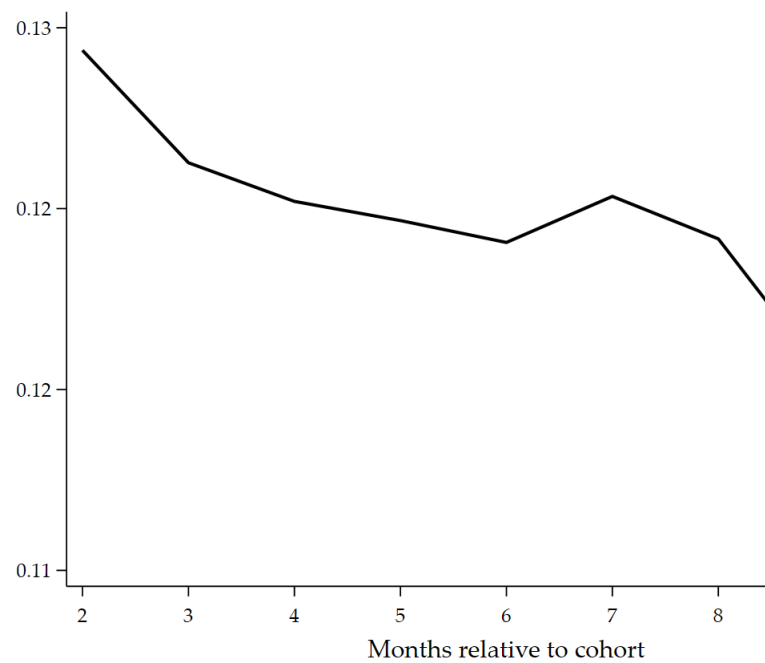
(a) Share of *borrowers nearing distress* with at least one card 90+ days overdue



(b) Share of *borrowers nearing distress* with at least one card 90+ days overdue and at least one in good standing



(c) Share of *borrowers nearing distress* with at least one card 90+ days overdue, at least one in good standing, and default on less than 50% of the total card balance

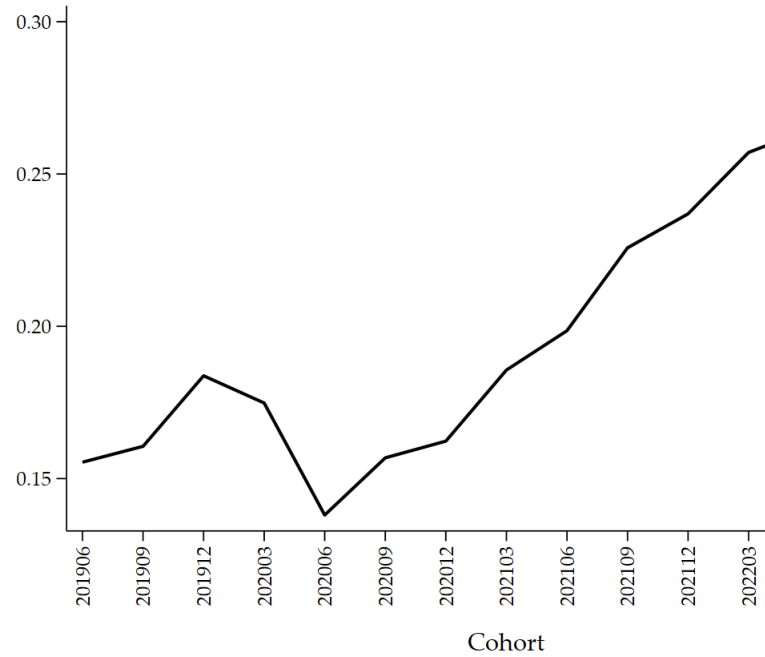


(d) Percentage of *borrowers nearing distress* with at least one card more than 90 days overdue and at least one card in good standing, and default on more than 50% of the total card balance

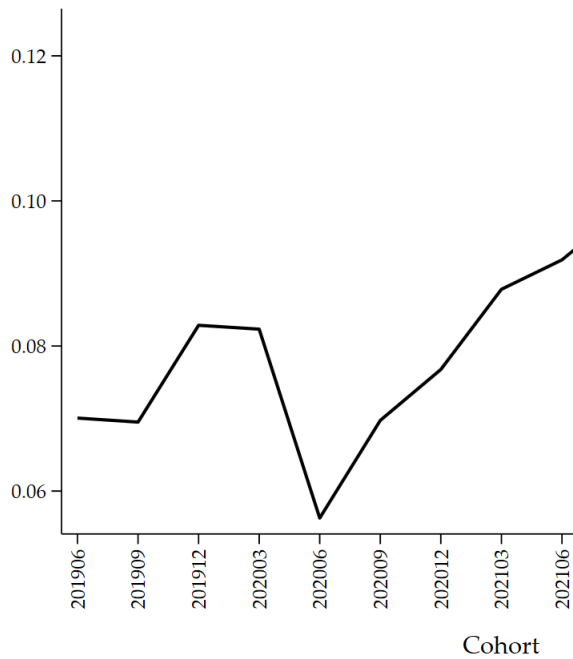
Figure A6: Default and selective default evolution



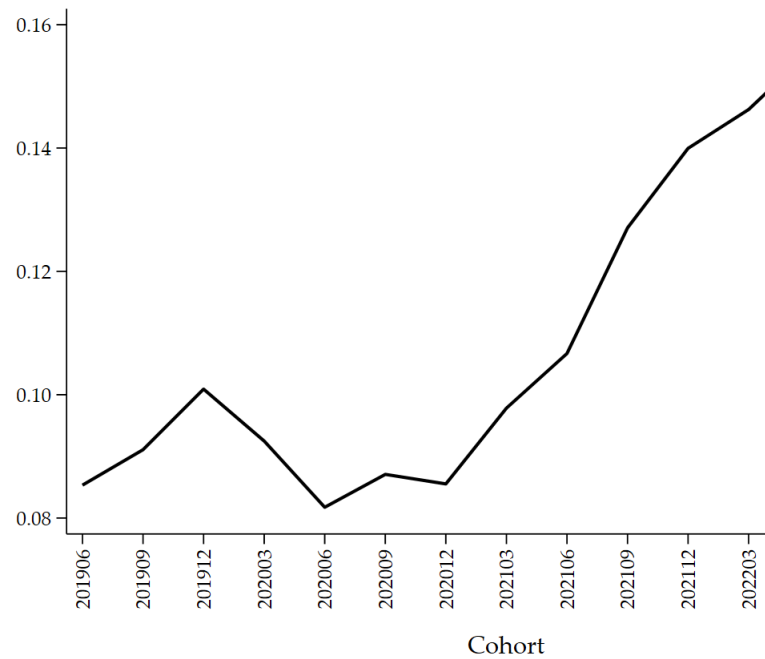
(a) Share of *borrowers nearing distress* with at least one card 90+ days overdue



(b) Share of *borrowers nearing distress* with at least one card 90+ days overdue and at least one in good standing



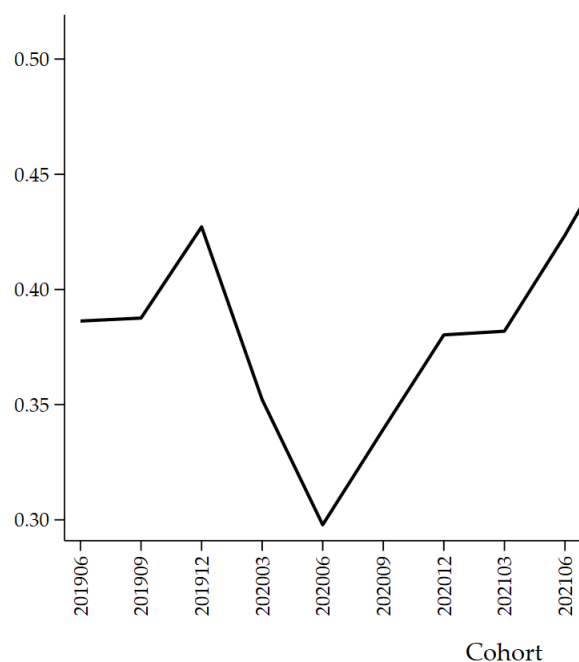
(c) Share of *borrowers nearing distress* with at least one card 90+ days overdue, at least one in good standing, and default on less than 50% of the total card balance



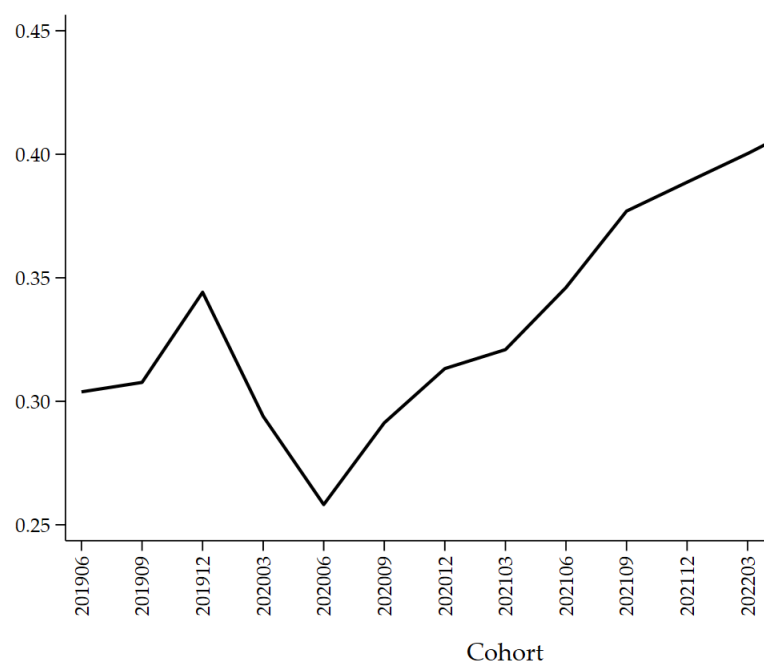
(d) Percentage of *borrowers nearing distress* with at least one card more than 90 days overdue and at least one card in good standing, and default on more than 50% of the total card balance

Notes: The sample comprises *borrowers nearing distress* who have one or more cards. We classify a card as in default if there is balance overdue for more than 90 days six months after the borrower becomes more than 15 days overdue.

Figure A7: Default and selective default evolution, borrowers with at least two cards



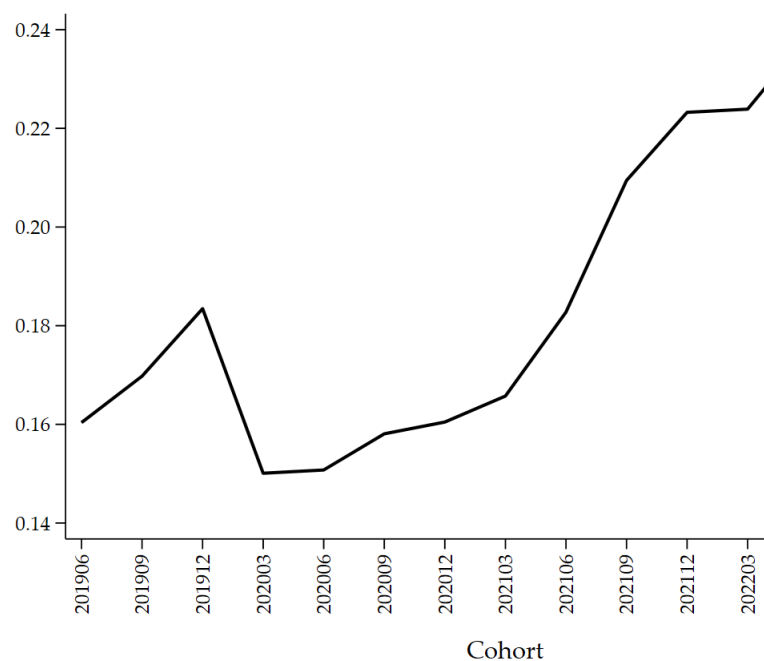
(a) Share of *borrowers nearing distress* with at least one card 90+ days overdue



(b) Share of *borrowers nearing distress* with at least one card 90+ days overdue and at least one in good standing



(c) Share of *borrowers nearing distress* with at least one card 90+ days overdue, at least one in good standing, and default on less than 50% of the total card balance

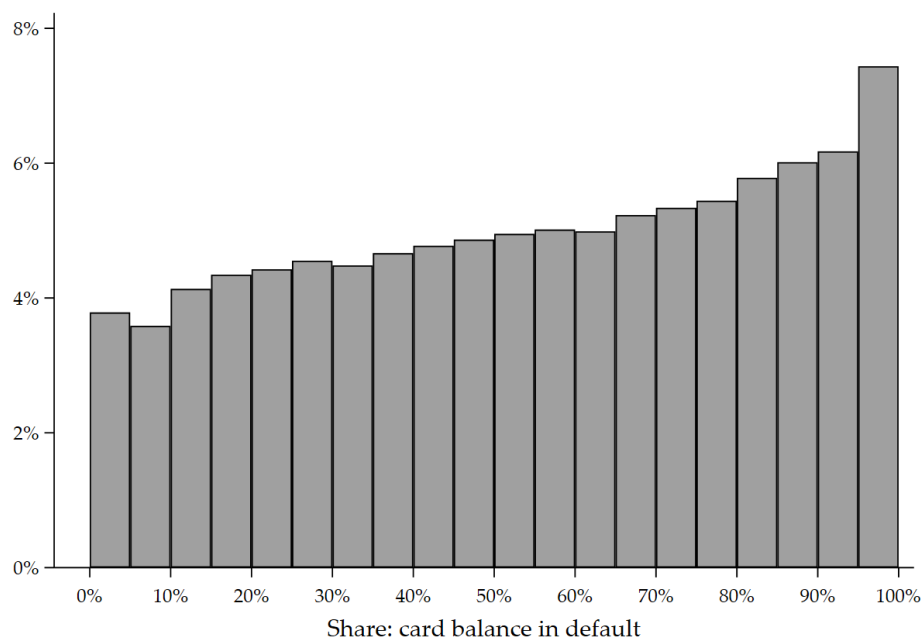


(d) Percentage of *borrowers nearing distress* with at least one card more than 90 days overdue and at least one card in good standing, and default on more than 50% of the total card balance

Notes: The sample comprises *borrowers nearing distress* who have two or more cards. We classify a card as in default if there is a balance overdue for more than 90 days six months after the borrower becomes more than 30 days overdue.

A4 Share of balance in default among selective defaulters

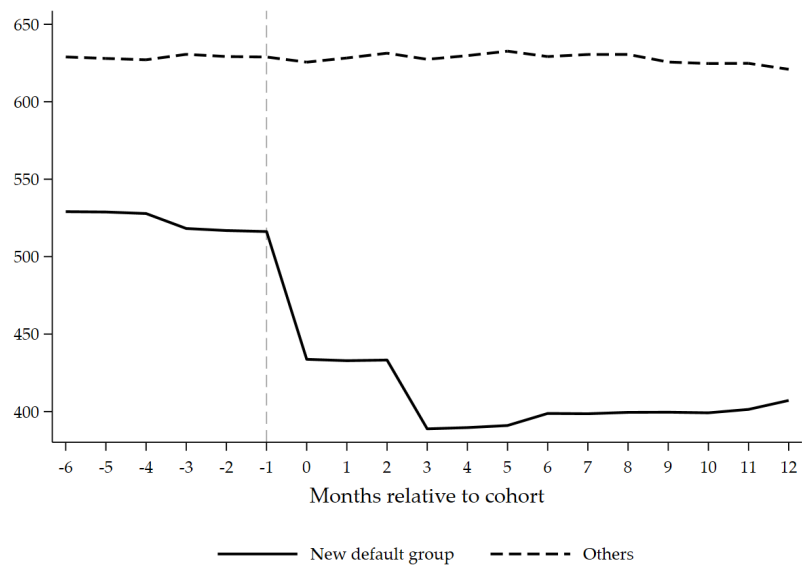
Figure A8: Share of balance in default



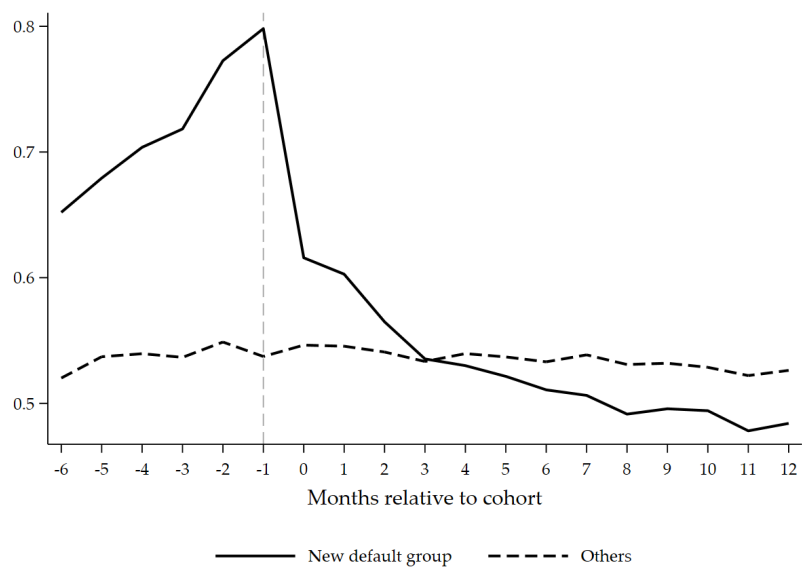
Notes: This figure plots the distribution of the share of total balance in default among selective defaulters. We measure default as payment more than 90 days past due six months after the cohort month. We compute the shares using the balances at the cohort month.

A5 Default and credit scores

Figure A9: Evolution of credit scores around default events



(a) Credit score



(b) Probability of obtaining a new loan

Notes: This figure plots the average credit score and probability of obtaining a new loan for borrowers who experience a new default in a given month (the cohort) relative to borrowers who do not experience a default in the same month.

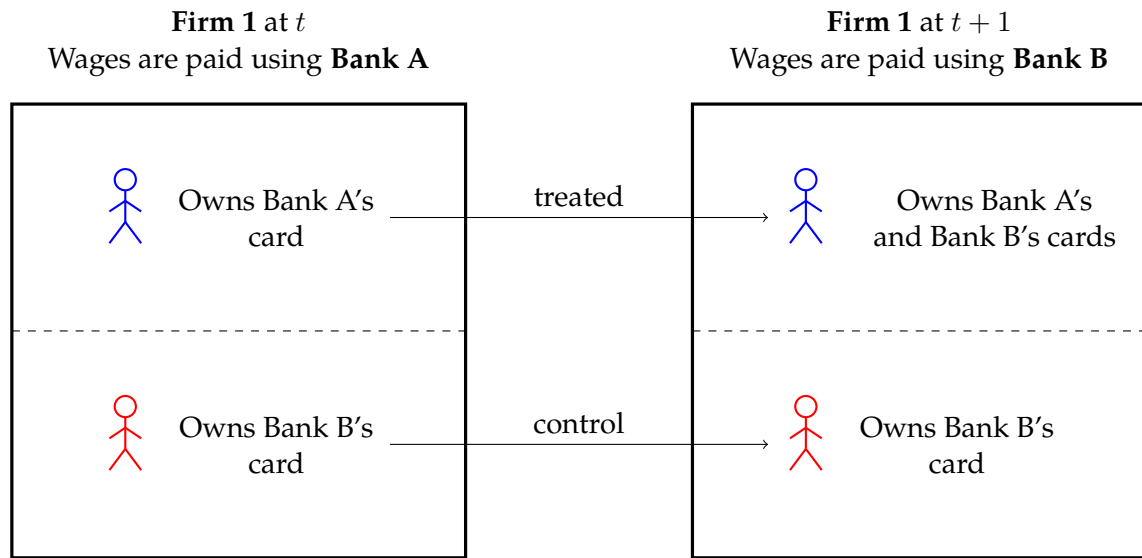
Table A5: Predictors of selective default

	(1)	(2)
	All cohorts	Post May 2021
Limit utilization 50% - 60%	0.11*** (0.01)	0.11*** (0.01)
Limit utilization 60% - 70%	0.12*** (0.01)	0.12*** (0.01)
Limit utilization 70% - 80%	0.13*** (0.01)	0.14*** (0.01)
Limit utilization 80% - 90%	0.20*** (0.01)	0.21*** (0.01)
Limit utilization 90% - 100%	0.23*** (0.01)	0.23*** (0.01)
Limit utilization 100%	0.46*** (0.01)	0.47*** (0.01)
Positive global limit	-0.28*** (0.01)	-0.26*** (0.02)
Digital bank	-0.19*** (0.00)	-0.15*** (0.01)
State-owned bank	0.20*** (0.01)	0.20*** (0.01)
Payroll loan	-0.03*** (0.01)	-0.07*** (0.01)
Mortgage	-0.06 (0.05)	-0.11** (0.05)
Mortgage x state-owned bank	0.10** (0.05)	0.08 (0.06)
Auto loan	-0.05*** (0.01)	-0.07*** (0.02)
Personal loan	0.05*** (0.01)	0.04*** (0.01)
Other loans	-0.03*** (0.01)	-0.03*** (0.01)
Default payroll	0.20*** (0.06)	0.25*** (0.07)
Default mortgage	0.15 (0.18)	0.14 (0.23)
Default auto loan	0.12 (0.08)	0.15 (0.09)
Default personal loan	0.15*** (0.03)	0.14*** (0.03)
Default other loans	0.24*** (0.03)	0.21*** (0.04)
Largest limit available	0.03*** (0.01)	0.03*** (0.01)
2nd largest limit available	0.02** (0.01)	0.02*** (0.01)
Highest interest rate	0.25*** (0.00)	0.25*** (0.00)
Constant	0.09*** (0.01)	0.09*** (0.01)
Observations	102,632	69,257
R-squared	0.321	0.344

Notes: The table reports coefficient estimates from Equation 1. The sample consists of *borrowers nearing distress* who have two or more credit card lenders, default on at least one card, and repay at least one card. The data are at the borrower-cohort-bank level. All variables are binary. All regressions include borrower-cohort fixed effects. Robust standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

A6 Difference-in-differences analysis

Figure A10: Identification strategy



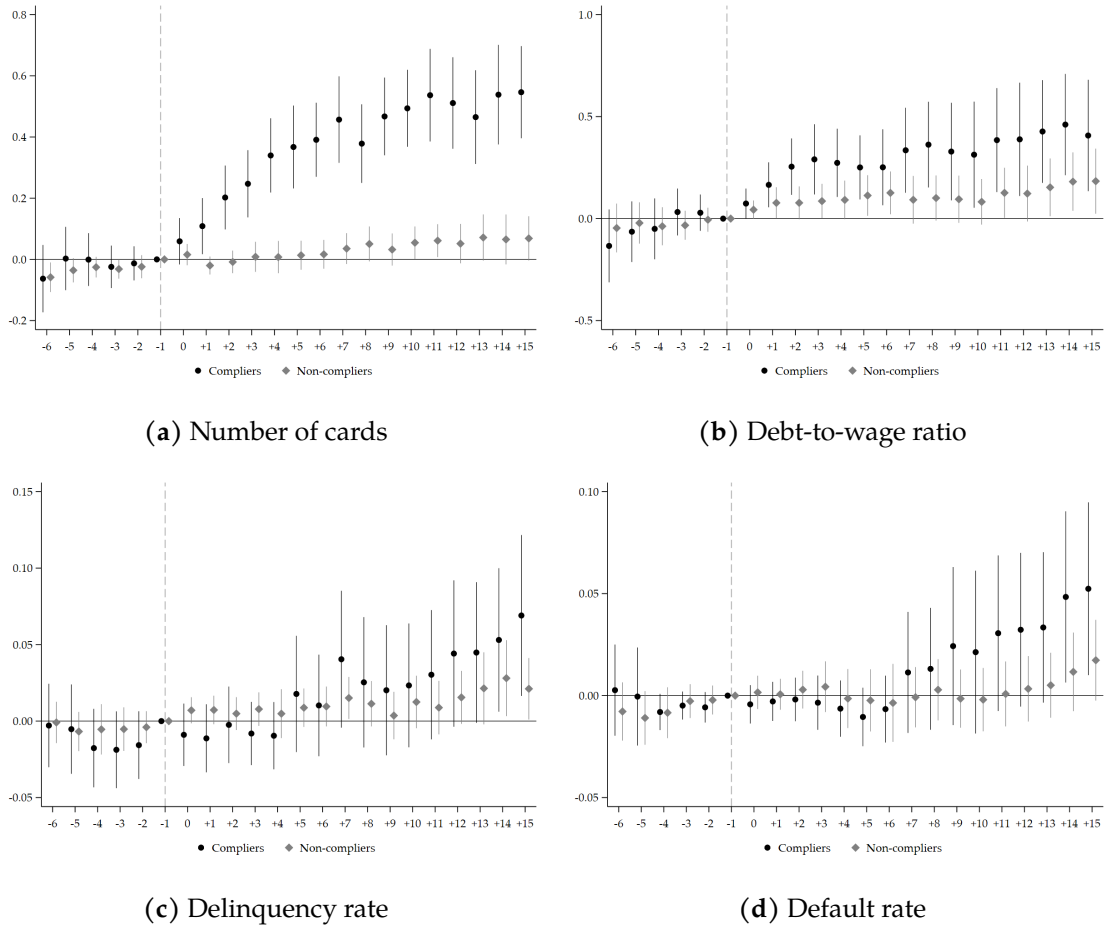
Notes: We compare the relative number of cards and default probability of the blue (treated) and red (control) groups of borrowers before and after the change in the bank that manages the payroll.

Table A6: Descriptive statistics

	Control	Treated		
		Complier	Non-complier	All
Number of card lenders	1.5	1.3	1.2	1.2
Debt-to-wage ratio	1.5	1.0	1.3	1.3
Delinquency rate	0.05	0.02	0.02	0.02
Default rate	0.04	0.01	0.02	0.02
Number of individuals	702	163	2,701	2,864

Notes: We report means of selected variables in the month prior to the new bank assuming management of payroll. Means are reported separately for three groups: individuals in the control group, compliers (treated borrowers who use a card from the new bank at some point after treatment), and non-compliers (treated borrowers who never use a card from the new bank in any month of the post-treatment period). Delinquency (default) is defined as a dummy variable equal to one if the individual has any debt payment overdue by more than 15 (90) days

Figure A11: Difference-in-Differences: Compliers versus Non-compliers



Notes: The figures show the results from estimating Equation 2 with the following modification: treatment dummies are interacted with post-treatment take-up status—compliers (treated borrowers who use a card from the new bank at some point after treatment) and non-compliers (treated borrowers who never use a card from the new bank in any month of the post-treatment period). We report 95% confidence intervals calculated using standard errors clustered at the firm level. Delinquency (default) is defined as a dummy variable equal to one if the individual has any debt payment overdue by more than 15 (90) days

B Proof Lenders FOCs

Taking the first order conditions with respect to an arbitrary characteritic x , we get:

$$\frac{\partial_x \pi}{\partial_x D} = -\frac{\pi}{D} \quad (24)$$

We have:

$$\partial_q \pi = -mc_q \quad (25)$$

$$\partial_R \pi = \theta_R R + \theta \quad (26)$$

$$\partial_L \pi = -mc_L \quad (27)$$

Using the envelope theorem, we have:

$$\partial_q D = D_U \cdot U_q = D_U \cdot \theta V_q \quad (28)$$

$$\partial_R D = D_U \cdot U_q = D_U \cdot \theta [1 + \Delta M] \quad (29)$$

$$\partial_L D = D_U \cdot U_q = D_U \cdot u'(\mathbf{L}) \quad (30)$$

With ΔM is the "shadow cost" of increasing loan size, derived below. When income is non-random, this cost is equal to 0 except exactly at points where an increase in R triggers an additional default in another card.

To derive ΔM , let us rewrite the borrower default choice:

$$\max_{r_i} \mathbf{1}_{r_i \geq R_i} [V - R_i + \epsilon_i] + (1 - \mathbf{1}_{r_i \geq R_i}) v^d + M(r_i, W, \epsilon_{-i}) \quad (31)$$

Where $M(r_i, W, \epsilon_{-i})$ is $\sum_{c \in \mathcal{C} \setminus i} \mathbf{1}_{r_c \geq R_c} [V - R_c + \epsilon_c] + (1 - \mathbf{1}_{r_i \geq R_i}) v^d$ s.t. $\sum_{c \in \mathcal{C} \setminus i} r_i \geq W - r_i$.

The above problem implies that, conditional on ϵ draws other than the one for credit card i (denoted by ϵ_{-i}), there exist an $\bar{\epsilon}_i$ so that the borrower chooses to repay the credit card.

The expected utility of the borrower conditional on ϵ_{-i} and income W is thus:

$$\int_{-\infty}^{\bar{\epsilon}_i} v^d + M(0) dF(\epsilon) + \int_{\bar{\epsilon}_i}^{\infty} V - R_i + \epsilon_i + M(R_i, W, \epsilon_{-i}) dF(\epsilon) \quad (32)$$

Let us now consider the case in which the lender cannot repay all the cards. There exist an ordering of the credit cards so that repaying the credit card i reduces the probability of repaying the card c which is just below in the repayment hierarchy. Using the budget constraint gives the repayment strategy $r_c(\epsilon) = W - \bar{R} - R_i$ where $\bar{R}$ is the sum of repayments for credit cards with a higher priority than card c and i . We thus have that the survival probability on credit card c when income is random is equal

to $\theta_c := E_W(\mathbf{1}_{W - \bar{R} - \bar{R} \geq R_c})$. Taking the derivative of equation 32 with respect to R_i , we get $\int_{\bar{\epsilon}_i}^{\infty} -1 + \Delta M(R_i, W, \epsilon_{-i}) dF(\epsilon)$. In that case, we have the partial derivative of M is $\Delta M = \partial_{R_i} \theta_c \cdot [V(q_c) - R_c + \epsilon_c - v^d]$.

The derivative of 32 with respect to R thus give $\theta + \Delta M$. When income is non-random, ΔM is equal to zero everywhere except when it crosses a threshold so that an additional card cannot be repaid due to the increase in R .

The first equation we report in the main text is $\frac{\partial_L \pi}{\partial_L D} = \frac{\partial_R \pi}{\partial_R D}$, the second is $\frac{\partial_L \pi}{\partial_L D} = \frac{\partial_q \pi}{\partial_q D}$ and the third is $\frac{\partial_R \pi}{\partial_R D} = -\frac{\pi}{D}$.

C Proof Linear model

Let us assume that lenders expect the borrowers to default on all but one credit card (i.e, when the debt face values of competitors' contracts are $R_c = W$). Then, given that the marginal rate of transformation $\frac{\alpha}{mc}$ on credit card loans is higher than the one on interest rates $\frac{1}{1+\theta_R} = 1$, the lender then optimally sets $W = R$.

Due to the non linearities in our model, depending on the parameters, this might not be the only equilibrium. In particular, there might be an equilibrium without default. For instance, let us assume that the borrower is at the budget constraint. Each lender might not have incentive to increase the credit limit and thus interest rates as it would create default (this is the case when $(\alpha - mc)\Delta L + \beta E[\sum \mathbf{1}_{\{r^*(\epsilon) > R\}} (V(q) - R + \epsilon - v^d) + v^d] < \beta E[V(q) - R + \epsilon]$). In addition, the borrower might not have incentives to get an additional card that would trigger default (e.g., abstracting from the love for variety and the cost of getting a credit card, this would happen when $\alpha L + \beta v^d < 0$).

We focus on an equilibrium with default as this is consistent with the data and makes competition for informal seniority relevant. Default would also appear naturally in a more general model when income is a random variable, making the equilibrium we pick relevant to understand selective default forces that would arise in the general model. We focus on the case where income is not random for tractability. While the assumption about default being constants affects the shape of the default function and default elasticity, the first order conditions would still be the same. The economic forces at play would thus remain the same.

One simple way to make the equilibrium with default and $R = W$ unique in our model is to assume that the cost of default is increasing in the share of credit card the borrower defaulted (denoted s). One can share those cost so that borrowers have incentives to default on all but one card (i.e., choosing $v^d(s)$ large enough when $s < 1$ so that ΔM , defined in section B is small or even negative when the borrowers default on less than all the credit cards). The assumption that defaulting on one card when you have, for instance two, is less costly than when you have one make sense empirically as the credit score impact is lower when you default. It is also arguably less costly for

borrowers as they can still use one of the card to make important transactions.

D Examples of micro-foundations for V_c

Let us now discuss different functional forms for V_c that can capture the stylized facts presented in Section 4.2. The empirical section suggests that three main channels can affect repayment behavior. First, the lenders can invest in an app or a reward system that the borrower values, making the threat of cancelling the card upon default more costly. Second, the lender can change the credit limit on the card depending on the repayment behavior. Third, the lender can offer another loan product in the second period if the borrower does not default on the card.

Those three channels can be captured using the following parametrization of V_c :

$$\text{App or reward system: } V_c(q_c, \mathcal{R}) = q_c(\mathcal{R}) \quad (33)$$

$$\text{Credit limit: } V_c(\tilde{L}_c, \tilde{R}_c, \mathcal{R}) = \tilde{L}_c(\mathcal{R}) - \beta \tilde{R}_c(\mathcal{R}) \quad (34)$$

$$\text{Other Loan product: } \sum_c V_c(\underbrace{\tilde{L}_c, \tilde{R}_c}_{q_c}, \mathcal{R}) = E_{\varepsilon_b}[\max_b \{\tilde{L}_b(\mathcal{R}) - \beta \tilde{R}_b(\mathcal{R}) + \varepsilon_b\}] \quad (35)$$

In Equation 73, q_c can be interpreted as the app quality or the generosity of the reward system. In Equations 74 and 75, $\tilde{L}_c$ and $\tilde{R}_c$ are the loan size and the debt face value for the credit card and the other loan product, respectively. The variables q_c , $\tilde{L}_c$ and $\tilde{R}_c$ are functions of $\mathcal{R}$. That is, lenders can condition their value based on the borrower's repayment behavior, potentially canceling access to the bank app or the rewards if the borrower does not repay the credit card.

The timing assumption behind Equations 74 and 75 is the following, and requires adding one period to the model. After seeing the repayment in period 2, the lender changes the credit card contract to offer a new loan product. The repayment $\tilde{R}$ of the new contract happens at the end of a third period. The benefit of borrowing $\tilde{L}$ happens in the second period. For simplicity, we assume that the borrower cannot get a new credit card contract in period 2 (for Equation 74), or that the other loan contracts are exclusive in the second period (for Equation 75). This assumption is in line with the empirical evidence that other contracts offered are mortgages or car loans, for which borrowers typically only get one. The fact that the borrower does not get a new card (for Equation 74) can be micro-founded by a model in which the repayment shocks in the third period are perfectly correlated with those in the second period. In that case, since the next section shows that the borrower repays only one contract in the second period, the lender learns that if they have seniority in the third period, and are thus the only lender.

The following section ?? shows that when lending generate a positive net present

value, there exist an equilibrium in which it is optimal to exclude borrowers in the second period (i.e., set $q_c = L_c = R_c = 0$ if $c \notin \mathcal{R}$) if they did not repay their debt, and to set $\tilde{R} = \tilde{W}$ if they repays, where $\tilde{W}$ is the borrower wage in the third period. In that case, all three equations are equivalent: They are all increasing functions of one variable (q_c or $\tilde{L}_c$) conditional on repayment. V_c can be normalized to zero conditional on defaulting on card c . This justifies the general approach used in Equation ??.

E Examples of Micro-foundations for c_c

Using the three micro-foundations discussed in Section ??, potential functional forms for the cost function are:

$$\text{Rewards: } c_c(q_c, \theta_c(R_c, q_c)) := c_c(\max\{q_c(\mathcal{R})\}) \text{ positive convex and increasing} \quad (36)$$

$$\text{Credit limit: } c_c(R, \tilde{R}, \tilde{L}, \theta_c(R, \tilde{R}, \tilde{L})) = -\theta_c(R, \tilde{R}, \tilde{L})\{\tilde{R} - mc\tilde{L}\} \quad (37)$$

$$\text{Other Loan: } c_c(R, \tilde{R}, \tilde{L}, \theta_c(R, \tilde{R}, \tilde{L})) = -\theta_c(R, \tilde{R}, \tilde{L})\tilde{D}(\tilde{R}, \tilde{L})\{\tilde{R} - mc\tilde{L}\} \quad (38)$$

In Equation 76, we consider that setting a generous reward system requires the lender to set aside some money in the first period $c(\max\{q_c(\mathcal{R})\})$. The operator $\max\{q_c(\mathcal{R})\}$ captures the fact that the bank pays the cost in the first period and should ensure to have invested enough to provide the maximum reward it committed to.

In equations 77 and 78, the costs are paid in the second period, after the repayment $\mathcal{R}$ is observed. We drop the $\mathcal{R}$ from the variables $(R, \tilde{R}, \tilde{L})$ to simplify the notation. The cost function can be positive or negative depending on the markup the lender sets for the other product, sold in the second period. In Equation 78, $D(\tilde{R}, \tilde{L})$ is the probability of choosing bank c in the second period. It is derived from Equation 75.

As in section ??, when lending is a positive NPV, it is optimal to set $\tilde{R} = \tilde{W}$, where $\tilde{W}$ is the borrower's income in the second period. All the equations are thus functions of one variable (q or $\tilde{L}$).

F Numerical Simulation

Utility : To perform the numerical simulation, we model the default choice as an effort choice problem:

$$U(\mathcal{C}) = \max_{e_c} \sum_{c \in \mathcal{C}} \{u(L_c) + \theta(e_c)[u(V(q_c) - R_c) - v^d(e)] + v^d(e)\} \quad (39)$$

$$s.t. \sum_c e_c = W \quad (40)$$

$v^d(e)$ captures the idea that the cost of defaulting is lower when the lender sees

that the borrower has put effort into repaying. This modelling assumption is done for tractability and does not change the key economic channels affecting default, they just affect the functional forms.

Default : Using the first order condition and denoting λ the shadow cost, we get:

$$\theta'[u(V(q_c) - R_c) - v^d] + (1 - \theta)v^{d'} = \lambda \quad (41)$$

Assuming $\theta(e_c) = \alpha + \beta e$ and $v^d := v + \nu e$, Using the budget constraint, the survival probability, denoted θ_c is:

$$e_c = \frac{\overbrace{\beta[u(V(q_c) - R_c) - v] + (1 - \alpha)\nu}^{U_c} - \lambda}{2\beta\nu} \quad (42)$$

$$\Rightarrow \sum_c \frac{\beta[u(V(q_c) - R_c) - v] + (1 - \alpha)\nu - \lambda}{2\beta\nu} = W \quad (43)$$

$$\Leftrightarrow \lambda = \frac{\sum_c \{\beta[u(V(q_c) - R_c) - v] + (1 - \alpha)\nu\} - W2\beta\nu}{|\mathcal{C}|} \quad (44)$$

$$\Rightarrow \theta(e_c) = \alpha + \frac{\nu^{-1}}{2} [U_c - E[U_c]] + \beta\nu \frac{W}{|\mathcal{C}|} \quad (45)$$

The survival probability is thus decreasing in the number of credit cards ($|\mathcal{C}|$) and increasing in how good the card is relative to other cards in the portfolio ($U_c - E[U_c]$)

Choice of cards: We consider that the lender observes the cards the borrower chooses $\mathcal{C}$ and offers a contract that depends on that

$$\max_c \overbrace{U(\mathcal{C}) - C(|\mathcal{C}|)}^{\tilde{U}(\mathcal{C})} + e_{U(\mathcal{C})} \quad (46)$$

Where $e_{U(\mathcal{C})}$ is extreme value distributed. We consider that the lender b observes the set of competitors' cards held by the borrowers ($\mathcal{C}_{-b}$) and then chooses the contract terms conditional on that set. The demand thus has a logit form:

$$D(U(\mathcal{C})) = \frac{\exp(\tilde{U}(\mathcal{C}))}{\sum_c \exp(\tilde{U}(c))} \quad (47)$$

Contract terms : Lenders set contract terms to maximise profits. Denoting $\mathcal{C}_b$ the set of credit card portfolios containing the lender b card, lender maximization problem

is:

$$\sum_{\mathcal{C} \in \mathcal{C}_b} D(U(\mathcal{C})) \underbrace{[\theta(\mathcal{C})R - mcL]}_{\pi(\mathcal{C})} \quad (48)$$

Using the envelope theorem, we get that the sensitivity of $D(U(\mathcal{C}))$ to R and q are $-\theta_c((\mathcal{C}))D(U(\mathcal{C}))$ and $\theta_c((\mathcal{C}))V'(q)D(U(\mathcal{C}))$.

Using the first-order conditions, we get:

$$R : \sum D(U(\mathcal{C}))[\theta_R(\mathcal{C})R + \theta(\mathcal{C})] = \sum D'(U(\mathcal{C}))\theta(\mathcal{C})\pi(\mathcal{C}) \quad (49)$$

$$L : mc \sum D(U(\mathcal{C})) = \sum D'(U(\mathcal{C}))u'(L)\pi(\mathcal{C}) \quad (50)$$

$$\iff u'(L) = mc \frac{\sum D(U(\mathcal{C}))}{\sum D'(U(\mathcal{C}))\pi(\mathcal{C})} \quad (51)$$

$$V : [c'(q) - \partial_q \theta_c R] \sum D(U(\mathcal{C})) = V'(L) \sum D'(U(\mathcal{C}))\theta(\mathcal{C})\pi(\mathcal{C}) \quad (52)$$

$$\iff V'(q) = [c'(q) - \partial_q \theta_c R] \frac{\sum D(U(\mathcal{C}))}{\sum D'(U(\mathcal{C}))\theta(\mathcal{C})\pi(\mathcal{C})} \quad (53)$$

In a symmetric equilibrium, we have:

$$\theta_c(\mathcal{C}) = \alpha + \beta \nu \frac{W}{|\mathcal{C}|} \quad (54)$$

$$\theta_X = \frac{\nu^{-1}}{2} [1 - \frac{1}{|\mathcal{C}|}] \partial_X U_c \quad (55)$$

Simplification: If the demand for cards is such that the borrowers get n cards (i.e., $C(|\mathcal{C}|)$ is equal to minus infinity whenever $|\mathcal{C}|$ is different than n), the problem is simpler. In that case, the demand and default functions are the same regardless of the chosen bundle. In that case, the system of equations can be written exactly as in the previous model:

$$R : \theta R = mc(L) + [1 - \tilde{\varepsilon}R]\varepsilon^{-1} \quad (56)$$

$$L : u'(L) = \frac{mc'(L)}{1 - \tilde{\varepsilon}R} \quad (57)$$

$$V : V'(q)\theta[\frac{mc}{u'(L)} - \tilde{\varepsilon}R] = c'(q) \quad (58)$$

with $\varepsilon^{-1} := \frac{\sum D(U(\mathcal{C}))}{\sum D'(U(\mathcal{C}))}$ which is equal to a constant in a symmetric equilibrium. $\tilde{\varepsilon} := \frac{\partial_R \theta(n)}{\theta(n)}$ is also a constant (equal to $\frac{\nu^{-1} [1 - \frac{1}{n}]}{\alpha + \beta \nu \frac{W}{n}}$).

Solving for R using equations 56 and 57, we get:

$$R = \frac{mc(L) + \varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}} \quad (59)$$

$$u'(L) = \frac{mc'(L)}{1 - \tilde{\varepsilon} \frac{mc(L) + \varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}}} \quad (60)$$

Considering that $u'(L) = \gamma \log(L)$ and $mc(L) = mc \cdot L$, we get:

$$L^* = \gamma \frac{1 - \tilde{\varepsilon} \frac{\varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}}}{mc(1 + \gamma \frac{\tilde{\varepsilon}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}})} \quad (61)$$

$$R^* = \frac{mcL^* + \varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}} \quad (62)$$

Using it into equation and assuming $V'(q) = V \cdot q$ and $c'(q) = cq$, we get:

$$V'(q)\theta[\frac{mcL^*}{\gamma} - \tilde{\varepsilon}R^*] = c'(q) \quad (63)$$

$$q^* = \frac{V\theta[\frac{mcL^*}{\gamma} - \tilde{\varepsilon} \frac{mcL^* + \varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}}]}{c} \quad (64)$$

$$q^* = \frac{V\theta[mcL^*[\frac{1}{\gamma} - \tilde{\varepsilon} \frac{1}{\theta + \tilde{\varepsilon}\varepsilon^{-1}}] - \tilde{\varepsilon} \frac{\varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}}]}{c} \quad (65)$$

The borrower welfare is:

$$W = n\{\gamma \log(L^*) + \theta[Vq^* - R^* - v^d(e^*)] + v^d(e^*)\} \quad (66)$$

with:

$$\theta = \alpha + \frac{\beta\nu}{n} \quad (67)$$

$$v^d(e^*) = v + \nu e^* \quad (68)$$

$$e^* = \nu\beta \frac{W}{n} \quad (69)$$

$$L^* = \gamma \frac{1 - \tilde{\varepsilon} \frac{\varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}}}{mc(1 + \gamma \frac{\tilde{\varepsilon}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}})} \quad (70)$$

$$R^* = \frac{mcL^* + \varepsilon^{-1}}{\theta + \tilde{\varepsilon}\varepsilon^{-1}} \quad (71)$$

$$q^* = \frac{V\theta[\frac{mcL^*}{\gamma} - \tilde{\varepsilon}R^*]}{c} \quad (72)$$

The survival probability θ is a parameter (parametrized by α and the number of cards). The markup ε^{-1} is also a free parameter. $\tilde{\varepsilon}$ is parametrized by ν and the effort

benefits by β . We can thus focus on changing each parameters ($\varepsilon^{-1}, \tilde{\varepsilon}, \theta$) holding effort e^* constant. This allows our comparative static not to depend on our modeling choice of e^* .

The figure below report the result of the simulation for parameters close to our estimates

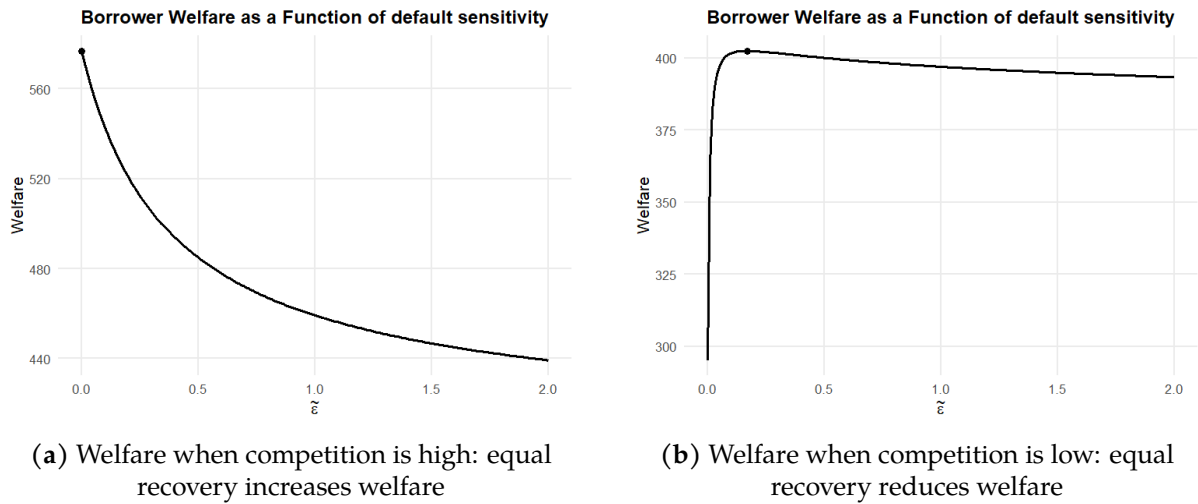


Figure A12: Borrower Welfare as a function of default sensitivities

G Cancelling card upon not repaying

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H Non-exclusive versus exclusive

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I Examples of micro-foundations for V_c

Let us now discuss different functional forms for V_c that can capture the stylized facts presented in [Section 4.2](#). The empirical section suggests that three main channels can affect repayment behavior. First, the lenders can invest in an app or a reward system that the borrower values, making the threat of cancelling the card upon default more costly. Second, the lender can change the credit limit on the card depending on the repayment behavior. Third, the lender can offer another loan product in the second period if the borrower does not default on the card.

Those three channels can be captured using the following parametrization of V_c :

$$\text{App or reward system: } V_c(q_c, \mathcal{R}) = q_c(\mathcal{R}) \quad (73)$$

$$\text{Credit limit: } V_c(\tilde{L}_c, \tilde{R}_c, \mathcal{R}) = \tilde{L}_c(\mathcal{R}) - \beta \tilde{R}_c(\mathcal{R}) \quad (74)$$

$$\text{Other Loan product: } \sum_c V_c(\underbrace{\tilde{L}_c, \tilde{R}_c}_{q_c}, \mathcal{R}) = E_{\varepsilon_b}[\max_b \{\tilde{L}_b(\mathcal{R}) - \beta \tilde{R}_b(\mathcal{R}) + \varepsilon_b\}] \quad (75)$$

In Equation 73, q_c can be interpreted as the app quality or the generosity of the reward system. In Equations 74 and 75, $\tilde{L}_c$ and $\tilde{R}_c$ are the loan size and the debt face value for the credit card and the other loan product, respectively. The variables q_c , $\tilde{L}_c$ and $\tilde{R}_c$ are functions of $\mathcal{R}$. That is, lenders can condition their value based on the borrower's repayment behavior, potentially canceling access to the bank app or the rewards if the borrower does not repay the credit card.

The timing assumption behind Equations 74 and 75 is the following, and requires adding one period to the model. After seeing the repayment in period 2, the lender changes the credit card contract to offer a new loan product. The repayment $\tilde{R}$ of the new contract happens at the end of a third period. The benefit of borrowing $\tilde{L}$ happens in the second period. For simplicity, we assume that the borrower cannot get a new credit card contract in period 2 (for Equation 74), or that the other loan contracts are exclusive in the second period (for Equation 75). This assumption is in line with the empirical evidence that other contracts offered are mortgages or car loans, for which borrowers typically only get one. The fact that the borrower does not get a new card (for Equation 74) can be micro-founded by a model in which the repayment shocks in the third period are perfectly correlated with those in the second period. In that case, since the next section shows that the borrower repays only one contract in the second period, the lender learns that if they have seniority in the third period, and are thus the only lender.

The following section ?? shows that when lending generate a positive net present value, there exist an equilibrium in which it is optimal to exclude borrowers in the second period (i.e., set $q_c = L_c = R_c = 0$ if $c \notin \mathcal{R}$) if they did not repay their debt, and to set $\tilde{R} = \tilde{W}$ if they repays, where $\tilde{W}$ is the borrower wage in the third period. In that case, all three equations are equivalent: They are all increasing functions of one variable (q_c or $\tilde{L}_c$) conditional on repayment. V_c can be normalized to zero conditional on defaulting on card c . This justifies the general approach used in Equation ??.

J Examples of Micro-foundations for c_c

Using the three micro-foundations discussed in Section ??, potential functional forms for the cost function are:

$$\text{Rewards: } c_c(q_c, \theta_c(R_c, q_c)) := c_c(\max\{q_c(\mathcal{R})\}) \text{ positive convex and increasing} \quad (76)$$

$$\text{Credit limit: } c_c(R, \tilde{R}, \tilde{L}, \theta_c(R, \tilde{R}, \tilde{L})) = -\theta_c(R, \tilde{R}, \tilde{L})\{\tilde{R} - mc\tilde{L}\} \quad (77)$$

$$\text{Other Loan: } c_c(R, \tilde{R}, \tilde{L}, \theta_c(R, \tilde{R}, \tilde{L})) = -\theta_c(R, \tilde{R}, \tilde{L})\tilde{D}(\tilde{R}, \tilde{L})\{\tilde{R} - mc\tilde{L}\} \quad (78)$$

In Equation 76, we consider that setting a generous reward system requires the lender to set aside some money in the first period $c(\max\{q_c(\mathcal{R})\})$. The operator $\max\{q_c(\mathcal{R})\}$ captures the fact that the bank pays the cost in the first period and should ensure to have invested enough to provide the maximum reward it committed to.

In equations 77 and 78, the costs are paid in the second period, after the repayment $\mathcal{R}$ is observed. We drop the $\mathcal{R}$ from the variables $(R, \tilde{R}, \tilde{L})$ to simplify the notation. The cost function can be positive or negative depending on the markup the lender sets for the other product, sold in the second period. In Equation 78, $D(\tilde{R}, \tilde{L})$ is the probability of choosing bank c in the second period. It is derived from Equation 75.

As in section ??, when lending is a positive NPV, it is optimal to set $\tilde{R} = \tilde{W}$, where $\tilde{W}$ is the borrower's income in the second period. All the equations are thus functions of one variable (q or $\tilde{L}$).

K Proofs

K1 Proof exclusive contracts equilibrium

With assumption A1, default is not optimal. In addition, A1 also implies that it is optimal to max out the borrower's borrowing capacity (i.e., $R = W$). To see this, one can use the change of variable $R = \alpha L - u + V_c(q)$, and allow the lender to choose (u, L, q) . The first-order condition with respect to L is always positive, so we must be at a corner solution.

More formally, let us denote u the borrower utility and $q := (\bar{q}, \underline{q})$ where $\bar{q}$ is the card quality conditional on repaying the card $\underline{q} \geq 0$ is the reward conditional on default:

$$u := \alpha L + \int_{-\infty}^{\bar{\epsilon}} V(\bar{q}) - R - \epsilon dF(\epsilon) + \int_{\bar{\epsilon}}^{\infty} V(\underline{q}) dF(\epsilon) \quad (79)$$

With $\bar{\epsilon} := V(\bar{q}) - V(\underline{q}) - R$. $\bar{\epsilon}$ is the threshold above which the borrower chooses to default.

The lender problem is:

$$\max_{\bar{q}, q, R, L} D(u) \left[\int_{-\infty}^{\bar{\epsilon}} R dF(\epsilon) - mcL - c(\bar{q}, \underline{q}) \right] \quad (80)$$

Using the change of variable $L = \frac{1}{\alpha} [u - (\int_{-\infty}^{\bar{\epsilon}} V(\bar{q}) - R - \epsilon dF(\epsilon) + \int_{\bar{\epsilon}}^{\infty} V(q) dF(\epsilon))]$, the lender problem becomes:

$$\max_{\bar{q} \geq 0, q \geq 0, W \geq R \geq 0, u} D(u) \left\{ \int_{-\infty}^{\bar{\epsilon}} R dF(\epsilon) - \frac{mc}{\alpha} [u - \int_{-\infty}^{\bar{\epsilon}} V(\bar{q}) - R - \epsilon dF(\epsilon) - \int_{\bar{\epsilon}}^{\infty} V(q) dF(\epsilon)] - c(\bar{q}, \underline{q}) \right\} \quad (81)$$

This setup is general enough to capture the credit limit and new loan cases by letting: $c(q) = \int_{-\infty}^{\bar{\epsilon}} \tilde{c}(\bar{q}) dF(\epsilon) + \int_{\bar{\epsilon}}^{\infty} \tilde{c}(q) dF(\epsilon)$.

The derivative with respect to $\underline{q}$ yields:

$$A := -V'(\underline{q}) R f(\bar{\epsilon}) + \frac{mc}{\alpha} V'(\underline{q}) [1 - F(\bar{\epsilon})] - c_2(\bar{q}, \underline{q}) \quad (82)$$

$$A < 0 \forall q, R > 0, \bar{\epsilon} \implies \underline{q} = 0 \quad (83)$$

The derivative with respect to $\bar{q}$ yields:

$$V'(\bar{q}) R f(\bar{\epsilon}) + \frac{mc}{\alpha} V'(\bar{q}) [F(\bar{\epsilon})] - c_1(\bar{q}, \underline{q}) \quad (84)$$

The derivative with respect to R yields:

$$B := F[1 - \frac{mc}{\alpha}] - R f(\bar{\epsilon}) \quad (85)$$

$$B > 0 \forall R, \bar{\epsilon} \implies R = W \quad (86)$$

The derivative with respect to u yields:

$$\pi := \left[\int_{-\infty}^{\bar{\epsilon}} R dF(\epsilon) - mcL - c(q, \bar{q}) \right] = \frac{D}{D'} \frac{mc}{\alpha} \quad (87)$$

The condition $\alpha - mc(\theta) > 0, \forall \theta$ implies that $F[1 - \frac{mc}{\alpha}] > 0$ in Equation 85. Using the first order conditions with respect to q , we have $\bar{q} > \underline{q}$. Together with the conditions $V_c(q, \{c\}) - \epsilon_{ir} - R > V_c(q, \emptyset), \forall R \leq W_i, \forall \epsilon_{ir} \in S$, we have $f(\bar{\epsilon}) = 0$ if $R < W$. Equation 86 is satisfied so that $R = W$.

The same conditions imply that Equation 83 is satisfied as well ($[1 - F(\bar{\epsilon})] = 0$ and $f(\bar{\epsilon}) = 0, c' > 0$ so that $B < 0$) so that the card is canceled conditional on not repaying.

Rearranging the first order conditions with respect to u (Equation 87) and $\bar{q}$ (Equation 84) using the fact that $F = 1$ and $f = 0, \frac{D}{D'} = \varepsilon^{-1}$ gives us the L^* and q^* conditions.

K2 Proof non-exclusive contracts equilibrium

We start by describing the behavior of borrowers and lenders when borrowers own multiple cards. Then, we show that in equilibrium, borrowers indeed use multiple cards.

The proof proceeds in four steps. We first show that lenders cancel the card when the borrower does not repay the debt entirely. Then, we prove that the borrower chooses to repay only one card. Third, we demonstrate that the borrower uses multiple credit cards in equilibrium. Finally, we derive the equilibrium contract characteristics.

Step 1: Lenders cancel the card conditional on not repaying the debt fully

We show that the lender cancels the credit card if the borrower does not repay the full debt face value. As in proof K1, I denote $q = (\bar{q}, \underline{q})$, where $\underline{q}$ is the card quality conditional on default. We show that lenders optimally set $\underline{q} = 0$ if the borrowers repay less than the debt face value.

We prove the result here for the credit reward case (i.e., the cost is paid in the first period as in Equation 76). Appendix K3 show that this result also holds for the situation in which the borrower also uses the credit card in period 2 (as in equations 77 and 78).

We now describe the timing in the lender problem ?? for the credit reward case. In period 2, the lenders can choose the card quality function $0 \leq q(R_c) \leq q^*$ at no cost. q^* is the maximum card quality, determined by the investment in the first period. R_c^* is the debt face value determined in period 1, R_c is the amount of debt c the borrower chooses to repay in the second period.

The lender maximisation problem ??, conditional on having originated a card, and having already invested in the card maximum quality (q^*) can be rewritten:

$$\max_{q(R_b)} E[R_b] \quad (88)$$

$$s.t. \text{ Borrower behavior : } \{R_b : \max_{R_c} \sum_c V_c(q_c(R_c)) - R_c, s.t. \sum_c R_c \leq W\} \quad (89)$$

$$\text{Repayment lower than debt face value : } R_b \leq R_b^* \quad (90)$$

$$\text{Quality cannot be negative : } q(R_b) \geq 0 \quad (91)$$

$$\text{Maximum quality is bounded by period 1 investment : } q(R_b) \leq \bar{q} \quad (92)$$

To maximize the probability of repayment, the lender sets: $q_c(R_c) = \bar{q}$ for $R_c = R_b^*$ and $q_c(R_c) = 0$ otherwise. This is because they want to maximize $V'_c(q_c(R_c))q'_c(R_c)$. One way to do this is to set $q_c(R_c) = q^*$ when repaying fully and $q_c(R_c) = 0$ otherwise. This is the best response when other lenders also follow this strategy. This equilibrium may not be unique. However, this implementation prevents borrowers from repaying

each credit card a bit, which is consistent with the data.

Step 2: *The borrower repays only one card when they borrow from multiple lenders*

Assumption A1 ensures that lenders set $R^* = W$ as long as $F[1 - \frac{mc}{\alpha}] - Rf(\bar{\epsilon}) > 0$ and that borrowers do not default on all their cards (see appendix K1).

Given that $R_c^* = W$ for all cards c , and the step 1 results, the borrower chooses to default on all but one card.

Step 3: *Borrowers get multiple credit cards*

Assumption A2 ensures that the borrower gets more than one card. To see that, let us show that borrowers using only one contract cannot be an equilibrium.

We show in steps 2 and 3 that the borrower chooses to repay one card at most. Indexing by b the lender from which the borrower got their first card and c any other lender, the borrower can benefit from getting a new card from c with a loan size dL if:

$$\underbrace{\alpha L_b + W + V_b(\{c\}, \{c, b\})}_{\text{utility not repaying lender } b} + \underbrace{\alpha dL - mc(\theta)dL + V_c(\{c\}, \{c, b\})}_{\text{utility new card}} > \underbrace{\alpha L_b + W - R_b + V_b(\{b\}, \{b\})}_{\text{utility repaying card } b \text{ and not getting new card}} \quad (93)$$

Let us now consider that lender b offers the first-best contract so that the right-hand side of Equation 93 is maximized,³³ which makes getting a new card less likely.

Lender b is unwilling to increase the loan size as this would trigger default, which increases the cost of lending and lowers the borrower's utility (see proof K1).

Yet, the borrower has incentives to get a new card from another lender. This happens because the new contract benefits both the borrower and the new lender, thanks to assumption A2 which makes Equation 93 hold.

Indeed, the new contract offers the borrower the possibility to default on the first credit card while still having access to a second credit card at the end of the second period. A2 ensure that this possibility is valuable to the borrower so that the second lender can generate some profits by originating a new card: Formally, the marginal surplus generated by a dollar lent ($\alpha L_b + \alpha + W - mc(\theta)$) plus the cost of defaulting on one card only ($V_c(\{c\}, \{c, b\}) + V_b(\{c\}, \{c, b\})$) is strictly greater than the utility of having only the first best contract ($\alpha L_b + W - R_b + V_b(\{b\}, \{b\})$).

The second lender offers a new contract if they expect the borrower to repay them with some strictly positive probability. In that case, there exists a repayment ($R = \frac{mcL + \epsilon}{\theta}$, $\epsilon > 0$) and loan size such that profits is positive ($D(u)[\theta R - mcL] > 0$) and borrower utility increases when taking an additional card. Step 4 shows that such

³³The same argument holds for any other contract offered. If $R < W$, then a competitor can make profits.

θ exists in a symmetric equilibrium.

Step 4: Deriving the optimal loan size and quality

The borrower utility net of the card specific shocks ϵ_{ic} is denoted:

$$u := \sum_c L_c + \beta \sum_c \{\theta_c(V(q))[W - R_c + V_c(q, c) - E[\epsilon_{ir}|\mathcal{R} = c]] \quad (94)$$

$$+ [1 - \theta_c(V(q))]V_c(0, \mathcal{R} \neq c)\} \quad (95)$$

Where $V(q) := (V_c(q_c))_c$.

We normalize $V_c(c, \mathcal{R} \neq c)$ to zero. The maximization problem is thus:

$$\max_{u, q} D(u)[\theta_c(V(q))W - \frac{mc}{\alpha}[u - \beta \sum_d \{\theta_d((V(q))V_d(q_d) - E[\epsilon_{ir}|\mathcal{R} = c]\})] - c(q)] \quad (96)$$

The first-order conditions with respect to q yield:

$$V'_c(q_c)\theta'_c(V(q))W + \frac{mc\beta}{\alpha}\theta_c V'_c(q_c) = c'(q) \quad (97)$$

I used the fact that $\sum_d \theta'_d[V_d + E[\epsilon|\mathcal{R} = c]] = 0$.

For simplicity, I also assumed that lenders do not internalize the effect of q on capital requirements. The rationale is that the government sets capital requirements, and those are taken as given by the lender.

K3 Repayment with other contracts in the second period

The section shows that it is optimal for the lender to cancel the card, conditional on the borrower not repaying the full debt face value. We focus on the case in which the card quality is interpreted as offering a loan in the second period as in equations 77 and 78.

Formally, let us look at the lender problem in period 2 before the borrower makes his repayment choice. Lender b conditions the credit card offer in the second period on repayment r_b made by the borrower. Denoting the credit card offer $(L(r_b), R(r_b))$, the lender objective function is:

$$\max_{L(r_b), R(r_b)} \int r_b + [R_b(r_b)\theta((r_c)_c) - mcL(r_b)]dF(r_b) \quad (98)$$

$$s.t. F(r_b) \text{ defined by problem 100} \quad (99)$$

Let us assume that the borrower repayment shocks ϵ are persistent for the first and second contract, so that lender b expects to lose seniority with probability one with the new loan contract if they lost it with the old one, that is, $\theta((r_c)_c)$ is zero if they repaid the other lender. This simplifies the exposition by making the second contract "exclusive"

in the second period, as only one lender is willing to lend.

The borrower in the second period chose which card to repay. Their maximization problem is:

$$\begin{aligned} \max_{r:=(r_c)_c} \quad & W + \sum_c \alpha L_c(r_c) - R_c(r_c) - r_c + \epsilon_c \\ \text{s.t.} \quad & \sum r_c \leq W \\ & 0 \leq r_c \leq W \end{aligned} \quad (100)$$

Suppose the other lenders offer $\bar{L}_c, R(r_b) = W$ if the borrower repays the debt in full and zero otherwise. We now show that doing this is the best response for lender b . The maximisation problem implies that the borrower repays only one card.

Using the borrower problem 100, the probability of lender b being repaid is thus: $Pr(\alpha \bar{L}_c + \epsilon_c \leq \max_{r_b \leq W} W - \alpha L(r_b) - r_b + \epsilon_b)$. For the lender, the problem is then equivalent to choosing r_b and L instead of the pricing schedule. Lender utility is maximized at $r_b = W$ as $\alpha > mc$. One way to implement the optimal contract is by offering $\bar{L}_c, R(r_b) = W$ if the borrower repays the debt in full and zero otherwise. The lender's problem is thus to choose the loan size conditional on repayment L_b . This problem is thus similar to the one analyzed in appendix K2. Formally, the lender problem is:

$$\max_{L_b} \overbrace{Pr(\alpha L_b + \epsilon_b \geq \alpha L_c + \epsilon_c \forall c)}^{\theta(L_b)} [W_1 + W_2 - mcL_b] \quad (101)$$

W_1 is the repayment from the first loan and W_2 from the second. The first order conditions give the optimal L_b .

L Cross-Selling

For the case in which a product is sold in the second period, we have:

$$c' := D'\pi + D[\sigma\tilde{W} - \tilde{m}\tilde{L} - \tilde{m}] \quad (102)$$

Plugging it into Equation ??, we have:

$$\frac{\beta}{\alpha} mc = D[\sigma(W - \tilde{m}L) - \tilde{m}] \quad (103)$$

$$\iff \tilde{L}^* = \frac{\tilde{W}}{\tilde{m}} - [1 + \frac{\beta mc}{\alpha \tilde{m}}] \frac{1}{\sigma} \quad (104)$$

$\frac{1}{\sigma}$ is the markup absent cross subsidy. $\frac{\beta mc}{\alpha \tilde{m}} \frac{1}{\sigma}$ is the cross-subsidy incentives: Having

a credit card puts the lender in the borrower consideration set to get another loan.

Using Equation ??, and the fact that $c(q^*) = [1 + \frac{\beta mc}{\alpha \tilde{m}}] \frac{D^*}{\sigma}$

$$L^* = \overbrace{\frac{W - c(q^*)}{mc}}^{\text{Lender break even condition}} - \underbrace{\varepsilon^{-1}}_{\text{inverse demand semi-elasticity}} \quad (105)$$

The borrower utility is thus:

$$u^* = L^* + \beta \tilde{L}^* = \alpha \left[\frac{W + [1 + \frac{\beta mc}{\alpha \tilde{m}}] \frac{D^*}{\sigma}}{mc} - \varepsilon^{-1} \right] + \beta \left[\frac{\tilde{W}}{\tilde{m}} - [1 + \frac{\beta mc}{\alpha \tilde{m}}] \frac{1}{\sigma} \right] \quad (106)$$

$$\Leftarrow u^* = \alpha \left[\frac{W + [1 + \frac{\beta mc}{\alpha \tilde{m}}] \frac{D^*}{\sigma}}{mc} - \varepsilon^{-1} \right] + \beta \left[\frac{\tilde{W}}{\tilde{m}} - [1 + \frac{\beta mc}{\alpha \tilde{m}}] \frac{1}{\sigma} \right] \quad (107)$$

The impact of cross-subsidy is:

$$\frac{\alpha}{mc} \frac{1}{N} \frac{1}{\sigma} + \frac{\beta}{\tilde{m}} \frac{1}{\sigma} \left[\frac{1}{N} - \frac{mc}{\alpha} \right] \quad (108)$$

Cross-subsidy can be a way to attract the borrower to make them on your choice set, which is bad for welfare. On the other hand, this can allow the borrower to get more credit card loans as this provides incentives to lower the markup to attract borrowers.

M Exclusive versus non-exclusive

Let us interpret q^{NE} as an additional loan contract taken in period 2 and use the extreme value random shocks as in Section 6.3.

M1 Exclusive case

The borrower's utility in the exclusive case is:

$$u_i^E = \alpha L + \beta E[W - R + \{\max_c V(q) + \tilde{\sigma} \epsilon_c\}] \quad (109)$$

$$= \alpha L + \beta [W - R + \ln(\sum_c \exp(\tilde{\sigma} V(q_c)))] \quad (110)$$

Assuming that lending a card in the first period puts the borrower in the consideration set for the second product, the profits are:

$$\max_{R, L, q} D(u) \left[R - \frac{mc}{\alpha} L + \tilde{D}(q) [W - mcq] \right] \quad (111)$$

$$\max_{u, q} D(u) \left[W - \frac{mc}{\alpha} (u - \beta \ln(\sum_c \exp(\tilde{\sigma} V(q_c)))) + \tilde{D}(q) [W - mcq] \right] \quad (112)$$

The first-order condition with respect to q yields:

$$\frac{mc}{\alpha} \beta \tilde{\sigma} V'(q_c) \tilde{D}(q) + \tilde{D}'[W - mcq] - \tilde{D}(q)mc = 0 \quad (113)$$

$$\Longleftrightarrow q^E = \frac{W}{mc} - \tilde{\varepsilon}^{-1} \left[1 - \frac{mc}{\alpha} \beta \tilde{\sigma} V'(q_c) \right] \quad (114)$$

The relevant functions at equilibrium are (when denoting N the number of lenders in the consideration set in the second period):

$$c(q) = -\frac{1}{N} [W - mcq] \quad (115)$$

$$V(q) := \gamma q - W - \tilde{R} = \gamma q \quad (116)$$

$$L^E = \frac{W - c(q^E)}{mc} - \epsilon^{-1} \quad (117)$$

$$q^E = \frac{W}{mc} - \tilde{\varepsilon}^{-1} \left[1 - \frac{mc}{\alpha} \beta \tilde{\sigma} \gamma \right] \quad (118)$$

$$u_i^E = \alpha L + \beta [\sigma V(q_c) + \ln(N)] \text{ in a symmetric equilibrium} \quad (119)$$

M2 Non-exclusive case

$$u_i^{NE} := \sum_{c \in B} \alpha L_c^{NE} + \beta E[\max_c \{ \overbrace{E[\max_d V(q|c) + \tilde{\sigma} \epsilon_d]}^{V_c} \} + \varepsilon_c] \quad (120)$$

$$= \sum_{c \in B} \alpha L_c^{NE} + \beta [\ln(\sum_d \exp(V(c)))] \quad (121)$$

The lender problem is:

$$\max_{R, L, q} D(u) \left[\theta R - \frac{mc}{\alpha} L + \theta_c \tilde{D}(q) [W - mcq] \right] \quad (122)$$

$$\max_{u, q} D(u) \left[\theta W - \frac{mc}{\alpha} (u - \beta [\ln(\sum_d \exp(V(c)))]) + \tilde{\theta}_c D(q) [W - mcq] \right] \quad (123)$$

Taking the first order condition with respect to q :

$$\theta' W + \frac{mc}{\alpha} \beta \tilde{\sigma} V'(q_c) \theta_c \tilde{D}(q) + [\theta(q) \tilde{D}'(q) + \theta'(q) \tilde{D}(q)] [W - mcq] - \theta(q) \tilde{D}(q) mc = 0 \quad (124)$$

$$\Longleftrightarrow q^{NE} = \frac{W}{mc} \underbrace{-\tilde{\varepsilon}_{NE}^{-1}}_{\text{lower markup}} + \underbrace{\varepsilon_{NE}^{-1} \frac{mc}{\alpha} \beta \tilde{\sigma} V'(q_c)}_{\text{under-investment}} + \underbrace{\frac{\theta'}{\delta} W}_{\text{over-investment}} \quad (125)$$

with $\delta := \theta(q) \tilde{D}'(q) + \theta'(q) \tilde{D}(q)$.

The markup tends to be lower because of the competition for loan seniority.

The relevant functions are now:

$$c(q) = -\theta \frac{1}{N} [W - mcq] \quad (126)$$

$$V(q) := \gamma q \quad (127)$$

$$L^{NE} = \frac{\theta W - c(q^E)}{mc} - \epsilon^{-1} \quad (128)$$

$$q^{NE} = \frac{W}{mc} - \tilde{\epsilon}_{NE}^{-1} [1 - \frac{mc}{\alpha} \beta \tilde{\sigma} \gamma] \quad (129)$$

$$u_i^E = \sum_c \alpha L_c + \beta [\sigma V(q_c)] + \ln(N) + \ln(|B|) \text{ in a symmetric equilibrium} \quad (130)$$

M3 Comparison exclusive versus non exclusive

We have:

$$u_i^{NE} = \alpha L_c^{NE} \cdot |B| + \beta [\tilde{\sigma} V_c(q^{NE}) + \ln(|B|)] \quad (131)$$

$$u_i^E := \alpha L_c^E + \beta \tilde{\sigma} V_c(q^E) \quad (132)$$

The welfare impact of non-exclusive contract are ambiguous. Non-exclusive contract benefits the borrower due to their love for variety (captured by $\ln(|B|)$). It also decreases the markup for the loan market in the second period due to the competition for loan seniority ($\epsilon_{NE}^{-1} < \epsilon^{-1}$). However, a non-exclusive contract creates a debt dilution problem, which can create distortions (i.e., the over-investment channel $\frac{\theta'}{\delta} W$).

N Identification of supply parameters and counterfactual

The key challenge with this type of model is tractability. This section presents some modelling assumptions that allow us to achieve tractability and uniqueness of the equilibrium.

N1 Borrowers

The borrower chose the number of cards, the card in their portfolio and which card to default on.

The choice of how many cards to own is:

$$Pr(\mathcal{C}_i, \mathcal{R}_i) = Pr(Card(\mathcal{C}_i)) \cdot Pr(\mathcal{C}_i | Card(\mathcal{C}_i)) \cdot Pr(\mathcal{R}_i | \mathcal{C}_i, Card(\mathcal{C}_i)) \quad (133)$$

$\mathcal{C}_i$ is the choice of credit cards the borrower gets. $\mathcal{R}_i$ is the set of cards the borrower

repays.

Borrower problem: The borrower's choice of $\mathcal{C}_i$ and $\mathcal{R}_i$ is driven by the following problem:

$$\max_{\mathcal{C}_i} U_{i\mathcal{C}_i} + \varepsilon_{i\mathcal{C}} + E\left[\max_{r \in P(\mathcal{C}_i)} V_r + \varepsilon_{ir}\right] \quad (134)$$

with $V_r := \sum_{c \in \mathcal{R}_i} v_{ic_j}$, and $U_{i\mathcal{C}} := \sum_{c \in \mathcal{C}} u_{ic_j}$

Assuming the shocks are extreme value distributed, we have closed-form expressions. Indeed, when ε_{ic_j} is extreme value distributed, we have $E[\max_{r \in P(\mathcal{C}_i)} V_r] := I(\mathcal{C}_i) := \ln(\sum_{r \in P(\mathcal{C}_i)} \exp(\frac{V_r}{\lambda}))$. In that case, the probability of seeing the set of cards $\mathcal{C}_i$ is:

$$Pr(\mathcal{C}_i | Card(\mathcal{C}_i)) = \frac{\exp(U_{i\mathcal{C}} + \lambda I(\mathcal{C}))}{\sum_{m \in M(n)} \exp(U_{im} + \lambda I(m))} \quad (135)$$

$M(n)$ is the set of all combinations of cards the borrower can choose from that have n cards.

N1.1 Default

The utility of borrower i of not defaulting on the card c_j offered by bank j is:

$$u_{ic_j} = \beta_i X_{c_j} + \xi_{c_j} + \varepsilon_{ic_j} \quad (136)$$

X_{c_j} are card characteristics and β_i is a function of borrower observable characteristics. ξ_{c_j} is a card bank fixed effect.

With probability $d(\mathcal{C}_i)$ they enter the default state. Conditional on default, they universally default with probability $1 - \lambda$, the lender j survival probability for borrower is thus:

$$\theta_{ic_j}(\mathcal{C}_i) = (1 - d(\mathcal{C}_i)) + d(\mathcal{C}_i) \lambda_i Pr(u_{ic_j} \geq u_{ic_h}, \forall h | \mathcal{C}_i) \quad (137)$$

N2 Lender

Lender maximizes static profits. We allow lenders to invest in marketing (ξ) at the cost ($mc^m(\xi)$) to increase their demand. This approach is used for tractability so that the model can be solved using a contraction mapping. This assumption is consistent with the empirical findings of [Drechsler et al. \(2025\)](#), which shows that marketing is a big

component of the cost of credit cards. The lender problem is:

$$\max_{X, \xi} D(f(X) + \xi) \overbrace{[\tau + p_r \cdot [\theta(X)r - mc]L - mc^m(\xi)]}^{\pi(X, u)} \quad (138)$$

$D(f(X) + \xi)$ is the probability that the borrower chooses the bank, conditional on the number of cards he has.

τ is the profits coming from fees, p_r is the probability of revolving.

$\theta(X)$ is the probability of repaying the loan, given the borrower's card choice set.

Using the change of variable $U = f(X) + \xi$, we have:

$$\max_{X, U} D(U) \overbrace{[\tau + p_r \cdot [\theta(X)r - mc]L + mc^m(U - \beta X)]}^{\pi(X)} \quad (139)$$

The first-order condition with respect to X yields (assuming $mc(d, U - \beta X)$ linear):

$$rL : p_r \cdot [\theta + \theta' rL] = mc^m \beta^r \quad (140)$$

$$L : \tau - p_r mc + L\theta' r + mc^m \beta^L = 0 \quad (141)$$

Those two equations allow us to recover the marginal costs conditional on having default elasticities and revolving probabilities (mc^m for Equation 140 and $\tau - p_r mc$ for Equation 141).

Let us now look at how to solve for counterfactuals.

Using Equation 141, we can express $L\theta' r$ as a function of parameters. Let's denote this γ . Plugging it into the Equation 140 and using the fact that $\theta' = \sigma\theta(1 - \theta)$, we can solve for R for each bank b . Using the formula for $\frac{\theta_b}{\theta_c}$, we can then recover L_c as a function of L_b . Using the θ_b we can then solve for L_b .

Let us consider that we can solve for R and L . In that case, the maximization problem is of the form:

$$\max_U D(U)[A - mcU] \quad (142)$$

Where A is a constant (derived below). The first-order condition yields:

$$D'(U)[A - mcU] = Dmc \quad (143)$$

When D has a logit form, we have:

$$[A - mcU] = D[A - mcU] + mc \quad (144)$$

Indeed, the function $F(A - mcU) := D(U)[A - mcU] + mc$ is a contraction mapping

as $D(U) < 1$.